SHARP TWO-SIDED SIZE OF CHAOS ESTIMATES FOR DIFFUSIVE GASES IN THE HILBERT–SCHMIDT REGIME

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Abstract. We identify the complete first-order correction to independence for equilibrium mean-field Gibbs ensembles and quantify its approximation uniformly in the marginal size. The correction is an explicit sum of singleton and Hoeffding-centered pair terms. An abstract theorem for symmetric interactions with square-integrable centered kernels, suitable row and source moments, and an amplified centered-pair exponential bound gives algebraic approximation rates without a sign condition on the centered interaction operator. For repulsive logarithmic and Riesz gases below the Hilbert–Schmidt threshold, these hypotheses hold at every fixed positive inverse temperature for bounded background densities, with a polynomial-moment condition in the logarithmic case and no additional spatial moment in the Riesz case. The approximation determines the leading constants in chi-square divergence, relative entropy, and total variation for fixed marginals, and the normalized limits for every growing marginal containing a vanishing fraction of the particles. Its proof combines quantitative Laplace comparison, correlated replicas, screened response, and a second-order cavity expansion, followed by orthogonal Hoeffding projection. Under a separate, more flexible cavity criterion, we also prove matching order bounds: total variation is of order k/N, and relative entropy and chi-square divergence are of order k^2/N^2, uniformly for $2 \le k \le N$; the upper bounds include $k = 1$. Under compatible moment and cavity assumptions, a subquadratic partition-function moment suffices for order-N^{-1} convergence of every fixed marginal in L^τ and total variation, for $1 < \tau < 2$. At the critical Riesz exponent $\mathsf{s} = \mathsf{d}/2$, logarithmically enhanced fixed-particle lower bounds and a sequential Hellinger argument obstruct the classical scales. In the strictly supercritical range $\mathsf{d}/2 < \mathsf{s} < \mathsf{d}$, depleted close pairs give the corresponding mesoscopic obstruction. The quantitative exponents are sufficient and are not claimed optimal.

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1. Introduction

We study the leading dependence among labeled particles in fixed-temperature mean-field Gibbs ensembles. What is the first correction to independence, and how accurately does it describe a marginal whose size grows with the number of particles? Which leading constants does this correction determine in statistical divergences? How does the singularity of the interaction limit these conclusions? Our principal results identify and quantify the marginal correction under abstract analytic hypotheses, with applications to logarithmic and Riesz interactions. A separate order theory and ultraviolet obstructions complement these asymptotic results.

Let $W : (\mathbb{R}^{\mathsf{d}})^2 \to \mathbb{R}$ be a symmetric two-body interaction potential and let $V : \mathbb{R}^{\mathsf{d}} \to \mathbb{R}$ be a confining potential. In the mean-field scaling, the Hamiltonian of an N-particle configuration $X_N = (x_1, \dots, x_N)$ is

$$\mathcal{H}_N(X_N) \coloneqq \frac{1}{2N} \sum_{1 \le i \ne j \le N} W(x_i, x_j) + \sum_{i=1}^{N} V(x_i), \tag{1.1}$$

and the associated canonical Gibbs measure at inverse temperature $\beta > 0$ is

$$\mathrm{d}\mathbb{P}_{N,\beta}(X_N) = \frac{1}{Z_{N,\beta}} e^{-\beta \mathcal{H}_N(X_N)} \, \mathrm{d}X_N. \tag{1.2}$$

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When the corresponding overdamped interacting diffusion is well posed, (1.2) is its equilibrium law. We work at fixed inverse temperature as $N \to \infty$, which we call the *diffusive temperature regime*.

An absolutely continuous thermal equilibrium measure μ is characterized by the Euler–Lagrange relation

$$\mathrm{d}\mu(x) = Z_\mu^{-1} e^{-\beta(V(x)+h_\mu(x))}\,\mathrm{d}x, \qquad h_\mu(x) := \int_{\mathbb{R}^{\mathsf{d}}} W(x,y)\,\mathrm{d}\mu(y). \tag{1.3}$$

When this thermal equilibrium measure is unique, qualitative chaos asserts that, for every fixed k, the labeled marginal $\mathbb{P}_{N,\beta}^{(k)}$ converges to $\mu^{\otimes k}$. Equivalently, the empirical measure converges in law to the deterministic measure μ; see Kac [Kac56], Sznitman [Szn91], and Hauray–Mischler [HM14]. The quantitative question is finer: how much dependence remains among k labeled particles when k is fixed or grows with N?

The principal positive result is an explicit singleton-plus-pair approximation of the marginal density, with algebraic errors uniform in the number of observed particles. It identifies the leading divergence constants for fixed marginals and, after normalization, for every growing marginal containing a vanishing fraction of the particles. The underlying abstract theorem allows general symmetric interactions satisfying row, source, and amplified exponential-moment bounds, with no sign condition on the centered interaction operator; the logarithmic and Riesz statements are applications. The precise results are given in Section 2.2, and the abstract theorem in Theorem 3.1.

Following Lacker [Lac22], it is useful to distinguish this *local* question from the *global* problem of controlling the full law or the empirical measure. The distinction is already visible in relative entropy. A global estimate

$$\mathsf{H}(\mathbb{P}_{N,\beta} \mid \mu^{\otimes N}) = O(1) \tag{1.4}$$

combined with entropy subadditivity gives only

$$\mathsf{H}(\mathbb{P}_{N,\beta}^{(k)} \mid \mu^{\otimes k}) = O(\frac{k}{N}). \tag{1.5}$$

For regular continuous Gibbs measures, Lacker showed that this loss is not intrinsic: under a perturbative force-level transport-information hypothesis, he proved a relative-entropy upper bound of quadratic order in k/N and exhibited a Gaussian model attaining that order, showing that the exponent cannot in general be improved [Lac22]. Ren and Wang subsequently obtained the same entropy scale nonperturbatively in several convex regimes by means of a hierarchy of conditional entropies [RW25]. Singular logarithmic and Riesz interactions lie outside both theories.

The independent order theorem Theorem 3.4 and its model verification in Section 7 give the following sharp scales. For every fixed $\beta > 0$, a bounded background density, and either $0 < \mathsf{s} < \mathsf{d}/2$ or the logarithmic moment criterion of Lemma 7.5, there are positive comparison constants and N_0, independent of k, such that, for every $N \geq N_0$ and $2 \leq k \leq N$,

$$\mathrm{TV}(\mathbb{P}_{N,\beta}^{(k)}, \mu^{\otimes k}) \asymp \frac{k}{N}, \qquad \mathsf{H}(\mathbb{P}_{N,\beta}^{(k)} \mid \mu^{\otimes k}) \asymp \frac{k^2}{N^2}, \qquad \chi^2(\mathbb{P}_{N,\beta}^{(k)} \mid \mu^{\otimes k}) \asymp \frac{k^2}{N^2}. \tag{1.6}$$

The corresponding upper bounds also hold for $k = 1$. The logarithmic order criterion is weaker than the hypothesis used for the quantitative marginal approximation. These conclusions do not assert a positive singleton lower bound: for a uniform background and a translation-invariant interaction on the flat torus, the one-point marginal is exactly the background.

For a Riesz singularity of order s, the centered pair interaction is locally square-integrable precisely when $2\mathsf{s} < \mathsf{d}$. At and above this threshold, near-diagonal configurations produce lower bounds that rule out the classical fixed-particle chi-square and entropy scales and the uniform total-variation scale. The latter obstruction occurs for a vanishing fraction of the particles. These are one-sided conclusions: we do not determine matching ultraviolet upper bounds or limiting profiles. The optimal convergence rates and extensions under weaker exponential-moment hypotheses remain open.

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The pair-correlation and quadratic-information mechanisms underlying these scales have antecedents in the Bogolyubov hierarchy for connected correlations [Bog62] and in classical multiparticle expansions of equilibrium entropy.[1]

2. Main results and proof strategy

2.1. **Modulated Gibbs formulation and divergence conventions.** We now introduce the centered formulation used throughout the paper. Let $\mu \in \mathcal{P}(\mathbb{R}^{\mathsf{d}})$ be a Borel probability measure, and let W be real, symmetric, measurable, and integrable with respect to $\mu^{\otimes 2}$. No absolute continuity or bounded-density assumption is needed for these definitions. When a measure is absolutely continuous, we use the same symbol for the measure and its Lebesgue density. The integrability assumption makes the potentials below finite almost everywhere; all identities are understood with respect to the appropriate product of μ. Set

$$h_\mu(x) \coloneqq \int_{\mathbb{R}^{\mathsf{d}}} W(x,y)\,\mathrm{d}\mu(y), \qquad I_\mu \coloneqq \frac{1}{2}\int_{(\mathbb{R}^{\mathsf{d}})^2} W(x,y)\,\mathrm{d}\mu(x)\mathrm{d}\mu(y), \tag{2.1}$$

and define the Hoeffding centering of a two-body kernel by

$$\widetilde{f}(x,y) \coloneqq f(x,y) - \int f(x,z)\,\mathrm{d}\mu(z) - \int f(z,y)\,\mathrm{d}\mu(z) + \int f(z,w)\,\mathrm{d}\mu^{\otimes 2}(z,w). \tag{2.2}$$

Thus,

$$\widetilde{W}(x,y) = W(x,y) - h_\mu(x) - h_\mu(y) + 2I_\mu, \qquad \int \widetilde{W}(x,y)\,\mathrm{d}\mu(y) = 0. \tag{2.3}$$

The modulated energy is

$$\mathsf{F}_N(X_N,\mu) = \frac{1}{2N^2}\sum_{1\le i\ne j\le N} \widetilde{W}(x_i,x_j) - \frac{1}{N^2}\sum_{i=1}^N h_\mu(x_i) + \frac{I_\mu}{N}, \tag{2.4}$$

and its partition function takes values in $(0,\infty]$; whenever it is finite, the associated Gibbs measure is defined by

$$\mathsf{K}_{N,\beta}(\mu) \coloneqq \int e^{-\beta N \mathsf{F}_N(X_N,\mu)}\,\mathrm{d}\mu^{\otimes N}(X_N), \tag{2.5}$$

$$\mathrm{d}\mathbb{Q}_{N,\beta}(\mu)(X_N) \coloneqq \frac{1}{\mathsf{K}_{N,\beta}(\mu)} e^{-\beta N \mathsf{F}_N(X_N,\mu)}\,\mathrm{d}\mu^{\otimes N}(X_N). \tag{2.6}$$

The hypotheses of each result below ensure finiteness of every partition function used to define a Gibbs measure in that result. The exact splitting formula

$$\frac{1}{2N}\sum_{1\le i\ne j\le N} W(x_i,x_j) = N\mathsf{F}_N(X_N,\mu) + \sum_{i=1}^N h_\mu(x_i) - NI_\mu \tag{2.7}$$

shows that, when μ satisfies (1.3), the modulated Gibbs measure $\mathbb{Q}_{N,\beta}(\mu)$ coincides with the physical Gibbs ensemble $\mathbb{P}_{N,\beta}$. More generally, $\mathbb{Q}_{N,\beta}(\mu)$ is the natural Gibbs tilt of the prescribed product background $\mu^{\otimes N}$. Its normalizing constant $\mathsf{K}_{N,\beta}(\mu)$ is the exact global cost of this change of reference.

[1] The Bogolyubov hierarchy predicts that the connected two-particle correction has order N^{-1}, and more generally that the connected $(m+1)$-particle correlation has order N^{-m}; see [Bog62]. Duerinckx proved this scaling, in suitable negative Sobolev norms, for smooth classical mean-field dynamics [Due21]. In the entropy expansions of H. S. Green, Stratonovich, and Nettleton–Green [Gre52, Str55, NG58], the two-body contribution is governed by the relative-entropy integrand $g\log g - g + 1$ and hence begins quadratically at the uncorrelated state $g = 1$. Taken together, these observations suggest the k^2/N^2 local entropy scale, but they do not provide a nonasymptotic relative-entropy estimate for k-particle marginals. The present paper shows that this k^2/N^2 prediction fails for sufficiently singular interactions; the older statistical-mechanics expansions did not isolate the relevant regularity threshold.

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Uniform control of this cost bounds the full Gibbs density, but it does not by itself yield the sharp local gain in (1.6). For the splitting formula and the modulated-energy formalism in Coulomb and Riesz gases, see [Ser24, Chapter 5, Section 5.1].

For $1 \le k \le N$, the labeled k-point marginal is

$$\mathbb{Q}^{(k)}_{N,\beta}(\mu) := (\pi_{1,\dots,k})_{\#}\mathbb{Q}_{N,\beta}(\mu), \qquad \pi_{1,\dots,k}(x_1,\dots,x_N) := (x_1,\dots,x_k). \tag{2.8}$$

For $\nu \ll \pi$, we respectively denote the chi-square divergence, relative entropy or Kullback–Leibler divergence, and total-variation distance by

$$\chi^2(\nu \mid \pi) := \int (\frac{\mathrm{d}\nu}{\mathrm{d}\pi} - 1)^2 \,\mathrm{d}\pi, \tag{2.9}$$

$$\mathsf{H}(\nu \mid \pi) := \int \frac{\mathrm{d}\nu}{\mathrm{d}\pi} \log \frac{\mathrm{d}\nu}{\mathrm{d}\pi} \,\mathrm{d}\pi, \tag{2.10}$$

$$\mathrm{TV}(\nu,\pi) := \frac{1}{2}\int |\frac{\mathrm{d}\nu}{\mathrm{d}\pi} - 1| \,\mathrm{d}\pi. \tag{2.11}$$

We write

$$f^{(k)}_{N,\beta} := \frac{\mathrm{d}\mathbb{Q}^{(k)}_{N,\beta}(\mu)}{\mathrm{d}\mu^{\otimes k}}, \tag{2.12}$$

and set $f_{N,\beta} = f^{(N)}_{N,\beta}$.

2.2. **Quantitative marginal approximation.** The next two theorems are applications of Theorem 3.1, which treats general real symmetric interactions under analytic row, source, and exponential-moment assumptions. No sign condition is imposed on the centered interaction operator. The first theorem gives the complete marginal approximation and its divergence errors; the second identifies the normalized divergence limits for growing marginals. Their proofs are common to the abstract class; Section 7.4 verifies the logarithmic and Riesz hypotheses.

For the model statements, use

$$\mathsf{g}_{\mathsf{s}}(z) = \begin{cases} -\log|z|, & \mathsf{s} = 0, \\ \mathsf{s}^{-1}|z|^{-\mathsf{s}}, & 0 < \mathsf{s} < \mathsf{d}. \end{cases} \tag{2.13}$$

We introduce the common profiles and rate before stating the two theorems.

For a symmetric centered kernel $\widetilde{W} \in L^2(\mu^{\otimes 2})$, we use the integral operator

$$T_\mu f(x) := \int_{\mathbb{R}^{\mathsf{d}}} \widetilde{W}(x,y) f(y) \,\mathrm{d}\mu(y). \tag{2.14}$$

Since $\widetilde{W} \in L^2(\mu^{\otimes 2})$, the operator T_μ is self-adjoint and Hilbert–Schmidt on $L^2(\mu)$. Canonicity gives $T_\mu \mathbf{1} = 0$, but no sign condition is imposed on the remaining spectrum.

Whenever $\mathrm{Id} + \beta T_\mu$ is strictly positive and boundedly invertible, set

$$R_{\beta,\mu} := (\mathrm{Id} + \beta T_\mu)^{-1}. \tag{2.15}$$

The operator

$$B_{\beta,\mu} := R_{\beta,\mu} - \mathrm{Id} = -\beta T_\mu R_{\beta,\mu} \tag{2.16}$$

is Hilbert–Schmidt and has a real symmetric canonical kernel $b_{\beta,\mu}$. For the repulsive log/Riesz models, the required invertibility holds at every $\beta \ge 0$ by nonnegativity of T_μ. Under the abstract quantitative hypotheses in Theorem 3.1, it follows instead from the amplified exponential bound. Define the centered one-body function and kernel sections by

$$\psi_\mu := h_\mu - 2I_\mu, \tag{2.17}$$

$$w_x(y) := \widetilde{W}(x,y). \tag{2.18}$$

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The row response and the singleton profile are

$$\omega_{\beta,\mu}(x) \coloneqq \beta^2 \langle w_x, R_{\beta,\mu} w_x \rangle_{L^2(\mu)}, \qquad \omega^\circ_{\beta,\mu} \coloneqq \omega_{\beta,\mu} - \int \omega_{\beta,\mu}\, \mathrm{d}\mu, \tag{2.19}$$

$$a_{\beta,\mu} \coloneqq R_{\beta,\mu}\left(\beta\psi_\mu + \frac{1}{2}\omega^\circ_{\beta,\mu}\right). \tag{2.20}$$

The estimates below show that these functions belong to $L^2(\mu)$ under either the model or the abstract quantitative hypotheses. Both ψ_μ and $\omega^\circ_{\beta,\mu}$ are centered, and $R_{\beta,\mu}$ preserves $L^2_0(\mu)$; hence, $a_{\beta,\mu}$ is centered. For each integer $k \geq 1$, define

$$G_{k,\beta,\mu}(x_1, \dots, x_k) \coloneqq \sum_{i=1}^{k} a_{\beta,\mu}(x_i) + \sum_{1 \leq i < j \leq k} b_{\beta,\mu}(x_i, x_j). \tag{2.21}$$

Canonicity makes distinct singleton and pair terms orthogonal. Consequently,

$$\|G_{k,\beta,\mu}\|^2_{L^2(\mu^{\otimes k})} = k\|a_{\beta,\mu}\|^2_{L^2(\mu)} + \binom{k}{2}\|b_{\beta,\mu}\|^2_{L^2(\mu^{\otimes 2})}. \tag{2.22}$$

The kernel/operator correspondence gives $\|b_{\beta,\mu}\|_2 = \|B_{\beta,\mu}\|_{\mathrm{HS}}$.

For the logarithmic model, we use the explicit cavity-envelope constants

$$C_{\mathrm{loc}} \coloneqq \|\mu\|_{L^\infty} \int_{|z|<1} -\log|z|\, \mathrm{d}z, \tag{2.23}$$

$$\mathsf{c}_{\log,\beta,\mu} \coloneqq \beta\left(3C_{\mathrm{loc}} + 4|I_\mu| + 1\right). \tag{2.24}$$

This agrees with (3.23) for the envelope constants specified in Lemma 7.2.

In the Riesz case, fix

$$0 < \alpha \leq 1, \qquad 2 + \alpha < \mathsf{d}/\mathsf{s}, \tag{2.25}$$

and, in the logarithmic case, set $\alpha = 1$. Define the model rate

$$\mathfrak{r}_N \coloneqq \begin{cases} N^{-\alpha/8}, & 0 < \mathsf{s} < \mathsf{d}/2, \\ N^{-1/8}, & \mathsf{s} = 0. \end{cases} \tag{2.26}$$

For example, $\alpha = \min\{1, (\mathsf{d}/\mathsf{s} - 2)/2\}$ is admissible in the Riesz case. Constants in all the estimates below may depend on this fixed choice of α, on the model parameters, and on the stated moment data, but not on N or k. The exponents and the logarithmic moment threshold are sufficient and are not claimed optimal.

Theorem 2.1 (Quantitative exact marginal profile)**.** *Assume one of the following model hypotheses:*

(i) $\mathsf{d} \geq 1$, $0 < \mathsf{s} < \mathsf{d}/2$, $W(x,y) = 1/(\mathsf{s}|x-y|^{\mathsf{s}})$, μ *is a probability measure on* $\mathbb{R}^{\mathsf{d}}$ *with bounded density, and* $\beta \geq 0$*;*

(ii) $\mathsf{d} \geq 1$, $\mathsf{s} = 0$, $W(x,y) = -\log|x-y|$, $\beta > 0$, μ *is a probability measure on* $\mathbb{R}^{\mathsf{d}}$ *with bounded density, and, for some* $q > 0$*,*

$$\int_{\mathbb{R}^{\mathsf{d}}} (e + |x|)^q\, \mathrm{d}\mu(x) < \infty, \tag{2.27}$$

$$q > \max\left\{4\beta,\, 36\mathsf{c}_{\log,\beta,\mu} + 8\right\}. \tag{2.28}$$

Then, for every $N \geq 2$ *and* $1 \leq k \leq N$*,*

$$\left\| \frac{N}{k}(f^{(k)}_{N,\beta} - 1) - \frac{1}{k} G_{k,\beta,\mu} \right\|_{L^2(\mu^{\otimes k})} \leq C\left(\mathfrak{r}_N + \sqrt{\frac{k}{N}}\right). \tag{2.29}$$

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Moreover,

$$\left|N^2\chi^2\left(\mathbb{Q}^{(k)}_{N,\beta}(\mu)\mid\mu^{\otimes k}\right)-\|G_{k,\beta,\mu}\|_2^2\right|\le C\left(k\mathfrak{r}_N+k^2\mathfrak{r}_N^2+\frac{k^3}{N}\right), \tag{2.30}$$

$$\left|N^2\mathsf{H}\left(\mathbb{Q}^{(k)}_{N,\beta}(\mu)\mid\mu^{\otimes k}\right)-\frac{1}{2}\|G_{k,\beta,\mu}\|_2^2\right|\le C_\alpha\left(k\mathfrak{r}_N+k^2\mathfrak{r}_N^2+k^2(k/N)^\alpha\right), \tag{2.31}$$

$$\left|N\,\mathrm{TV}\left(\mathbb{Q}^{(k)}_{N,\beta}(\mu),\mu^{\otimes k}\right)-\frac{1}{2}\|G_{k,\beta,\mu}\|_1\right|\le C\left(k\mathfrak{r}_N+\frac{k^{3/2}}{\sqrt{N}}\right). \tag{2.32}$$

For fixed k, the estimates identify the strong first-order marginal limit and the leading constants in all three divergences; the constants in the two quadratic divergences are explicit in (2.22). The logarithmic moment threshold is sufficient and nonoptimal, while no additional moment assumption is imposed in the Riesz case. The logarithmic conclusion at $\beta=0$ is immediate, since the Gibbs law is the product law and $a_{0,\mu}$, $b_{0,\mu}$, and $G_{k,0,\mu}$ vanish.

Let $(\lambda_j(B_{\beta,\mu}))_j$ be the nonzero eigenvalues of $B_{\beta,\mu}$, counted with multiplicity, and let $(Z_j)_j$ be independent standard real Gaussian variables. Define the centered Gaussian quadratic form

$$\mathcal{Q}_{\beta,\mu}\coloneqq\sum_j\lambda_j(B_{\beta,\mu})(Z_j^2-1)\quad\text{as a limit in } L^2. \tag{2.33}$$

This definition uses only the Hilbert–Schmidt property; $B_{\beta,\mu}$ need not be trace class. In particular,

$$\mathbb{E}\left[\mathcal{Q}^2_{\beta,\mu}\right]=2\|B_{\beta,\mu}\|^2_{\mathrm{HS}}. \tag{2.34}$$

Theorem 2.2 (Quantitative divergence limits for growing marginals)**.** *Under the same hypotheses, for every $N\ge 2$ and $1\le k\le N$,*

$$\left|\frac{N^2}{k^2}\chi^2\left(\mathbb{Q}^{(k)}_{N,\beta}(\mu)\mid\mu^{\otimes k}\right)-\frac{1}{2}\|B_{\beta,\mu}\|^2_{\mathrm{HS}}\right|\le C\left(\mathfrak{r}_N^2+\frac{1}{k}+\frac{k}{N}\right), \tag{2.35}$$

$$\left|\frac{N^2}{k^2}\mathsf{H}\left(\mathbb{Q}^{(k)}_{N,\beta}(\mu)\mid\mu^{\otimes k}\right)-\frac{1}{4}\|B_{\beta,\mu}\|^2_{\mathrm{HS}}\right|\le C_\alpha\left(\mathfrak{r}_N^2+\frac{1}{k}+(k/N)^\alpha\right), \tag{2.36}$$

$$\left|\frac{N}{k}\,\mathrm{TV}\left(\mathbb{Q}^{(k)}_{N,\beta}(\mu),\mu^{\otimes k}\right)-\frac{1}{4}\mathbb{E}\left[|\mathcal{Q}_{\beta,\mu}|\right]\right|\le C_\alpha\left(\mathfrak{r}_N+\sqrt{\frac{k}{N}}+k^{-\alpha/(2(2+\alpha))}\right). \tag{2.37}$$

Thus, all three normalized divergences converge to the displayed constants along every sequence $k_N\to\infty$ with $k_N=o(N)$. In the logarithmic case, the last error in (2.37) is $k^{-1/6}$, and the right-hand sides of (2.35) and (2.36) are both bounded by $C(\mathfrak{r}_N^2+k^{-1}+k/N)$.

The normalized limits require only $k_N\to\infty$ and $k_N/N\to 0$. Requiring the undivided density error $\|N(f^{(k)}_{N,\beta}-1)-G_{k,\beta,\mu}\|_2$ and the divergence errors in (2.30)–(2.32) to vanish is stronger; Corollary 6.6 gives sufficient windows. On their common hypotheses, the quantitative estimates recover the order bounds. The separate criterion in Theorem 3.4 retains the weaker logarithmic hypothesis and its broader weighted cavity formulation; its model specializations are recorded in Section 7.

2.3. **Ultraviolet obstructions.** The exponent $\mathsf{s}=\mathsf{d}/2$ is the exact Hilbert–Schmidt threshold. For bounded densities, the centered Riesz kernel belongs to $L^2(\mu^{\otimes 2})$ precisely when $2\mathsf{s}<\mathsf{d}$. The authors' companion paper [DGR26, Proposition 5.10] also proves that the modulated partition function diverges for every fixed $\beta>0$ throughout $\mathsf{d}/2\le\mathsf{s}<\mathsf{d}$. Below the threshold, its boundedness is an input to our upper bounds. At and above the threshold, however, partition-function divergence does not imply a local marginal lower bound: the results below are proved independently from the near-diagonal geometry of the Riesz interaction. The two papers therefore identify global and local

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manifestations of the same ultraviolet obstruction, but neither supercritical statement is a formal consequence of the other.

The upper endpoint $\mathsf{s} < \mathsf{d}$ in the next theorem is imposed by the modulated mean-field formulation, not by the existence of the bare finite-particle repulsive Gibbs law. The latter may remain meaningful in the hypersingular regime $\mathsf{s} \geq \mathsf{d}$, but the continuum potential h_μ and energy I_μ are then generally infinite for an absolutely continuous background. Consequently, the Hoeffding centering and splitting formula used here are not defined without a different scaling and renormalization.

Theorem 2.3 (Ultraviolet obstructions at and above the Hilbert–Schmidt threshold)**.** *Let $\beta > 0$, let $\mathsf{d}/2 \leq \mathsf{s} < \mathsf{d}$, set $W(x,y) = \mathsf{g}_{\mathsf{s}}(x-y)$, and let μ be a bounded probability density. Set*

$$\mathfrak{a}_N := \begin{cases} N^{-2}\log N, & \mathsf{s} = \mathsf{d}/2, \\ N^{-\mathsf{d}/\mathsf{s}}, & \mathsf{d}/2 < \mathsf{s} < \mathsf{d}. \end{cases} \tag{2.38}$$

At the critical exponent $\mathsf{s} = \mathsf{d}/2$, assume additionally that $\mu \geq a > 0$ on some ball.

(i) For every fixed $k \geq 2$, there are positive constants $c_\chi, c_H, c_{\mathrm{TV}}$ such that, for all sufficiently large N,

$$\chi^2(\mathbb{Q}^{(k)}_{N,\beta}(\mu) \mid \mu^{\otimes k}) \geq c_\chi \mathfrak{a}_N, \tag{2.39a}$$

$$\mathsf{H}(\mathbb{Q}^{(k)}_{N,\beta}(\mu) \mid \mu^{\otimes k}) \geq c_H \mathfrak{a}_N, \tag{2.39b}$$

$$\mathrm{TV}(\mathbb{Q}^{(k)}_{N,\beta}(\mu), \mu^{\otimes k}) \geq c_{\mathrm{TV}} \mathfrak{a}_N. \tag{2.39c}$$

In particular, the fixed-particle N^{-2} chi-square and entropy scales fail at and above $\mathsf{s} = \mathsf{d}/2$.

(ii) There are $c_0, c_1 > 0$ such that, for all sufficiently large N,

$$\mathrm{TV}(\mathbb{Q}^{(k)}_{N,\beta}(\mu), \mu^{\otimes k}) \geq c_1 k^2 \mathfrak{a}_N \tag{2.40}$$

for every integer

$$2 \leq k \leq c_0 \mathfrak{a}_N^{-1/2}. \tag{2.41}$$

Consequently, for every sufficiently small fixed $\kappa > 0$, the sequence

$$k_N := \left\lfloor \kappa \mathfrak{a}_N^{-1/2} \right\rfloor \tag{2.42}$$

satisfies

$$\liminf_{N\to\infty} \mathrm{TV}(\mathbb{Q}^{(k_N)}_{N,\beta}(\mu), \mu^{\otimes k_N}) > 0, \tag{2.43}$$

$$\frac{k_N}{N} \longrightarrow 0. \tag{2.44}$$

Thus, the uniform k/N total-variation bound fails throughout $\mathsf{d}/2 \leq \mathsf{s} < \mathsf{d}$. At criticality, $\mathfrak{a}_N^{-1/2} = N/\sqrt{\log N}$; in the strictly supercritical range, $\mathfrak{a}_N^{-1/2} = N^{\mathsf{d}/(2\mathsf{s})}$.

By Pinsker's inequality and the elementary bound $\chi^2(\nu \mid \pi) \geq 4\,\mathrm{TV}(\nu,\pi)^2$, part (ii) also implies that the corresponding k_N^2/N^2 entropy and chi-square scales fail. Indeed, $k_N^2/N^2 \asymp 1/\log N \to 0$ at $\mathsf{s} = \mathsf{d}/2$, while $k_N^2/N^2 \asymp N^{\mathsf{d}/\mathsf{s}-2} \to 0$ when $\mathsf{d}/2 < \mathsf{s} < \mathsf{d}$. Thus, the Hilbert–Schmidt threshold is the exact boundary for the classical local scales in all three divergences: the fixed-particle chi-square and entropy scales fail at the threshold, and the uniform total-variation scale fails at and above it.

The lower comparison in Corollary 3.5 shows, in particular, that the critical sequence $k_N \asymp N/\sqrt{\log N}$ also rules out the classical uniform $(k/N)^\tau$ scale for every centered L^τ-divergence, $1 \leq \tau \leq 2$.

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2.4. **Proof strategy and logical structure.** The proof of Theorem 3.1 identifies the first two Hoeffding components of the Gibbs density and controls the remaining components together. We first explain why this suffices for the marginal approximation, then describe how the two leading kernels are obtained. Figure 1 distinguishes this argument from the complementary order lower bounds and ultraviolet obstructions.

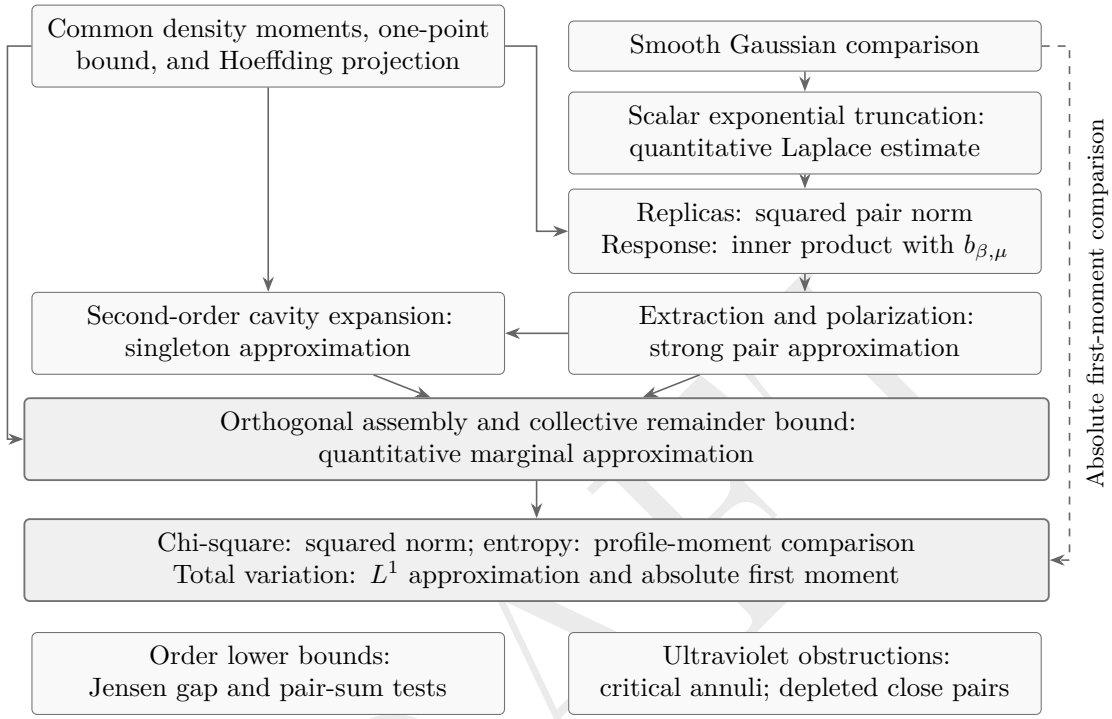


FIGURE 1. The common estimates enter pair extraction, the cavity expansion, and marginal projection. The dashed arrow is the absolute-first-moment application of the smooth Gaussian comparison, needed for growing-marginal total variation. The lower boxes are not inputs to the quantitative theorem.

Why two components suffice. Marginalization is conditional expectation under the product law: it removes every Hoeffding component involving an unobserved coordinate. By exchangeability, the components involving r coordinates have equal squared norms. The fraction of their total squared norm that survives in the k-particle marginal is therefore

$$\frac{\binom{k}{r}}{\binom{N}{r}} \le \left(\frac{k}{N}\right)^r, \qquad 1 \le r \le k \le N. \tag{2.45}$$

The density-moment estimate bounds the full N-particle chi-square divergence and hence the sum of these squared norms. After the first two components are removed, (2.45) bounds the squared marginal remainder by $C(k/N)^3$; see (6.26). At the density normalization N/k, its norm is at most $C\sqrt{k/N}$. Thus, quantitative identification of the first two kernels suffices throughout every sublinear marginal range. Higher components are present, but no separate estimate for each degree and no hierarchy of coupled marginal or correlation equations is required.

Common upper estimates. The same projection argument gives the marginal upper bounds once the first component is bounded at order N^{-1}. For this estimate, we condition on the other $N-1$ particles and expand the normalized conditional exponential. Writing $u_N = f^{(1)}_{N,\beta} - 1$, averaging

over their actual Gibbs marginal law gives

$$(\mathrm{Id} + \beta\frac{N-1}{N}T_\mu)u_N = \frac{\beta}{N}(h_\mu - 2I_\mu) + r_N, \qquad \|r_N\|_{L^2(\mu)} \lesssim N^{-1}. \tag{2.46}$$

The order hypotheses supply a uniform inverse bound through the modulated partition moment at 2β; the quantitative hypotheses supply it through their amplified centered-pair exponential moment. The two assumptions are compared after Theorem 3.1. In either case, the mean response of the reservoir remains on the left-hand side of (2.46), rather than being bounded as a perturbative error. The inverse bound yields $\|u_N\|_2 \le C/N$. Combining this with the full-density bound and projection proves the all-marginal upper estimates; see Section 4.

Strong pair approximation. For the two-variable Hoeffding kernel v_N defined in (5.13), we estimate both $\|Nv_N\|_2^2$ and $\langle Nv_N, b_{\beta,\mu}\rangle$. Their limits determine the squared error through

$$\|Nv_N - b_{\beta,\mu}\|_2^2 = \|Nv_N\|_2^2 + \|b_{\beta,\mu}\|_2^2 - 2\langle Nv_N, b_{\beta,\mu}\rangle. \tag{2.47}$$

The two measurements play different roles: convergence against fixed tests would identify a candidate limit but would not exclude an additional squared norm. Replicas determine the norm, and a perturbation in the direction of the candidate kernel determines the inner product.

Both computations use the quantitative Laplace estimate of Theorem 5.2. A one-coordinate Gaussian replacement, in the spirit of [Cha06, Section 2], controls smooth tests of the canonical quadratic statistic. We apply it to a truncated scalar exponential. The amplified exponential bound controls the discarded part under the original product law and the limiting Gaussian law, so no exponential estimate at the intermediate replacement laws is needed. The truncation is applied to the test function, not to obtain a quantitative approximation rate for the interaction kernel. This gives the determinant comparison without a sign condition on the centered operator.

For the norm calculation, sample a configuration under $\mu^{\otimes N}$ and form a second configuration by independently keeping each coordinate with probability t or resampling it from μ. The underlying identity is the usual noise-stability identity for product resampling; its finite-space formulation is given in [O'D14, Proposition 8.28 and Section 8.3]. Canonicity makes resampling act by t^r on an order-r Hoeffding component. Consequently, the expected product of the two Gibbs density ratios is a polynomial whose degree-r coefficient is the total squared norm of that level, including its subset multiplicity. The configurations themselves have product marginals; they are not independent Gibbs samples. The Gibbs exponential expresses this overlap as a paired partition function, to which the scalar Laplace estimate applies. Extracting its quadratic coefficient gives the pair squared norm, with the full-density bound controlling the extraction error; see Section 5.3.

For the inner product, perturb the centered interaction in the direction $b_{\beta,\mu}$ while leaving the one-body source unchanged. Differentiation of the logarithmic partition function at zero gives $-(N-1)\langle v_N, b_{\beta,\mu}\rangle/2$. Quantitative control for both signs of the perturbation permits comparison of this derivative with that of the limiting determinant. The resulting inner product and the replica norm estimate give strong $L^2(\mu^{\otimes 2})$ convergence by (2.47); see Sections 5.4 and 5.5.

The singleton correction. The first-order cavity bound leaves an error of the same order as u_N, so identifying the singleton profile requires the quadratic term of the normalized exponential. Its averaged contribution is one half of the spatially centered cavity second moment q_N defined in (6.6). This is a second moment under the actual Gibbs reservoir law, not a variance: centering in the trial coordinate does not make the reservoir expectation vanish. Separating equal and distinct reservoir labels expresses it through the row square and the two-variable Hoeffding kernel. The strong pair estimate then identifies $\omega_{\beta,\mu}$ as the $L^2(\mu)$ limit of Nq_N, with the quantitative bound (6.20). Applying the screened inverse in (2.46) identifies the profile $a_{\beta,\mu}$ in (2.20). The estimates take the Gibbs expectation before the spatial norm and combine row integrability with Hilbert-valued moment bounds; no fourth pointwise moment of the singular interaction is required.

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Marginal approximation and the three divergences. Orthogonality combines the two kernel errors with the collective remainder already described. This is the density approximation in Theorem 2.1; the proof is given in Section 6.4. For chi-square divergence, the squared norm is exact. The leading term follows from the profile norm identity (2.22), while the sharper quantitative error uses the separately proved pair squared-norm estimate, not just the strong pair bound.

For relative entropy, a quadratic comparison produces one half of the chi-square leading term. Its justification uses a moment above two only for the explicit profile $G_{k,\beta,\mu}/k$, uniformly in k; it requires neither pointwise closeness of the density to one nor a higher moment of the density error. The comparison preserves the squared approximation error. This is the role of Lemmas 6.3 and 6.4.

For total variation, the L^2 density approximation gives an L^1 approximation. When k grows, the normalized singleton sum vanishes and the remaining constant is the absolute first moment of the canonical pair sum. Its comparison with $\mathcal{Q}_{\beta,\mu}/2$ uses the same smooth Gaussian-comparison lemma as the Laplace estimate, now applied to a smooth approximation of absolute value; see Proposition 6.5. The limiting quadratic form need not be Gaussian. Section 6.5 completes these three evaluations of the growing-marginal constants.

The complementary order theory. The order theorem has its own analytic hypotheses, including a weighted lower-tail alternative. Its lower bound follows from nonzero product variance, higher-moment control, and a strict Jensen gap for the scalar log-Laplace transform. Differentiation gives a negative pair response of order N^{-1}. Testing with the centered pair sum gives the chi-square lower bound; truncating that sum gives total variation, and Pinsker's inequality gives entropy. Under the common hypotheses and a nontrivial interaction, the quantitative estimates and contraction under marginalization also give the order lower bounds for every $2 \le k \le N$, once N is sufficiently large independently of k. We prove them directly in Section 9 to establish Theorem 3.4 without the quantitative theorem's additional assumptions.

The subquadratic theorem instead pairs a full-density L^τ bound with canonical ordered U-statistics by duality. A refined degree-two moment estimate preserves the order-N^{-1} contribution to the norm; higher fixed orders are negligible at that scale. This proves convergence of every fixed marginal under the separate hypotheses of Theorem 3.6.

Ultraviolet obstructions. Square-integrability of the centered pair kernel is the finite-variance condition behind the pair-statistic and Hilbert–Schmidt estimates above. For a Riesz singularity, it holds precisely below $2\mathsf{s} = \mathsf{d}$. It is not, by itself, a substitute for the positive results' other analytic assumptions. At and above this threshold, the lower obstructions use instead the explicit Boltzmann factor of an observed pair. For $\mathsf{s} > \mathsf{d}/2$, repulsion depletes sufficiently close pairs, and a union over observed pairs, with their overlaps controlled, gives the mesoscopic obstruction.

At $\mathsf{s} = \mathsf{d}/2$, the logarithmic enhancement accumulates across transition annuli. A marked-reservoir representation gives conditional Hellinger-affinity deficits near suitable isolated observed particles. Their expected number is bounded below by a fixed multiple of the number of observed labels, uniformly over the stated marginal window. The exact affinity recursion accumulates these deficits to prove the critical lower bound; no high-probability assertion about that count is required. Section 10 gives the argument under the respective hypotheses of Theorem 2.3. The conclusions are one-sided obstructions and do not follow from divergence of the partition function.

Remark 2.4 (Relation with Gaussian fluctuations). Under the hypotheses of the companion's static linear-statistics theorem [DGR26, Proposition 2.11], the limiting covariance for bounded empirical tests is

$$\Sigma_\beta(\phi,\psi) = \left\langle \bar{\phi}, (\mathrm{Id} + \beta T_\mu)^{-1} \bar{\psi} \right\rangle_{L^2(\mu)}, \qquad \bar{\phi} = \phi - \int \phi \, \mathrm{d}\mu. \tag{2.48}$$

This is the same screened resolvent as in the marginal correction. Covariance convergence alone does not give a strong pair estimate or identify the singleton correction; neither is deduced from that fluctuation theorem in the present proof.

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2.5. **Relation to previous work.** Classical chaos results are primarily qualitative. Large-deviation methods identify the mean-field equilibrium and yield convergence of the empirical measure, hence convergence of every fixed marginal. Ben Arous and Zeitouni obtained an increasing-propagation-of-chaos result under uniqueness and nondegeneracy of the mean-field minimizer [BAZ99]. At the level relevant here, such global arguments typically give an $O(1)$ full relative entropy and therefore only the local estimate (1.5).

Local divergence estimates. For local divergence orders, Lacker's work [Lac22] is the principal direct predecessor. Its central theorem introduces a nonasymptotic hierarchy relating the k- and $(k+1)$-particle relative Fisher informations. Under additional transport-information, Poincaré, or convexity hypotheses, this yields corresponding relative-entropy and squared-Wasserstein bounds. To the best of our knowledge, this was the first general nonasymptotic theorem to prove an upper bound at the quadratic local relative-entropy scale for a broad class of continuous mean-field Gibbs measures. The abstract assumptions are force-level: a square-integrability bound for the centered interaction force and a transport-information inequality with constant $\gamma < \beta^{-2}$. Bounded- or Lipschitz-gradient conditions furnish sufficient regimes. A Gaussian model attains the quadratic order in k/N, showing that the exponent in the upper bound cannot in general be improved. This is a model-based optimality statement, rather than a matching lower-bound theorem throughout a class of interactions. Singular interactions are explicitly left outside the physical scope of the result.

Ren and Wang [RW25] replace the hierarchy of marginal Fisher informations by a hierarchy of conditional entropies. Comparing neighboring conditional laws produces the second difference of the marginal entropy sequence and introduces a nonlinear Fisher-information dissipation tied to the mean-field free-energy landscape. This yields nonperturbative entropy upper bounds of order k^2/N^2 under suitable nonlinear and conditional transport inequalities. In their flat semiconvex setting, the interaction is decomposed as $W = W_+ - W_-$, with both W_+ and W_- of positive type—that is, conditionally positive definite on zero-mass signed measures—and either the relevant interaction forces are bounded or $W_\pm$ are twice continuously differentiable with globally bounded mixed Hessians. In their displacement-convex setting, $W(x,y) = W'(x-y)$, where W' is even, convex, twice continuously differentiable, and has globally bounded Hessian. These hypotheses exclude the diagonal logarithmic and Riesz singularities considered here.

Related recent work of Bernou and Duerinckx [BD25] obtains quantitative correlation estimates, Gibbs-relaxation bounds, and cross estimates for Brownian particle systems with bounded interaction forces, initially in a weak-interaction regime and with partial extensions beyond it. Duerinckx and Jabin [DJ26] prove high-temperature partition-function estimates in a homogeneous setting and use the resulting weighted-L^p control of the Gibbs equilibrium as an input to the derivation of the linearized Vlasov dynamics near equilibrium. These works provide one-sided quantitative control of Gibbs equilibria and their correlations, but do not give matching lower bounds for local marginal divergences.

Marginal and correlation approximations. The terminology and systematic dynamical study of the size of chaos go back in particular to Paul, Pulvirenti, and Simonella [PPS19], who made the pair-counting mechanism explicit and proved an $O(k^2/N)$ marginal estimate in their correlation-error framework. Duerinckx subsequently established the Bogolyubov scaling $G_N^{m+1} = O(N^{-m})$ for connected $(m+1)$-particle correlations, in suitable negative Sobolev norms, for smooth classical mean-field dynamics by Glauber calculus [Due21]. Lacker developed the corresponding local entropy hierarchy for mean-field diffusions [Lac23]. Subsequent sharp dynamical results include the uniform-in-time estimates of Lacker and Le Flem [LLF23], the higher-order L^2 expansion of Hess-Childs and Rowan [HCR25], the coupled entropy–Fisher-information hierarchy of Grass, Guillin, and Poquet [GGP25], and the extension by Arnese and Lacker to coefficients depending nonlinearly on the measure argument, with applications to mean-field Langevin dynamics, control, and games [AL26]. For singular diffusions, Jabin and Wang obtained quantitative propagation of

chaos in global relative entropy for $W^{-1,\infty}$ interaction forces [JW18]; see also Guillin, Le Bris, and Monmarché [GLBM25] for subsequent uniform-in-time improvements. Building on this line, Songbo Wang proved sharp local entropy and weighted-L^2 estimates in the same rough-force class [Wan26].

Several complementary approaches control singular dynamical marginals or correlations in norms other than statistical divergences. Bresch, Jabin, and Soler [BJS25] developed uniform-in-N weighted-L^p estimates for the hierarchy of marginals in Vlasov–Fokker–Planck dynamics and also recorded a simpler first-order counterpart, for which no velocity weight is required. Bresch, Duerinckx, and Jabin [BDJ24] introduced a dual-correlation method for first- and second-order systems with square-integrable interaction forces, allowing possibly vanishing temperature. For Brownian systems with square-integrable singular interactions, Duerinckx and Jabin [DJ25] obtained systematic correlation estimates and derived the Bogolyubov correction and a central limit theorem. Béjar-López, Blaustein, Jabin, and Soler [BLBJS26] proved uniform-in-time and uniform-in-N L^2 bounds for marginals in singular Liouville dynamics, including configurations far from chaos. These results are complementary to, but distinct from, local estimates in statistical divergences.

Nonexchangeable systems form a separate direction. Jabin, Poyato, and Soler [JPS25] established mean-field limits for nonexchangeable systems on sparse, possibly directed graphs through an extended-graphon framework. Lacker, Yeung, and Zhou [LYZ24] proved sharp relative-entropy estimates for the marginals associated with arbitrary particle subsets in heterogeneous diffusions, extending Lacker's local entropy hierarchy by means of first-passage-percolation estimates. These works address different questions: the former develops a broad nonexchangeable mean-field framework, whereas the latter gives sharp quantitative control of subset marginals.

A stationary result of Zhang gives sharp one-particle Rényi-divergence upper bounds under strong isoperimetry and weak interaction assumptions [Zha26]. The results considered here instead concern singular equilibrium marginals of every size, together with their quantitative leading profiles and ultraviolet obstructions.

Partition functions and the present argument. The partition-function input has a more specific recent lineage. Delgadino and Gvalani [DG25, Theorem 2.5] proved an all-temperature, N-uniform bound for the repulsive logarithmic gas under bounded-background assumptions. The companion paper [DGR26] extends this result to logarithmic and Riesz interactions throughout the Hilbert–Schmidt range and additionally identifies the Carleman–Fredholm determinant normalization and the static fluctuation law.

Concurrent work of Wang and Zhao [WZ26, Assumption A and Theorems 1.1 and 1.3] proves all-temperature uniform partition-function estimates for logarithmic, Riesz, and Bessel–Riesz kernels throughout $0 \le \mathsf{s} < \mathsf{d}/2$, together with the second-Wiener-chaos and Carleman–Fredholm determinant limit. Their logarithmic hypothesis includes a stretched-exponential spatial moment, whereas the companion uniform-bound theorem used here assumes finite absolute logarithmic energy. Their proof uses a connected-graph expansion and heat-kernel truncation, while [DGR26] uses positive-definite truncations and a low/high-frequency decomposition. The two near-simultaneous works overlap at the level of the companion partition-function branch; no priority claim between the two approaches is intended.

Our equilibrium argument works at the level of the interaction rather than its derivatives. It combines the explicit Gibbs density with square-integrability of the Hoeffding-centered interaction, cavity estimates, and partition-function control of the screened response. This permits the logarithmic and Riesz singularities in the Hilbert–Schmidt regime. Beyond the upper bounds, the abstract order theorem gives matching local-divergence lower bounds for every nontrivial interaction satisfying its hypotheses. The abstract quantitative theorem identifies the first-order marginal profile and its growing-marginal divergence constants under row, source, and amplified centered-pair exponential bounds, without a sign condition on the centered operator. The logarithmic and Riesz models satisfy these assumptions at every fixed inverse temperature under the stated background hypotheses.

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The ultraviolet theorem supplies lower obstructions to extending the classical scales beyond this regularity range.

2.6. **Organization of the paper.** Section 3 states the abstract quantitative theorem, followed by the separate order and subquadratic results. Section 4 collects the exact Gibbs identities, density and screened one-point estimates, and orthogonal marginal projection. Sections 5 and 6 prove the strong pair and singleton approximations and the statistical-divergence conclusions. Section 7 verifies the respective logarithmic and Riesz hypotheses. Sections 8 and 9 give the subquadratic argument and the matching order lower bounds. Section 10 proves the ultraviolet obstructions. Appendices A to C contain the weighted cavity proof, the constant ledger, and the shared probability estimates.

2.7. **Statement on AI use.** This working manuscript was generated with the assistance of generative AI tools, including OpenAI's ChatGPT and Codex, for literature identification, the development and testing of mathematical arguments, exposition, and consistency checks. The authors have not yet reviewed and independently verified all AI-generated content. They are in the process of formalizing the results in Lean and revising the natural-language presentation. The manuscript is therefore being shared as a draft, and its statements, proofs, references, and exposition remain subject to correction. Any completed formal verification applies only to the statements and versions explicitly identified in the accompanying verification records.

3. Abstract results

We first state the abstract theorem giving the quantitative marginal and divergence estimates of Section 2.2. The order and subquadratic results that follow have separate moment and cavity hypotheses. The comparison after the first theorem explains this distinction. All model-specific verification is deferred to Section 7.

3.1. **Quantitative marginal approximation.** The assumptions in this subsection are independent of those in Section 3.2. Let μ be a Borel probability measure on $\mathbb{R}^{\mathsf{d}}$, let $W \in L^1(\mu^{\otimes 2})$ be real and symmetric, and use its centering and one-body function from (2.3) and (2.17). Fix $\beta > 0$, $p > 2$, and

$$0 < \vartheta < \min\{1, p-2\}. \tag{3.1}$$

Assume that, for some finite nonnegative C_-, C_+,

$$\widetilde{W} \geq -C_-, \qquad M_p := \operatorname*{ess\,sup}_x \|\widetilde{W}(x,\cdot)\|_{L^p(\mu)} < \infty, \tag{3.2}$$

$$\psi_\mu \in L^6(\mu), \qquad \psi_\mu \leq C_+. \tag{3.3}$$

All essential bounds are relative to the appropriate product of μ. For a canonical kernel c and independent coordinates $x_1, \dots, x_N$ with law μ, write

$$\mathsf{W}_N(c) := \frac{1}{N} \sum_{i \neq j} c(x_i, x_j). \tag{3.4}$$

Set

$$r_* := \max\left\{12, \frac{p}{p-2-\vartheta}, \frac{2(2+\vartheta)}{\vartheta}\right\}, \qquad \Gamma := \beta r_*, \tag{3.5}$$

and assume the centered-pair exponential bound

$$\sup_{N \geq 2} \mathbb{E}_{\mu^{\otimes N}}\left[e^{-\Gamma \mathsf{W}_N(\widetilde{W})/2}\right] < \infty. \tag{3.6}$$

This bound does not include the one-body term in the modulated Gibbs density. Its implication for the density moments is recorded in Section 4.5. No sign condition is imposed on T_μ, and no nontriviality assumption is imposed on $\widetilde{W}$.

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Theorem 3.1 (Abstract quantitative marginal approximation)**.** *Assume* (3.1)–(3.3) *and* (3.6)*, with* r_*, Γ *as in* (3.5)*. Fix* $0 < \alpha \le 1$ *with* $2 + \alpha \le p$*, and set*

$$\mathfrak{r}_N := N^{-\alpha/8} + N^{-\vartheta/2}. \tag{3.7}$$

The profiles in (2.15)–(2.21) *are well defined. With this rate and these profiles, the estimates* (2.29)*,* (2.30)–(2.32)*, and* (2.35)–(2.37) *hold for every* $N \ge 2$ *and* $1 \le k \le N$*. Their constants depend only on the fixed exponents and hypothesis data, not on* N *or* k*. In particular, the normalized density error vanishes uniformly on every sublinear marginal range, and all three normalized divergences converge to the displayed constants whenever* $k_N \to \infty$ *and* $k_N/N \to 0$*. At* $\beta = 0$*, the conclusions hold trivially whenever the modulated definitions are well defined.*

Both Theorems 3.1 and 3.4 allow positive and negative spectrum of the centered interaction operator. Their hypotheses serve different purposes. The order theorem permits the weighted logarithmic lower-tail condition and uses modulated partition bounds at the exponents in (3.27). The quantitative theorem instead assumes uniform row integrability, a uniformly lower-bounded centered kernel, and the stronger centered-pair exponential bound (3.6); its source may be unbounded below but must belong to $L^6(\mu)$. These assumptions give quantitative Laplace estimates for the replica and response families, not merely an inverse bound for the screened operator. Neither theorem inherits the hypotheses of the other. The proof of Theorem 3.1 is completed in Section 6, after the strong pair estimate in Section 5.

Remark 3.2 (Spectral stability and exponential control)*.* The amplified exponential bound cannot be replaced by spectral stability. Let $\epsilon = e^{-4}$ and let a centered function ξ equal 1 on a set of μ-measure ϵ, and $-\epsilon/(1-\epsilon)$ otherwise. For $W = -\xi \otimes \xi$, one has $\widetilde{W} = W$, $\psi_\mu = 0$, and the only nonzero eigenvalue is $\lambda = -\epsilon/(1-\epsilon)$. At $\beta = 10$, $1 + 2\beta\lambda > 0$, but the event on which every coordinate lies in the first set gives

$$\mathbb{E}_{\mu^{\otimes N}}\left[e^{-\beta \mathsf{W}_N(\widetilde{W})/2}\right] \ge \epsilon^N e^{5(N-1)} = e^{N-5}. \tag{3.8}$$

Conversely, the assumptions admit genuinely mixed-sign interactions. For independent symmetric signs σ, τ under μ, take $W = a\sigma \otimes \sigma - b\tau \otimes \tau$, with $a, b > 0$. Its nonzero eigenvalues are $a, -b$, and $\psi_\mu = 0$. Gaussian integration and $\cosh u \le e^{u^2/2}$ give, whenever $\Gamma b < 1$,

$$\mathbb{E}_{\mu^{\otimes N}}\left[e^{-\Gamma \mathsf{W}_N(W)/2}\right] \le e^{\Gamma(a-b)/2}\mathbb{E}_Z\left[\cosh\left(\sqrt{\Gamma b/N}\, Z\right)^N\right] \le \frac{e^{\Gamma(a-b)/2}}{\sqrt{1-\Gamma b}}, \tag{3.9}$$

where Z is a standard Gaussian. The other assumptions are immediate, and no restriction on a is needed.

3.2. **Hypotheses for the order bounds.** The structural upper-bound assumption is square-integrability of the centered interaction,

$$\widetilde{W} \in L^2(\mu^{\otimes 2}). \tag{3.10}$$

No sign condition is imposed on the associated Hilbert–Schmidt operator. The two-sided theorem below separately assumes that the centered interaction is nontrivial, because its lower bounds are generated by a nonzero pair fluctuation.

For $\gamma \ge 0$, define the normalized partition-function constant by

$$\mathfrak{P}_\gamma(\mu) := e^{-\gamma I_\mu} \sup_{M \ge 1} \mathsf{K}_{M,\gamma}(\mu). \tag{3.11}$$

Jensen's inequality gives $\mathsf{K}_{M,\gamma}(\mu) \ge e^{\gamma I_\mu}$, and therefore $1 \le \mathfrak{P}_\gamma(\mu) < \infty$ whenever the corresponding partition-function bound holds.

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Lemma 3.3 (Interpolation of normalized partition functions)**.** *Let $\beta \geq 0$ and $1 \leq s \leq t < \infty$. Then, for every $N \geq 1$,*

$$\mathsf{K}_{N,s\beta}(\mu) \leq \mathsf{K}_{N,t\beta}(\mu)^{s/t}. \tag{3.12}$$

Consequently,

$$\mathfrak{P}_{s\beta}(\mu) \leq \mathfrak{P}_{t\beta}(\mu)^{s/t}, \qquad \mathfrak{P}_{s\beta}(\mu)^{1/s} \leq \mathfrak{P}_{t\beta}(\mu)^{1/t}. \tag{3.13}$$

Proof. Apply Hölder's inequality to $Y_N = e^{-\beta N \mathsf{F}_N}$:

$$\mathbb{E}_{\mu^{\otimes N}}\left[Y_N^s\right] \leq \mathbb{E}_{\mu^{\otimes N}}\left[Y_N^t\right]^{s/t}. \tag{3.14}$$

This is (3.12). Multiplication by $e^{-s\beta I_\mu}$, followed by the supremum over N, gives (3.13). □

Fix

$$p > 2, \qquad r \coloneqq \frac{p}{p-2}, \tag{3.15}$$

and assume

$$\|h_\mu - 2I_\mu\|_{L^2(\mu)} + \|h_\mu - 2I_\mu\|_{L^p(\mu)} + \|\widetilde{W}\|_{L^2(\mu^{\otimes 2})} + \|\widetilde{W}\|_{L^p(\mu^{\otimes 2})} < \infty. \tag{3.16}$$

For the lower bounds only, we additionally impose the nontriviality condition

$$\|\widetilde{W}\|_{L^2(\mu^{\otimes 2})} > 0. \tag{3.17}$$

For later use, we write

$$\|\widetilde{W}\|_{L^\infty_x L^p_y(\mu^{\otimes 2})} \coloneqq \sup_{x \in \mathbb{R}^{\mathsf{d}}} \|\widetilde{W}(x,\cdot)\|_{L^p(\mu)}. \tag{3.18}$$

The joint condition $\widetilde{W} \in L^p(\mu^{\otimes 2})$ and the mixed-norm condition $\widetilde{W} \in L^\infty_x L^p_y(\mu^{\otimes 2})$ below serve different purposes as assumptions: the former controls product-space moments, whereas the latter gives a fixed-x moment bound for the section $y \mapsto \widetilde{W}(x,y)$.

For $N \geq 2$, write $\xi_1, \dots, \xi_N$ for the coordinate variables on $(\mathbb{R}^{\mathsf{d}})^N$, set $\hat{\Xi}_1 = (\xi_2, \dots, \xi_N)$, and define the centered one-particle cavity field

$$B_N(x; \hat{\Xi}_1) \coloneqq \frac{\beta}{N}(h_\mu(x) - 2I_\mu) - \frac{\beta}{N}\sum_{j=2}^N \widetilde{W}(x, \xi_j). \tag{3.19}$$

We impose one of the following two cavity conditions.

Bounded lower-tail condition. Assume

$$\|h_\mu - 2I_\mu\|_{L^\infty} + \|\widetilde{W}\|_{L^\infty_x L^p_y(\mu^{\otimes 2})} < \infty, \qquad \operatorname*{ess\,inf}_{(x,y)\in(\mathbb{R}^{\mathsf{d}})^2} \widetilde{W}(x,y) > -\infty. \tag{3.20}$$

This condition gives a deterministic upper bound on B_N, uniformly in N and in the configuration of the remaining $N-1$ particles.

Logarithmic lower-tail condition. Let $L(x) \coloneqq \log(e + |x|)$. Assume that there are finite constants a_0, a_1, a_h such that

$$(\widetilde{W})_-(x,y) \leq a_0 + a_1(L(x) + L(y)), \qquad (h_\mu - 2I_\mu)_+(x) \leq a_h(1 + L(x)) \tag{3.21}$$

for $\mu^{\otimes 2}$-a.e. (x,y) and μ-a.e. x, respectively. Define

$$\mathcal{L}_N(\hat{\Xi}_1) \coloneqq \frac{1}{N}\sum_{j=2}^N L(\xi_j), \tag{3.22}$$

$$\mathsf{c}_{\log,\beta,\mu} \coloneqq \beta(a_0 + a_1 + a_h), \tag{3.23}$$

$$\Phi_N(x; \hat{\Xi}_1) \coloneqq \mathsf{c}_{\log,\beta,\mu}(1 + L(x) + \mathcal{L}_N(\hat{\Xi}_1)). \tag{3.24}$$

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Then the positive part of the cavity field satisfies

$$B_N(x;\hat{\Xi}_1)_+ \le \Phi_N(x;\hat{\Xi}_1). \tag{3.25}$$

Under this condition, assume the finiteness of the following weighted cavity constants:

$$\mathsf{E}_{\log,\beta,\mu} := \frac{1}{2}\sup_{N\ge 2} N\|\mathbb{E}_{\mathbb{Q}_{N,\beta}(\mu)}\left[e^{\Phi_N(\cdot;\hat{\Xi}_1)}B_N(\cdot;\hat{\Xi}_1)^2\right]\|_{L^2(\mu)} < \infty, \tag{3.26a}$$

$$\mathsf{D}_{\log,\beta,\mu} := \frac{1}{2}\sup_{N\ge 2} N\mathbb{E}_{\mathbb{Q}_{N,\beta}(\mu)}\left[\int e^{\Phi_N(x;\hat{\Xi}_1)}B_N(x;\hat{\Xi}_1)^2\,\mathrm{d}\mu(x)\right] < \infty, \tag{3.26b}$$

$$\mathsf{V}_{\log,\beta,\mu} := \sup_{N\ge 2} N\mathbb{E}_{\mathbb{Q}_{N,\beta}(\mu)}\left[\left(\int e^{\Phi_N(x;\hat{\Xi}_1)}|B_N(x;\hat{\Xi}_1)|\,\mathrm{d}\mu(x)\right)^2\right] < \infty, \tag{3.26c}$$

$$\mathsf{B}_{\log,\beta,\mu} := \sup_{N\ge 2} N\mathbb{E}_{\mathbb{Q}_{N,\beta}(\mu)}\left[\int B_N(x;\hat{\Xi}_1)^2\,\mathrm{d}\mu(x)\right] < \infty. \tag{3.26d}$$

The bound (3.25) introduces polynomial weights into the nonlinear cavity estimate. Lemma 7.4 gives an explicit sufficient condition for the four estimates above.

3.3. **Abstract two-sided chaos theorem.** With the structural, moment, cavity, and partition-function assumptions now fixed, we can state the abstract result. The interaction remains arbitrary in this theorem; the logarithmic and Riesz kernels enter only through the verification in Section 7.

Theorem 3.4 (Abstract two-sided chaos bounds)**.** *Fix $\beta \ge 0$. Assume (3.10), (3.16), and (3.17) for some $p > 2$. Assume either the bounded lower-tail cavity condition (3.20), or the logarithmic lower-tail condition (3.21) together with the weighted estimates (3.26); in the latter case the weighted estimates may instead be verified through Lemma 7.4. With $r = p/(p-2)$, assume*

$$\sup_{N\ge 1}\mathsf{K}_{N,r\beta}(\mu) < \infty, \qquad \sup_{N\ge 1}\mathsf{K}_{N,2\beta}(\mu) < \infty. \tag{3.27}$$

Then there are finite constants $\mathsf{C}^{\mathrm{up}}_{\mathrm{TV},\beta,\mu}$, $\mathsf{C}^{\mathrm{up}}_{\mathsf{H},\beta,\mu}$, and $\mathsf{C}^{\mathrm{up}}_{\chi,\beta,\mu}$ such that, for every $1 \le k \le N$,

$$\mathrm{TV}(\mathbb{Q}^{(k)}_{N,\beta}(\mu),\mu^{\otimes k}) \le \mathsf{C}^{\mathrm{up}}_{\mathrm{TV},\beta,\mu}\frac{k}{N}, \tag{3.28}$$

$$\mathsf{H}(\mathbb{Q}^{(k)}_{N,\beta}(\mu)\mid\mu^{\otimes k}) \le \mathsf{C}^{\mathrm{up}}_{\mathsf{H},\beta,\mu}\frac{k^2}{N^2}, \tag{3.29}$$

$$\chi^2(\mathbb{Q}^{(k)}_{N,\beta}(\mu)\mid\mu^{\otimes k}) \le \mathsf{C}^{\mathrm{up}}_{\chi,\beta,\mu}\frac{k^2}{N^2}. \tag{3.30}$$

If $\beta > 0$, then there are $\mathsf{N}_{\beta,\mu} < \infty$ and positive constants $\mathsf{c}^{\mathrm{low}}_{\mathrm{TV},\beta,\mu}$, $\mathsf{c}^{\mathrm{low}}_{\mathsf{H},\beta,\mu}$, and $\mathsf{c}^{\mathrm{low}}_{\chi,\beta,\mu}$ such that, for every $N \ge \mathsf{N}_{\beta,\mu}$ and $2 \le k \le N$,

$$\mathrm{TV}(\mathbb{Q}^{(k)}_{N,\beta}(\mu),\mu^{\otimes k}) \ge \mathsf{c}^{\mathrm{low}}_{\mathrm{TV},\beta,\mu}\frac{k}{N}, \tag{3.31}$$

$$\mathsf{H}(\mathbb{Q}^{(k)}_{N,\beta}(\mu)\mid\mu^{\otimes k}) \ge \mathsf{c}^{\mathrm{low}}_{\mathsf{H},\beta,\mu}\frac{k^2}{N^2}, \tag{3.32}$$

$$\chi^2(\mathbb{Q}^{(k)}_{N,\beta}(\mu)\mid\mu^{\otimes k}) \ge \mathsf{c}^{\mathrm{low}}_{\chi,\beta,\mu}\frac{k^2}{N^2}. \tag{3.33}$$

These three quantities belong to the Csiszár–Morimoto class of f-divergences [AS66, Csi67]. For a convex function $f : [0,\infty) \to (-\infty,\infty]$ with $f(1) = 0$, define

$$\mathscr{D}_f(\nu\mid\pi) := \int f\left(\frac{\mathrm{d}\nu}{\mathrm{d}\pi}\right)\mathrm{d}\pi, \qquad \nu \ll \pi. \tag{3.34}$$

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Adding a multiple of $t-1$ to f does not change $\mathscr{D}_f$, so one may impose $f'(1)=0$ when this derivative exists. The three quantities above are generated by

$$f_{\mathrm{TV}}(t)=\frac{1}{2}|t-1|, \qquad f_{\mathrm{KL}}(t)=t\log t-t+1, \qquad f_{\chi^2}(t)=(t-1)^2. \tag{3.35}$$

Near the product law,

$$f_{\mathrm{TV}}(1+u)=\frac{1}{2}|u|, \tag{3.36}$$

$$f_{\mathrm{KL}}(1+u)=\frac{1}{2}u^2+O(|u|^3), \qquad f_{\chi^2}(1+u)=u^2. \tag{3.37}$$

Thus, total variation is first order in the density perturbation, whereas relative entropy and chi-square divergence are quadratic; this accounts for the two scales in (1.6).

Corollary 3.5 (Consequences for f-divergences)**.** *Assume the hypotheses of Theorem 3.4.*

(i) For $1\le\tau\le 2$, define the centered L^τ-divergence by

$$\mathscr{D}_\tau(\nu\mid\pi):=\int\left|\frac{\mathrm{d}\nu}{\mathrm{d}\pi}-1\right|^\tau\,\mathrm{d}\pi, \qquad \nu\ll\pi. \tag{3.38}$$

Its endpoints are

$$\mathscr{D}_1(\nu\mid\pi)=2\,\mathrm{TV}(\nu,\pi), \qquad \mathscr{D}_2(\nu\mid\pi)=\chi^2(\nu\mid\pi). \tag{3.39}$$

For every $1\le k\le N$,

$$\mathscr{D}_\tau\Big(\mathbb{Q}^{(k)}_{N,\beta}(\mu)\mid\mu^{\otimes k}\Big)\le\Big(\mathsf{C}^{\mathrm{up}}_{\chi,\beta,\mu}\Big)^{\tau/2}\left(\frac{k}{N}\right)^\tau. \tag{3.40}$$

If $\beta>0$, then, for every $N\ge\mathsf{N}_{\beta,\mu}$ and $2\le k\le N$,

$$\mathscr{D}_\tau\Big(\mathbb{Q}^{(k)}_{N,\beta}(\mu)\mid\mu^{\otimes k}\Big)\ge\Big(2\mathsf{c}^{\mathrm{low}}_{\mathrm{TV},\beta,\mu}\Big)^{\tau}\left(\frac{k}{N}\right)^\tau. \tag{3.41}$$

(ii) Let $f:[0,\infty)\to(-\infty,\infty]$ be convex with $f(1)=0$, and let $\mathscr{D}_f$ be as in (3.34). Suppose that, for some $\tau\in[1,2]$ and constants $a_f,b_f>0$, the global comparisons

$$a_f\,\mathrm{TV}(\nu,\pi)^\tau\le\mathscr{D}_f(\nu\mid\pi)\le b_f\chi^2(\nu\mid\pi)^{\tau/2} \tag{3.42}$$

hold for every pair of probability measures $\nu\ll\pi$. Then, for every $1\le k\le N$,

$$\mathscr{D}_f\Big(\mathbb{Q}^{(k)}_{N,\beta}(\mu)\mid\mu^{\otimes k}\Big)\le b_f\Big(\mathsf{C}^{\mathrm{up}}_{\chi,\beta,\mu}\Big)^{\tau/2}\left(\frac{k}{N}\right)^\tau. \tag{3.43}$$

If $\beta>0$, then, for every $N\ge\mathsf{N}_{\beta,\mu}$ and $2\le k\le N$,

$$\mathscr{D}_f\Big(\mathbb{Q}^{(k)}_{N,\beta}(\mu)\mid\mu^{\otimes k}\Big)\ge a_f\Big(\mathsf{c}^{\mathrm{low}}_{\mathrm{TV},\beta,\mu}\Big)^{\tau}\left(\frac{k}{N}\right)^\tau. \tag{3.44}$$

Total variation is recovered with $\tau=1$; relative entropy is recovered with $\tau=2$, using Pinsker's inequality and $\mathsf{H}\le\chi^2$; and chi-square divergence is recovered with $\tau=2$.

Proof. Set $g=\mathrm{d}\nu/\mathrm{d}\pi-1$. Since π is a probability measure, for $1\le\tau\le 2$,

$$\|g\|_{L^1(\pi)}\le\|g\|_{L^\tau(\pi)}\le\|g\|_{L^2(\pi)}. \tag{3.45}$$

Consequently,

$$(2\,\mathrm{TV}(\nu,\pi))^\tau\le\mathscr{D}_\tau(\nu\mid\pi)\le\chi^2(\nu\mid\pi)^{\tau/2}. \tag{3.46}$$

The estimates in part (i) follow from the total-variation lower bound and the chi-square upper bound in Theorem 3.4. Part (ii) follows in the same way from (3.42). The final identifications use (3.39) and the standard inequalities $\mathsf{H}(\nu\mid\pi)\ge 2\,\mathrm{TV}(\nu,\pi)^2$ and $\mathsf{H}(\nu\mid\pi)\le\chi^2(\nu\mid\pi)$. □

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Theorem 3.6 (Fixed marginals under a subquadratic partition-function moment)**.** *Fix $\beta > 0$ and $1 < \tau < 2$, and set*

$$q := \frac{\tau}{\tau - 1}, \qquad p \geq 2q, \qquad r := \frac{p}{p-2} \leq \tau. \tag{3.47}$$

Assume (3.10) and the finiteness conditions in (3.16) with this exponent p. Assume either the bounded lower-tail condition (3.20), or the logarithmic lower-tail condition (3.21) together with the weighted estimates (3.26); in the latter case the weighted estimates may instead be verified through Lemma 7.4. Finally, assume

$$\sup_{N \geq 1} \mathsf{K}_{N,\tau\beta}(\mu) < \infty. \tag{3.48}$$

No sign condition is imposed on T_μ or on its quadratic form, and the nontriviality condition (3.17) is not assumed.

Let

$$\mathsf{M}_{\tau,\beta,\mu} := \mathfrak{P}_{\tau\beta}(\mu)^{1/\tau}, \qquad \mathsf{U}_{\tau,\beta,\mu} := \frac{\tau}{\tau - 1}\left(\beta \|h_\mu - 2I_\mu\|_{L^2(\mu)} + \mathsf{R}_{\mathrm{cav},\beta,\mu}\right), \tag{3.49}$$

where $\mathsf{R}_{\mathrm{cav},\beta,\mu}$ is the cavity remainder constant in (4.66), with the needed L^r density bound supplied by Lemma 3.3. Then, for every $N \geq 2$ and $2 \leq k \leq N$,

$$\begin{aligned} \|f_{N,\beta}^{(k)} - 1\|_{L^\tau(\mu^{\otimes k})} \leq \frac{\mathsf{U}_{\tau,\beta,\mu}\sqrt{k}}{N} + (\mathsf{M}_{\tau,\beta,\mu} + 1)\mathsf{B}_q \left[\frac{\sqrt{\binom{k}{2}}}{N-1} + \frac{4\binom{k}{2}N^{1/q-1/2}}{N-1}\right] \\ + (\mathsf{M}_{\tau,\beta,\mu} + 1)\sum_{j=3}^{k} \binom{k}{j} \frac{2^j \mathsf{A}_{j,q}}{\sqrt{(N)_j}}. \end{aligned} \tag{3.50}$$

Here, the empty sum is zero, and the constants $\mathsf{A}_{j,q}$ and B_q are those in Lemmas 8.1 and 8.2. For $k = 1$,

$$\|f_{N,\beta}^{(1)} - 1\|_{L^\tau(\mu)} \leq \frac{\mathsf{U}_{\tau,\beta,\mu}}{N}. \tag{3.51}$$

Consequently, for every fixed $k \geq 2$,

$$\limsup_{N\to\infty} N\|f_{N,\beta}^{(k)} - 1\|_{L^\tau(\mu^{\otimes k})} \leq \mathsf{U}_{\tau,\beta,\mu}\sqrt{k} + (\mathsf{M}_{\tau,\beta,\mu} + 1)\mathsf{B}_q\sqrt{\binom{k}{2}}. \tag{3.52}$$

In particular, for every fixed $k \geq 1$,

$$\|f_{N,\beta}^{(k)} - 1\|_{L^\tau(\mu^{\otimes k})} = O_{k,\tau,\beta,\mu}(N^{-1}), \tag{3.53}$$

$$\mathrm{TV}(\mathbb{Q}_{N,\beta}^{(k)}(\mu), \mu^{\otimes k}) = O_{k,\tau,\beta,\mu}(N^{-1}). \tag{3.54}$$

The corresponding centered divergence and relative entropy satisfy

$$\mathscr{D}_\tau(\mathbb{Q}_{N,\beta}^{(k)}(\mu) \mid \mu^{\otimes k}) = O_{k,\tau,\beta,\mu}(N^{-\tau}), \tag{3.55}$$

$$\mathsf{H}(\mathbb{Q}_{N,\beta}^{(k)}(\mu) \mid \mu^{\otimes k}) = O_{k,\tau,\beta,\mu}(N^{-\tau}). \tag{3.56}$$

The entropy estimate in (3.56) is only a nonsharp consequence. No quantitative growing-k range is asserted.

For the logarithmic and Riesz models below, whenever the remaining moment and cavity hypotheses of Theorem 3.6 hold for a given τ, its only external partition-function input is supplied by the companion uniform-bound theorem [DGR26, Theorem 2.2(i)] applied at the single inverse temperature $\tau\beta$. Equivalently, this follows from the compact-temperature statement in [DGR26,

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Corollary 5.7]; in the logarithmic case, finite absolute logarithmic energy is the relevant additional hypothesis.

4. Gibbs identities and marginal upper bounds

The exact identities below separate the observed particles from the reservoir. Orthogonal projection then reduces the marginal upper bound to the first Hoeffding component and the full density. We obtain the required one-point estimate from the conditional exponential and spectral stability. The order and quantitative assumptions are invoked separately when verifying the cavity remainders. These common inputs are reused in the strong pair and singleton proofs.

4.1. Exact marked identities.

This section isolates the exact finite-N identity that turns a marginal estimate into a marked partition-function estimate. The point is to separate the k observed variables from the remaining $N-k$ variables before any approximation is made.

With h_μ, I_μ, and $\widetilde{W}$ as in (2.1)–(2.2), for $X_k = (x_1, \dots, x_k)$, set

$$\Phi_{X_k}(y) \coloneqq \sum_{a=1}^{k} \widetilde{W}(x_a, y), \tag{4.1}$$

$$D_k(X_k) \coloneqq \frac{1}{2} \sum_{1 \le a \ne b \le k} \widetilde{W}(x_a, x_b) - \sum_{a=1}^{k} h_\mu(x_a) + kI_\mu. \tag{4.2}$$

Lemma 4.1 (Exact marked splitting)**.** *Let $1 \le k < N$, $n = N-k$, $X_k = (x_1, \dots, x_k)$, and $Y_n = (y_1, \dots, y_n)$. Then*

$$N\mathsf{F}_N((X_k, Y_n), \mu) = \frac{n}{N}\, n\mathsf{F}_n(Y_n, \mu) + \frac{1}{N} \sum_{j=1}^{n} \Phi_{X_k}(y_j) + \frac{1}{N} D_k(X_k). \tag{4.3}$$

Proof. By definition,

$$N\mathsf{F}_N(z_1, \dots, z_N; \mu) = \frac{1}{2N} \sum_{1 \le i \ne j \le N} \widetilde{W}(z_i, z_j) - \frac{1}{N} \sum_{i=1}^{N} h_\mu(z_i) + I_\mu. \tag{4.4}$$

Taking $(z_1, \dots, z_N) = (X_k, Y_n)$, split the off-diagonal sum into bulk–bulk, mark–bulk, and mark–mark pieces. This gives

$$\begin{aligned} N\mathsf{F}_N((X_k, Y_n), \mu) = {} & \frac{1}{2N} \sum_{1 \le i \ne j \le n} \widetilde{W}(y_i, y_j) - \frac{1}{N} \sum_{j=1}^{n} h_\mu(y_j) + \frac{n}{N} I_\mu \\ & + \frac{1}{N} \sum_{a=1}^{k} \sum_{j=1}^{n} \widetilde{W}(x_a, y_j) + \frac{1}{2N} \sum_{1 \le a \ne b \le k} \widetilde{W}(x_a, x_b) \\ & - \frac{1}{N} \sum_{a=1}^{k} h_\mu(x_a) + \frac{k}{N} I_\mu. \end{aligned} \tag{4.5}$$

The first line is $(n/N) n\mathsf{F}_n(Y_n, \mu)$, and the remaining terms are the final two terms in (4.3). □

We now package the three factors in the splitting: the bulk partition function, the interaction of the marks with the bulk, and the purely marked remainder. This notation is designed so that the marginal density is just a normalized version of one marked weight.

For $n = N - k$, define

$$\gamma_{N,k} \coloneqq \beta \frac{n}{N}, \tag{4.6}$$

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$$\mathsf{M}_{N,k}(X_k) \coloneqq \mathbb{E}_{Y_n\sim\mathbb{Q}_{n,\gamma_{N,k}}(\mu)}\left[\exp\left(-\frac{\beta}{N}\sum_{j=1}^{n}\Phi_{X_k}(y_j)\right)\right], \tag{4.7}$$

$$\Psi_{N,k}(X_k) \coloneqq \exp\left(-\frac{\beta}{N}D_k(X_k)\right)\mathsf{M}_{N,k}(X_k). \tag{4.8}$$

We use the marginal densities defined in (2.12). For $k < N$, the marginal density is a normalized marked partition function.

Lemma 4.2 (Normalized conditional identity)**.** *For $1 \le k < N$,*

$$f^{(k)}_{N,\beta}(X_k) = \frac{\Psi_{N,k}(X_k)}{\int \Psi_{N,k}\,\mathrm{d}\mu^{\otimes k}}. \tag{4.9}$$

Equivalently,

$$\int \Psi_{N,k}\,\mathrm{d}\mu^{\otimes k} = \frac{\mathsf{K}_{N,\beta}(\mu)}{\mathsf{K}_{n,\gamma_{N,k}}(\mu)}. \tag{4.10}$$

Proof. By Lemma 4.1,

$$\begin{aligned} f^{(k)}_{N,\beta}(X_k) &= \frac{1}{\mathsf{K}_{N,\beta}(\mu)}\int \exp\{-\beta N \mathsf{F}_N((X_k,Y_n),\mu)\}\,\mathrm{d}\mu^{\otimes n}(Y_n) \\ &= \frac{\mathsf{K}_{n,\gamma_{N,k}}(\mu)}{\mathsf{K}_{N,\beta}(\mu)}\exp\left(-\frac{\beta}{N}D_k(X_k)\right)\mathbb{E}_{\mathbb{Q}_{n,\gamma_{N,k}}(\mu)}\left[\exp\left(-\frac{\beta}{N}\sum_{j=1}^{n}\Phi_{X_k}(y_j)\right)\right] \\ &= \frac{\mathsf{K}_{n,\gamma_{N,k}}(\mu)}{\mathsf{K}_{N,\beta}(\mu)}\Psi_{N,k}(X_k). \end{aligned} \tag{4.11}$$

Integrating in X_k gives (4.10); substituting the resulting constant gives (4.9). □

In particular,

$$\chi^2(\mathbb{Q}^{(k)}_{N,\beta}(\mu)\mid\mu^{\otimes k}) = \left\|\frac{\Psi_{N,k}}{\int\Psi_{N,k}\,\mathrm{d}\mu^{\otimes k}} - 1\right\|^2_{L^2(\mu^{\otimes k})}. \tag{4.12}$$

For the upper bound, only the case $k = 1$ will be estimated directly.

4.2. **Orthogonal marginal projection.** The following exact decomposition separates the first Hoeffding level from all higher levels. It explains which estimates the Gibbs argument must supply: a sharp one-point bound and a bounded full-law chi-square divergence.

Lemma 4.3 (Hoeffding projection reduction)**.** *Let g_N be an exchangeable probability density on E^N with respect to $\mu^{\otimes N}$, and let $g^{(k)}_N$ be its labeled k-point marginal density. Then*

$$\chi^2(g^{(k)}_N\mu^{\otimes k}\mid\mu^{\otimes k}) \le k\,\chi^2(g^{(1)}_N\mu\mid\mu) + \left(\frac{k}{N}\right)^2\chi^2(g_N\mu^{\otimes N}\mid\mu^{\otimes N}). \tag{4.13}$$

Proof. Let

$$g_N = 1 + \sum_{\emptyset\neq A\subset\{1,\dots,N\}} h_A \tag{4.14}$$

be the orthogonal Hoeffding decomposition of g_N in $L^2(\mu^{\otimes N})$. Thus, each h_A depends only on $(x_i)_{i\in A}$ and is degenerate in each of its variables. By exchangeability, if $[r] \coloneqq \{1,\dots,r\}$, then every h_A with $|A| = r$ has the same L^2-norm as $h_{[r]}$. Conditional expectation onto the first k variables gives

$$g^{(k)}_N - 1 = \sum_{\emptyset\neq A\subset\{1,\dots,k\}} h_A. \tag{4.15}$$

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For $1 \le k \le N$, orthogonality gives

$$\chi^2(g_N^{(k)}\mu^{\otimes k} \mid \mu^{\otimes k}) = \sum_{r=1}^{k} \binom{k}{r} \|h_{[r]}\|^2_{L^2(\mu^{\otimes r})}. \tag{4.16}$$

The $r = 1$ contribution to (4.16) is $k\|h_{[1]}\|^2_{L^2(\mu)} = k\chi^2(g_N^{(1)}\mu \mid \mu)$. For $r \ge 2$,

$$\frac{\binom{k}{r}}{\binom{N}{r}} \le \left(\frac{k}{N}\right)^r \le \left(\frac{k}{N}\right)^2. \tag{4.17}$$

This proves (4.13). □

4.3. **Density moments and spectral stability.** The Gibbs density supplies the full-law moment bound. A partition-function moment also gives spectral stability of the screened operator. These two statements do not require the quantitative marginal hypotheses or a sign condition on the interaction.

We use the operator T_μ defined in (2.14). The following two consequences of the partition-function bound apply without a sign assumption.

For later use, introduce the centered energy

$$\mathcal{X}_N := N\mathsf{F}_N(X_N, \mu) + I_\mu = \frac{1}{N} \sum_{1 \le i < j \le N} \widetilde{W}(x_i, x_j) - \frac{1}{N}\sum_{i=1}^{N}(h_\mu(x_i) - 2I_\mu). \tag{4.18}$$

Its expectation under $\mu^{\otimes N}$ is zero.

Lemma 4.4 (Uniform L^r bounds for the full density)**.** *Recall from (2.12) that $f_{N,\beta} = f_{N,\beta}^{(N)}$, and recall the normalized partition-function constant $\mathfrak{P}_\gamma(\mu)$ from (3.11). For every $r \in [1, \infty)$, the following bound holds for the full density and, with the same constant, for every marginal density $f_{N,\beta}^{(j)}$, $1 \le j \le N$:*

$$\sup_{N \ge 1} \|f_{N,\beta}\|_{L^r(\mu^{\otimes N})} \le \mathfrak{P}_{r\beta}(\mu)^{1/r}. \tag{4.19}$$

Proof. For $r = 1$ there is nothing to prove. For $r > 1$,

$$\|f_{N,\beta}\|^r_{L^r(\mu^{\otimes N})} = \frac{\mathsf{K}_{N,r\beta}(\mu)}{\mathsf{K}_{N,\beta}(\mu)^r}. \tag{4.20}$$

The numerator is bounded by $e^{r\beta I_\mu}\mathfrak{P}_{r\beta}(\mu)$, while Jensen's inequality gives

$$\mathsf{K}_{N,\beta}(\mu) \ge \exp\left(\beta I_\mu\right). \tag{4.21}$$

Thus, $\|f_{N,\beta}\|^r_{L^r} \le \mathfrak{P}_{r\beta}(\mu)$, which proves (4.19). Marginalization is conditional expectation under the product law and is contractive in L^r, giving the marginal assertion. □

The quadratic-statistic limit used in the next lemma is classical; see Hoeffding [Hoe48] and Dynkin and Mandelbaum [DM83]. We retain its spectral truncation proof to track the partition-function obstruction.

Lemma 4.5 (Spectral stability from a partition-function moment)**.** *Fix $\beta > 0$ and $\theta > 1$. Assume*

$$\widetilde{W} \in L^2(\mu^{\otimes 2}), \qquad h_\mu - 2I_\mu \in L^2(\mu), \qquad \sup_{N \ge 1} \mathsf{K}_{N,\theta\beta}(\mu) < \infty. \tag{4.22}$$

Then every spectral value λ of T_μ satisfies

$$1 + \theta\beta\lambda > 0. \tag{4.23}$$

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Consequently, for every $N \geq 2$,

$$\left\| \left(I + \beta \frac{N-1}{N} T_\mu \right)^{-1} \right\|_{L^2(\mu) \to L^2(\mu)} \leq \frac{\theta N}{(\theta-1)N+1} \leq \frac{\theta}{\theta-1}. \tag{4.24}$$

Proof. Let $(\lambda_m, e_m)_{m \geq 1}$ be an orthonormal spectral representation of the nonzero part of T_μ. Since $T_\mu \mathbf{1} = 0$, each e_m is centered. Use the centered energy in (4.18). For $\widetilde{W}_M = \sum_{m=1}^M \lambda_m e_m \otimes e_m$,

$$\frac{1}{N} \sum_{1 \leq i < j \leq N} \widetilde{W}_M(\xi_i, \xi_j) = \frac{1}{2} \sum_{m=1}^M \lambda_m \left[\left(N^{-1/2} \sum_{i=1}^N e_m(\xi_i) \right)^2 - N^{-1} \sum_{i=1}^N e_m(\xi_i)^2 \right]. \tag{4.25}$$

The multivariate central limit theorem and the law of large numbers therefore give convergence in distribution, for each fixed M, to

$$\frac{1}{2} \sum_{m=1}^M \lambda_m (Z_m^2 - 1), \tag{4.26}$$

where $(Z_m)_{m \geq 1}$ are independent standard Gaussian random variables. If $R_M = \widetilde{W} - \widetilde{W}_M$, canonicity gives the exact variance identity

$$\mathbb{E}_{\mu^{\otimes N}} \left[\left| \frac{1}{N} \sum_{1 \leq i < j \leq N} R_M(\xi_i, \xi_j) \right|^2 \right] = \frac{N-1}{2N} \|R_M\|^2_{L^2(\mu^{\otimes 2})}. \tag{4.27}$$

For clarity, write

$$\mathcal{X}_{N,M} := \frac{1}{N} \sum_{1 \leq i < j \leq N} \widetilde{W}_M(\xi_i, \xi_j). \tag{4.28}$$

Since $(\lambda_m)_{m \geq 1} \in \ell^2$, define

$$\mathcal{X}_\infty := \frac{1}{2} \sum_{m \geq 1} \lambda_m (Z_m^2 - 1), \tag{4.29}$$

$$\mathcal{X}_{\infty,M} := \frac{1}{2} \sum_{m=1}^M \lambda_m (Z_m^2 - 1). \tag{4.30}$$

The first series converges in L^2. If $\Phi : \mathbb{R} \to \mathbb{R}$ is bounded and Lipschitz, then (4.27) and the centeredness of $h_\mu - 2I_\mu$ give

$$\begin{aligned} &\left| \mathbb{E}_{\mu^{\otimes N}} [\Phi(\mathcal{X}_N)] - \mathbb{E}_{\mu^{\otimes N}} [\Phi(\mathcal{X}_{N,M})] \right| \\ &\quad \leq \mathrm{Lip}(\Phi) \left[\left(\frac{N-1}{2N} \right)^{1/2} \|R_M\|_{L^2(\mu^{\otimes 2})} + \frac{1}{\sqrt{N}} \|h_\mu - 2I_\mu\|_{L^2(\mu)} \right]. \end{aligned} \tag{4.31}$$

Moreover, independence of the Gaussian coordinates yields

$$\mathbb{E}\left[|\mathcal{X}_\infty - \mathcal{X}_{\infty,M}|^2 \right] = \frac{1}{2} \sum_{m > M} \lambda_m^2 = \frac{1}{2} \|R_M\|^2_{L^2(\mu^{\otimes 2})} \longrightarrow 0. \tag{4.32}$$

For fixed M, (4.25) gives $\mathcal{X}_{N,M} \Longrightarrow \mathcal{X}_{\infty,M}$. Combining this convergence with (4.31) and (4.32), then letting first $N \to \infty$ and afterward $M \to \infty$, proves convergence of bounded Lipschitz expectations. Hence,

$$\mathcal{X}_N \Longrightarrow \mathcal{X}_\infty. \tag{4.33}$$

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The identity $\mathsf{K}_{N,\theta\beta}(\mu) = e^{\theta\beta I_\mu}\mathbb{E}[e^{-\theta\beta\mathcal{X}_N}]$ and the assumed partition-function bound imply

$$\sup_{N\geq 1}\mathbb{E}_{\mu^{\otimes N}}\left[e^{-\theta\beta\mathcal{X}_N}\right] < \infty. \tag{4.34}$$

By the Portmanteau theorem,

$$\mathbb{E}\left[e^{-\theta\beta\mathcal{X}_\infty}\right] < \infty. \tag{4.35}$$

Fix m. The complementary series

$$R_m \coloneqq \frac{1}{2}\sum_{n\neq m}\lambda_n(Z_n^2 - 1) \tag{4.36}$$

converges in L^2 and almost surely and is independent of Z_m. Tonelli's theorem and independence reduce the corresponding one-coordinate factor in (4.35) to

$$e^{\theta\beta\lambda_m/2}(2\pi)^{-1/2}\int_{\mathbb{R}} e^{-(1+\theta\beta\lambda_m)z^2/2}\,\mathrm{d}z. \tag{4.37}$$

This integral is finite exactly when $1 + \theta\beta\lambda_m > 0$; equality gives a constant integrand after combination with the Gaussian density, and a negative coefficient gives exponential growth. Since the independent complementary factor is strictly positive, (4.35) proves (4.23) for every nonzero eigenvalue. The zero spectral value satisfies the same inequality trivially.

For a negative spectral value, (4.23) gives

$$1 + \beta\frac{N-1}{N}\lambda > 1 - \frac{N-1}{\theta N} = \frac{(\theta-1)N+1}{\theta N}, \tag{4.38}$$

whereas a nonnegative spectral value gives a lower bound of one. The functional calculus proves (4.24). □

Until Section 4.5, let $\theta > 1$ be such that the partition-function bound at $\theta\beta$ holds, in addition to the quantitative moment and cavity hypotheses of Section 3.2. In the all-marginal theorem we take $\theta = 2$, while in Theorem 3.6 we take $\theta = \tau$. The case $\beta = 0$ is immediate, so the screened argument below is needed only for $\beta > 0$.

As in Section 3, write $\xi_1, \dots, \xi_N$ for the coordinate random variables on $(\mathbb{R}^{\mathsf{d}})^N$, reserving X_m for the configuration tuple $(x_1, \dots, x_m)$.

Lemma 4.6 (A one-dimensional moment input)**.** *This lemma is used only under the bounded lower-tail condition. Here, R_p denotes any admissible constant in the scalar Rosenthal inequality. Under (3.20), with p as in (3.15),*

$$\sup_{N\geq 2}\sup_{x\in\mathbb{R}^{\mathsf{d}}}\mathbb{E}_{\Xi_N\sim\mathbb{Q}_{N,\beta}(\mu)}\left[\left|\frac{1}{N}\sum_{j=2}^{N}\widetilde{W}(x,\xi_j)\right|^2\right] \leq \frac{\mathfrak{P}_{r\beta}(\mu)^{1/r}(2\mathsf{R}_p)^{2/p}\|\widetilde{W}\|^2_{L^\infty_x L^p_y(\mu^{\otimes 2})}}{N}. \tag{4.39}$$

Proof. The mixed-norm bound $\widetilde{W} \in L^\infty_x L^p_y(\mu^{\otimes 2})$ is part of the bounded lower-tail condition (3.20).

The Hölder conjugate of $r = p/(p-2)$ is $r' = p/2$. By Lemma 4.4, $\|f_{N,\beta}\|_{L^r} \leq \mathfrak{P}_{r\beta}(\mu)^{1/r}$. For fixed x, the random variables $\widetilde{W}(x,\xi_j)$ are centered under μ, because $\widetilde{W}$ is centered in each variable. The scalar Rosenthal inequality [Ros70] and (3.20) imply

$$\mathbb{E}_{\mu^{\otimes N}}\left[\left|\frac{1}{N}\sum_{j=2}^{N}\widetilde{W}(x,\xi_j)\right|^p\right] \leq 2\mathsf{R}_p\|\widetilde{W}\|^p_{L^\infty_x L^p_y(\mu^{\otimes 2})}N^{-p/2}. \tag{4.40}$$

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Therefore, by Hölder's inequality,

$$\mathbb{E}_{\mathbb{Q}_{N,\beta}(\mu)}\left[\left|\frac{1}{N}\sum_{j=2}^{N}\widetilde{W}(x,\xi_j)\right|^2\right] \le \|f_{N,\beta}\|_{L^r(\mu^{\otimes N})}\left(\mathbb{E}_{\mu^{\otimes N}}\left[\left|\frac{1}{N}\sum_{j=2}^{N}\widetilde{W}(x,\xi_j)\right|^p\right]\right)^{2/p}$$
$$\le \frac{\mathfrak{P}_{r\beta}(\mu)^{1/r}(2\mathsf{R}_p)^{2/p}\|\widetilde{W}\|^2_{L^\infty_x L^p_y(\mu^{\otimes 2})}}{N}, \tag{4.41}$$

which proves (4.39). □

The joint $L^p(\mu^{\otimes 2})$ assumption in (3.16) and the $L^\infty_x L^p_y(\mu^{\otimes 2})$ bound in (3.20) serve different purposes as assumptions. The present fixed-x Rosenthal estimate uses the latter, whereas the partition-function estimate controls the L^r-norm of the full Gibbs density appearing in Hölder's inequality.

4.4. **Conditional density and cancellation.** Recall that $f^{(1)}_{N,\beta}$ denotes the density of $\mathbb{Q}^{(1)}_{N,\beta}(\mu)$ with respect to μ. Initially, we use only that it belongs to $L^1(\mu)$. Set

$$u_N := f^{(1)}_{N,\beta} - 1. \tag{4.42}$$

For $N \ge 2$, set $\hat{\Xi}_1 := (\xi_2, \dots, \xi_N)$ and condition on $\sigma(\hat{\Xi}_1)$. Up to a multiplicative factor depending only on $\hat{\Xi}_1$, the x-dependent part of the Gibbs weight is

$$\exp\left\{\frac{\beta}{N}h_\mu(x) - \frac{\beta}{N}\sum_{j=2}^{N}\widetilde{W}(x,\xi_j)\right\}. \tag{4.43}$$

Subtracting the μ-mean of the exponent, which is $2\beta I_\mu/N$, gives exactly the cavity field B_N defined in (3.19). In particular,

$$\int B_N(x;\hat{\Xi}_1)\,\mathrm{d}\mu(x) = 0, \qquad \forall \hat{\Xi}_1, \tag{4.44}$$

and the conditional density of ξ_1 given $\hat{\Xi}_1$ is

$$\pi_N(x \mid \hat{\Xi}_1) = \frac{e^{B_N(x;\hat{\Xi}_1)}}{\int e^{B_N(z;\hat{\Xi}_1)}\,\mathrm{d}\mu(z)}. \tag{4.45}$$

We now prove the nonlinear cavity estimates under the two conditions stated in Section 3.2.

Bounded lower-tail condition. Under (3.20),

$$\widetilde{W}(x,y) \ge \operatorname*{ess\,inf}_{(z,w)\in(\mathbb{R}^{\mathsf{d}})^2}\widetilde{W}(z,w), \tag{4.46}$$

and therefore

$$B_N(x;\hat{\Xi}_1) \le \frac{\beta}{N}\|h_\mu - 2I_\mu\|_{L^\infty} - \beta\frac{N-1}{N}\operatorname*{ess\,inf}_{(z,w)\in(\mathbb{R}^{\mathsf{d}})^2}\widetilde{W}(z,w). \tag{4.47}$$

Thus, with

$$\mathsf{b}_{\beta,\mu} := \beta(\|h_\mu - 2I_\mu\|_{L^\infty} - \operatorname*{ess\,inf}_{(z,w)\in(\mathbb{R}^{\mathsf{d}})^2}\widetilde{W}(z,w)), \tag{4.48}$$

we have the deterministic upper bound

$$B_N(x;\hat{\Xi}_1) \le \mathsf{b}_{\beta,\mu} \tag{4.49}$$

for all $N \ge 2$, for μ-a.e. x, and for $\mu^{\otimes(N-1)}$-a.e. $\hat{\Xi}_1$. Combining (3.19) with Lemma 4.6, define

$$\mathsf{B}_{2,\beta,\mu} := 2\beta^2\|h_\mu - 2I_\mu\|^2_{L^\infty} + 2\beta^2\mathfrak{P}_{r\beta}(\mu)^{1/r}(2\mathsf{R}_p)^{2/p}\|\widetilde{W}\|^2_{L^\infty_x L^p_y(\mu^{\otimes 2})}. \tag{4.50}$$

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Then

$$\sup_{x\in\mathbb{R}^{\mathsf{d}}} \mathbb{E}_{\mathbb{Q}_{N,\beta}(\mu)}\left[|B_N(x;\hat{\Xi}_1)|^2\right] \le \frac{\mathsf{B}_{2,\beta,\mu}}{N}. \tag{4.51}$$

Logarithmic lower-tail condition. Under (3.21), the positive part of the cavity field satisfies (3.25), and the nonlinear estimates below are controlled by the weighted constants (3.26). The sufficient condition stated in Lemma 7.4 is proved in Appendix A.

The next lemma is the only nonlinear estimate in the one-point argument. After centering, it identifies the normalized exponential conditional density with its exponent up to an $O(N^{-1})$ error in $L^2(\mu)$. For $b \ge 0$, set

$$\Lambda_2(b) := \sup_{t\le b,\ t\neq 0} \frac{e^t - 1 - t}{t^2}, \tag{4.52}$$

$$\Lambda_1(b) := \sup_{t\le b,\ t\neq 0} \frac{e^t - 1 - t}{|t|}. \tag{4.53}$$

We refer to these two definitions collectively as (4.52)–(4.53).

Lemma 4.7 (Abstract nonlinear cavity estimate)**.** *For notational simplicity, write* $B(x) = B_N(x;\hat{\Xi}_1)$*, and set*

$$Z_B := \int e^{B(z)}\,\mathrm{d}\mu(z), \tag{4.54}$$

$$\epsilon(b) := e^b - 1 - b, \tag{4.55}$$

$$\delta_B := \int \epsilon(B(z))\,\mathrm{d}\mu(z). \tag{4.56}$$

Suppose that, for finite constants $\mathsf{E}_2, \mathsf{D}_1, \mathsf{D}_2, \mathsf{B}_2$*,*

$$\|\mathbb{E}\left[\epsilon(B(\cdot))\right]\|_{L^2(\mu)} \le \frac{\mathsf{E}_2}{N}, \tag{4.57a}$$

$$\mathbb{E}\left[\delta_B\right] \le \frac{\mathsf{D}_1}{N}, \tag{4.57b}$$

$$\mathbb{E}\left[\delta_B^2\right] \le \frac{\mathsf{D}_2}{N}, \tag{4.57c}$$

$$\mathbb{E}\left[\int B(x)^2\,\mathrm{d}\mu(x)\right] \le \frac{\mathsf{B}_2}{N}. \tag{4.57d}$$

Then r_N *defined by* (4.63) *satisfies*

$$\|r_N\|_{L^2(\mu)} \le \frac{\mathsf{E}_2 + \mathsf{D}_1 + \sqrt{\mathsf{B}_2\mathsf{D}_2}}{N}. \tag{4.58}$$

Proof. Since $\epsilon \ge 0$, (4.44) gives

$$Z_B = 1 + \delta_B, \tag{4.59}$$

$$\delta_B \ge 0. \tag{4.60}$$

Moreover,

$$\begin{aligned}\frac{e^{B(x)}}{Z_B} - 1 - B(x) &= \frac{1 + B(x) + \epsilon(B(x))}{1+\delta_B} - 1 - B(x)\\ &= \frac{\epsilon(B(x)) - \delta_B - B(x)\delta_B}{1+\delta_B}.\end{aligned} \tag{4.61}$$

Thus, $1 + \delta_B \ge 1$ and Cauchy's inequality yield

$$\|r_N\|_{L^2(\mu)} \le \|\mathbb{E}[\epsilon(B(\cdot))]\|_{L^2(\mu)} + \mathbb{E}[\delta_B] + \|\mathbb{E}[|B(\cdot)|\delta_B]\|_{L^2(\mu)}$$

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$$\leq \frac{\mathsf{E}_2 + \mathsf{D}_1}{N} + \mathbb{E}\left[\delta_B \|B\|_{L^2(\mu)}\right]$$
$$\leq \frac{\mathsf{E}_2 + \mathsf{D}_1}{N} + (\mathbb{E}[\delta_B^2])^{1/2}(\mathbb{E}\left[\int B(x)^2 \,\mathrm{d}\mu(x)\right])^{1/2}. \tag{4.62}$$

This proves (4.58). □

Lemma 4.8 (Cavity linearization)**.** *Define*

$$r_N(x) := \mathbb{E}_{\mathbb{Q}_{N,\beta}(\mu)}\left[\frac{e^{B_N(x;\hat{\Xi}_1)}}{\int e^{B_N(z;\hat{\Xi}_1)}\,\mathrm{d}\mu(z)} - 1 - B_N(x;\hat{\Xi}_1)\right]. \tag{4.63}$$

Under the bounded lower-tail condition, set

$$\mathsf{R}^{\mathrm{bd}}_{\mathrm{cav},\beta,\mu} := \mathsf{B}_{2,\beta,\mu}\left(2\Lambda_2(\mathsf{b}_{\beta,\mu}) + \sqrt{2\Lambda_1(\mathsf{b}_{\beta,\mu})\mathsf{b}_{\beta,\mu}\Lambda_2(\mathsf{b}_{\beta,\mu})}\right). \tag{4.64}$$

Under the logarithmic lower-tail condition, set

$$\mathsf{R}^{\log}_{\mathrm{cav},\beta,\mu} := \mathsf{E}_{\log,\beta,\mu} + \mathsf{D}_{\log,\beta,\mu} + \sqrt{\mathsf{B}_{\log,\beta,\mu}\mathsf{V}_{\log,\beta,\mu}}. \tag{4.65}$$

Let $\mathsf{R}_{\mathrm{cav},\beta,\mu}$ *denote the corresponding one of these two constants. Then, for every* $N \geq 2$*,*

$$\|r_N\|_{L^2(\mu)} \leq \frac{\mathsf{R}_{\mathrm{cav},\beta,\mu}}{N}. \tag{4.66}$$

Proof. It suffices to verify the hypotheses of Lemma 4.7.

Step 1: Bounded lower-tail condition. Under this condition, (4.49) and (4.52)–(4.53) give

$$0 \leq \epsilon(B(x)) \leq \Lambda_2(\mathsf{b}_{\beta,\mu})B(x)^2, \tag{4.67a}$$
$$0 \leq \epsilon(B(x)) \leq \Lambda_1(\mathsf{b}_{\beta,\mu})|B(x)|. \tag{4.67b}$$

Together with (4.51), this gives

$$\|\mathbb{E}[\epsilon(B(\cdot))]\|_{L^2(\mu)} \leq \frac{\Lambda_2(\mathsf{b}_{\beta,\mu})\mathsf{B}_{2,\beta,\mu}}{N}, \tag{4.68a}$$
$$\mathbb{E}[\delta_B] \leq \frac{\Lambda_2(\mathsf{b}_{\beta,\mu})\mathsf{B}_{2,\beta,\mu}}{N}, \tag{4.68b}$$
$$\mathbb{E}\left[\int B(x)^2 \,\mathrm{d}\mu(x)\right] \leq \frac{\mathsf{B}_{2,\beta,\mu}}{N}. \tag{4.68c}$$

Furthermore, (4.44) implies $\int B_+ \,\mathrm{d}\mu = \int B_- \,\mathrm{d}\mu$, and therefore

$$\int |B| \,\mathrm{d}\mu = 2\int B_+ \,\mathrm{d}\mu \leq 2\mathsf{b}_{\beta,\mu}. \tag{4.69}$$

Using (4.67b), we get $\delta_B \leq 2\Lambda_1(\mathsf{b}_{\beta,\mu})\mathsf{b}_{\beta,\mu}$, whence

$$\mathbb{E}[\delta_B^2] \leq \frac{2\Lambda_1(\mathsf{b}_{\beta,\mu})\mathsf{b}_{\beta,\mu}\Lambda_2(\mathsf{b}_{\beta,\mu})\mathsf{B}_{2,\beta,\mu}}{N}. \tag{4.70}$$

Lemma 4.7 gives (4.66) with $\mathsf{R}_{\mathrm{cav},\beta,\mu} = \mathsf{R}^{\mathrm{bd}}_{\mathrm{cav},\beta,\mu}$.

Step 2: Logarithmic lower-tail condition. Under this condition, the elementary inequalities

$$0 \leq e^t - 1 - t \leq \frac{1}{2}e^{t_+}t^2, \tag{4.71a}$$
$$0 \leq e^t - 1 - t \leq e^{t_+}|t| \tag{4.71b}$$

and the bound (3.25) imply

$$\epsilon(B(x)) \leq \frac{1}{2}e^{\Phi_N(x;\hat{\Xi}_1)}B(x)^2, \tag{4.72a}$$

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$$\epsilon(B(x)) \le e^{\Phi_N(x;\hat{\Xi}_1)}|B(x)|. \tag{4.72b}$$

The weighted estimates (3.26) therefore give

$$\|\mathbb{E}[\epsilon(B(\cdot))]\|_{L^2(\mu)} \le \frac{\mathsf{E}_{\log,\beta,\mu}}{N}, \tag{4.73a}$$

$$\mathbb{E}[\delta_B] \le \frac{\mathsf{D}_{\log,\beta,\mu}}{N}, \tag{4.73b}$$

$$\mathbb{E}[\delta_B^2] \le \frac{\mathsf{V}_{\log,\beta,\mu}}{N}, \tag{4.73c}$$

$$\mathbb{E}\left[\int B(x)^2 \,\mathrm{d}\mu(x)\right] \le \frac{\mathsf{B}_{\log,\beta,\mu}}{N}. \tag{4.73d}$$

Another application of Lemma 4.7 gives (4.66) with $\mathsf{R}_{\mathrm{cav},\beta,\mu} = \mathsf{R}^{\log}_{\mathrm{cav},\beta,\mu}$. □

We can now close the one-point estimate. By (4.45),

$$f^{(1)}_{N,\beta}(x) = \mathbb{E}_{\mathbb{Q}_{N,\beta}(\mu)}\left[\pi_N(x \mid \hat{\Xi}_1)\right]. \tag{4.74}$$

Therefore, by Lemma 4.8,

$$u_N(x) = \mathbb{E}_{\mathbb{Q}_{N,\beta}(\mu)}\left[B_N(x;\hat{\Xi}_1)\right] + r_N(x), \tag{4.75a}$$

$$\|r_N\|_{L^2(\mu)} \le \frac{\mathsf{R}_{\mathrm{cav},\beta,\mu}}{N}. \tag{4.75b}$$

The bounded branch estimate (4.51) and the logarithmic estimate (4.73d) imply, respectively, that there is a finite constant $\mathsf{B}_{\mathrm{cav},\beta,\mu}$ such that

$$\mathbb{E}_{\mathbb{Q}_{N,\beta}(\mu)}\left[\|B_N(\cdot;\hat{\Xi}_1)\|^2_{L^2(\mu)}\right] \le \frac{\mathsf{B}_{\mathrm{cav},\beta,\mu}}{N}. \tag{4.76}$$

The section map is strongly measurable by approximation with rectangular simple functions. It is therefore Bochner integrable in $L^2(\mu)$, and Jensen's inequality gives

$$\left\|\mathbb{E}_{\mathbb{Q}_{N,\beta}(\mu)}\left[B_N(\cdot;\hat{\Xi}_1)\right]\right\|^2_{L^2(\mu)} \le \frac{\mathsf{B}_{\mathrm{cav},\beta,\mu}}{N}. \tag{4.77}$$

Together with (4.75b), this proves $u_N \in L^2(\mu)$ before the screened equation is written.

Since $f^{(1)}_{N,\beta} = 1 + u_N \in L^2(\mu)$ and $\widetilde{W} \in L^2(\mu^{\otimes 2})$,

$$\int_{(\mathbb{R}^{\mathsf{d}})^2} |\widetilde{W}(x,y)| f^{(1)}_{N,\beta}(y)\,\mathrm{d}\mu(x)\mathrm{d}\mu(y) \le \|\widetilde{W}\|_{L^2(\mu^{\otimes 2})}\|f^{(1)}_{N,\beta}\|_{L^2(\mu)} < \infty. \tag{4.78}$$

Thus, Fubini's theorem and exchangeability justify

$$\begin{aligned}
\mathbb{E}_{\mathbb{Q}_{N,\beta}(\mu)}\left[B_N(x;\hat{\Xi}_1)\right] &= \frac{\beta}{N}(h_\mu(x) - 2I_\mu) - \frac{\beta}{N}\sum_{j=2}^N \mathbb{E}_{\mathbb{Q}_{N,\beta}(\mu)}\left[\widetilde{W}(x,\xi_j)\right] \\
&= \frac{\beta}{N}(h_\mu(x) - 2I_\mu) - \beta\frac{N-1}{N}\int \widetilde{W}(x,y) f^{(1)}_{N,\beta}(y)\,\mathrm{d}\mu(y) \\
&= \frac{\beta}{N}(h_\mu(x) - 2I_\mu) - \beta\frac{N-1}{N} T_\mu u_N(x).
\end{aligned} \tag{4.79}$$

Here, the last equality follows from $\int \widetilde{W}(x,y)\,\mathrm{d}\mu(y) = 0$. Combining (4.75a) and (4.79),

$$\left(I + \beta\frac{N-1}{N}T_\mu\right)u_N = \frac{\beta}{N}(h_\mu - 2I_\mu) + r_N. \tag{4.80}$$

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By Lemma 4.5, the screened operator is invertible and satisfies (4.24). Define

$$\mathsf{U}_{\theta,\beta,\mu} \coloneqq \frac{\theta}{\theta-1}\left(\beta\|h_\mu - 2I_\mu\|_{L^2(\mu)} + \mathsf{R}_{\mathrm{cav},\beta,\mu}\right). \tag{4.81}$$

Then (4.80) gives

$$\|u_N\|_{L^2(\mu)} \le \frac{\mathsf{U}_{\theta,\beta,\mu}}{N}, \qquad N \ge 2. \tag{4.82}$$

For the all-marginal theorem, set

$$\mathsf{U}_{\beta,\mu} \coloneqq \mathsf{U}_{2,\beta,\mu} = 2\left(\beta\|h_\mu - 2I_\mu\|_{L^2(\mu)} + \mathsf{R}_{\mathrm{cav},\beta,\mu}\right). \tag{4.83}$$

Thus, under the hypotheses of Theorem 3.4,

$$\chi^2(\mathbb{Q}^{(1)}_{N,\beta}(\mu) \mid \mu) = \|f^{(1)}_{N,\beta} - 1\|^2_{L^2(\mu)} \le \frac{\mathsf{U}^2_{\beta,\mu}}{N^2}. \tag{4.84}$$

By Lemma 4.2 with $k = 1$, this is equivalently the one-mark estimate

$$\left\| \frac{\Psi_{N,1}}{\int \Psi_{N,1}\,\mathrm{d}\mu} - 1 \right\|_{L^2(\mu)} \le \frac{\mathsf{U}_{\beta,\mu}}{N}. \tag{4.85}$$

Remark 4.9. The seemingly dangerous term $N^{-1}\sum_{j\neq 1}\widetilde{W}(x,\xi_j)$ has a nonzero Gibbs average, and this average is exactly $\frac{N-1}{N}T_\mu u_N$. Rather than estimating this contribution as an error, we absorb it into the left-hand side of (4.80). The partition-function moment excludes spectrum at or beyond the singular threshold and gives the uniform inverse bound (4.24), even when T_μ has both positive and negative spectral components. Thus, no smallness condition on β or on $\|T_\mu\|$, and no positivity assumption, is needed.

4.5. **Moment bounds for quantitative marginal approximation.** In this subsection, assume the hypotheses of Theorem 3.1, independently of the order-bound hypotheses used above. We verify the same first-order cavity estimate when the source is only bounded above and belongs to $L^6(\mu)$.

With $\mathcal{X}_N$ as in (4.18), set

$$\mathsf{K}^{\mathrm{c}}_{N,\gamma} \coloneqq e^{-\gamma I_\mu}\mathsf{K}_{N,\gamma}(\mu) = \mathbb{E}_{\mu^{\otimes N}}\left[e^{-\gamma\mathcal{X}_N}\right], \tag{4.86}$$

$$f_{N,\beta} = \frac{e^{-\beta\mathcal{X}_N}}{\mathsf{K}^{\mathrm{c}}_{N,\beta}}. \tag{4.87}$$

The exponent is $-\beta\mathsf{W}_N(\widetilde{W})/2 + \beta N^{-1}\sum_i \psi_\mu(x_i)$ and has product mean zero. Hence,

$$\mathsf{K}^{\mathrm{c}}_{N,\gamma} \ge 1, \qquad N \ge 1, \quad 0 \le \gamma \le \Gamma. \tag{4.88}$$

Hölder extends (3.6) to the interval $[0,\Gamma]$. The upper bound for ψ_μ and Lemma 4.4 therefore give

$$\sup_{N\ge 2}\|f_{N,\beta}\|^r_{L^r(\mu^{\otimes N})} \le e^{r\beta C_+}\sup_{N\ge 2}\mathbb{E}_{\mu^{\otimes N}}\left[e^{-r\beta\mathsf{W}_N(\widetilde{W})/2}\right] < \infty, \qquad 1 \le r \le r_*. \tag{4.89}$$

The same bound holds for every marginal by conditional expectation. The $N = 1$ normalizer is finite because $\psi_\mu \le C_+$, so Lemma 4.5 applies with $\theta = r_*$. In particular,

$$0 < R_{\beta,\mu} \le \frac{r_*}{r_*-1}\mathrm{Id}, \qquad \left\|\left(\mathrm{Id} + \beta\frac{N-1}{N}T_\mu\right)^{-1}\right\|_{\mathrm{op}} \le \frac{r_*}{r_*-1}, \qquad N \ge 2. \tag{4.90}$$

Lemma 4.10 (Cavity moments under the quantitative hypotheses)**.** *Use the field B_N from (3.19). It satisfies $B_N \le \beta(C_+ + C_-)$ outside a product-null set. Uniformly for $N \ge 2$,*

$$\mathbb{E}_{\mathbb{Q}_{N,\beta}(\mu)}\left[|B_N(x)|^{2+\vartheta}\right] \le C\left(N^{-1-\vartheta/2} + N^{-2-\vartheta}|\psi_\mu(x)|^{2+\vartheta}\right) \quad \textit{for } \mu\textit{-almost every } x. \tag{4.91}$$

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Moreover, for every finite $q \geq 2$,

$$\mathbb{E}_{\mathbb{Q}_{N,\beta}(\mu)}\left[\|B_N\|_{L^2(\mu)}^q\right] \leq C_q N^{-q/2}. \tag{4.92}$$

With δ_B defined in (4.56), for every finite $q \geq 1$,

$$0 \leq \delta_B \leq C\|B_N\|_2^2, \qquad \mathbb{E}_{\mathbb{Q}_{N,\beta}(\mu)}[\delta_B^q] \leq C_q N^{-q}. \tag{4.93}$$

Proof. The upper bound for the field follows from (3.2)–(3.3). For fixed x, the variables $w_x(x_j)$ are independent and centered under the product law. Rosenthal's inequality [Ros70] and the row bound give

$$\mathbb{E}_{\mu^{\otimes N}}\left[\left|\frac{1}{N}\sum_{j=2}^{N} w_x(x_j)\right|^p\right] \leq C_p M_p^p N^{-p/2}. \tag{4.94}$$

Take $r_0 = p/(p-2-\vartheta)$ and its conjugate $r_0' = p/(2+\vartheta)$. The full-density bound (4.89) applies at r_0. Separating the deterministic source in B_N and applying Hölder proves (4.91). Lyapunov's inequality gives the corresponding estimate at every exponent between 2 and $2+\vartheta$.

By symmetry, the row map $x \mapsto w_x$ takes values in $L^2(\mu)$, has norm at most M_p, and has Bochner mean zero. Strong measurability follows by approximation of the measurable square-integrable kernel with rectangular simple functions. For independent centered Hilbert-valued variables Y_j with $\|Y_j\| \leq M_p$, symmetrization and the Hilbert-space Khintchine–Kahane inequality [Kah85] give

$$\mathbb{E}\left[\left\|\sum_{j=2}^{N} Y_j\right\|^q\right] \leq C_q \mathbb{E}\left[\left(\sum_{j=2}^{N}\|Y_j\|^2\right)^{q/2}\right] \leq C_q M_p^q N^{q/2}. \tag{4.95}$$

Apply this at exponent $2q$ to $Y_j = w_{x_j}$. Hölder with the full-density L^2 bound transfers the estimate to Gibbs expectation. The deterministic term requires only $\psi_\mu \in L^2(\mu)$. This proves (4.92). Finally, on the upper-bounded range of B_N, $0 \leq e^{B_N} - 1 - B_N \leq CB_N^2$. Integration and (4.92) prove (4.93). □

Proposition 4.11 (The first component under the quantitative hypotheses)**.** *Under the hypotheses of Theorem 3.1,*

$$\|r_N\|_{L^2(\mu)} + \|u_N\|_{L^2(\mu)} \leq \frac{C}{N}, \qquad N \geq 2, \tag{4.96}$$

where u_N and r_N are defined in (4.42) and (4.63).

Proof. We verify Lemma 4.7. The exponent-2 consequence of (4.91) and $\psi_\mu \in L^4(\mu)$ give

$$\left\|\mathbb{E}_{\mathbb{Q}_{N,\beta}(\mu)}[B_N(\cdot)^2]\right\|_2 \leq C/N. \tag{4.97}$$

The upper field bound implies $\epsilon(B_N) \leq CB_N^2$. The other three inputs follow from (4.93) and (4.92); in particular, $\mathbb{E}[\delta_B^2] \leq CN^{-2} \leq CN^{-1}$. Thus, $\|r_N\|_2 \leq C/N$. The full-density L^2 bound already gives $u_N \in L^2(\mu)$. The conditional-density and Fubini computation leading to (4.80) uses only this integrability, canonicity, and the exact field, so the same equation holds here. Applying (4.90) proves the asserted estimate for u_N. □

4.6. **Proof of the marginal upper bounds.** The full-space χ^2-divergence is uniformly bounded by the partition-function estimate. Indeed,

$$1 + \chi^2(\mathbb{Q}_{N,\beta}(\mu) \mid \mu^{\otimes N}) = \frac{\mathsf{K}_{N,2\beta}(\mu)}{\mathsf{K}_{N,\beta}(\mu)^2} \leq \mathfrak{P}_{2\beta}(\mu), \tag{4.98}$$

using the definition of $\mathfrak{P}_{2\beta}(\mu)$ and Jensen's lower bound at β.

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Proof of the upper bound in Theorem 3.4. The case $\beta = 0$ is immediate. If $N = 1$, then necessarily $k = 1$, and

$$1 + \chi^2(\mathbb{Q}_{1,\beta}(\mu) \mid \mu) = \frac{\mathsf{K}_{1,2\beta}(\mu)}{\mathsf{K}_{1,\beta}(\mu)^2} \le \mathfrak{P}_{2\beta}(\mu). \tag{4.99}$$

so the asserted bound follows directly. We may therefore assume $N \ge 2$. Apply the one-point estimate with $\theta = 2$, and then apply Lemma 4.3 with $g_N = f_{N,\beta}$. Combining (4.84) and (4.98), we obtain

$$\chi^2(\mathbb{Q}^{(k)}_{N,\beta}(\mu) \mid \mu^{\otimes k}) \le k\frac{\mathsf{U}^2_{\beta,\mu}}{N^2} + (\mathfrak{P}_{2\beta}(\mu) - 1)\frac{k^2}{N^2}. \tag{4.100}$$

The χ^2-upper constant used here is

$$\mathsf{C}^{\mathrm{up}}_{\chi,\beta,\mu} := \mathsf{U}^2_{\beta,\mu} + (\mathfrak{P}_{2\beta}(\mu) - 1). \tag{4.101}$$

Since $k \le k^2$ for $k \ge 1$, this proves the χ^2-line of (3.28) with the constant $\mathsf{C}^{\mathrm{up}}_{\chi,\beta,\mu}$. □

5. Strong approximation of the pair kernel

The scalar Laplace comparison in Section 5.1 applies to signed canonical kernels. Under the quantitative Gibbs hypotheses, it estimates the correlated-replica and screened-response families. These estimates identify the squared norm and cross term of the pair kernel; polarization gives strong convergence. No interacting fluctuation theorem or weak-covariance identification is used.

5.1. Quantitative Laplace comparison without a sign condition.

We first prove a scalar estimate with assumptions independent of the Gibbs hypotheses. Let (S, ρ) be a standard Borel probability space, and let $c \in L^{2+\alpha}(\rho^{\otimes 2})$, where $0 < \alpha \le 1$, be real, symmetric, and canonical. Write C for its integral operator and $\mathsf{W}_N(c) = N^{-1}\sum_{i\neq j} c(x_i, x_j)$ for independent coordinates with law ρ. Let

$$\mathcal{Q}_C := \sum_j \lambda_j(C)(Z_j^2 - 1) \quad \text{in } L^2, \tag{5.1}$$

where the Z_j are independent standard Gaussians. The common construction and Hilbert–Schmidt continuity of these forms are given in Section C.3.1.

Lemma 5.1 (Smooth comparison for a canonical quadratic statistic)**.** *If $h \in C^2(\mathbb{R})$ has bounded first and second derivatives and globally α-Hölder second derivative, then*

$$\begin{aligned}&\left|\mathbb{E}\left[h\left(\frac{1}{2}\mathsf{W}_N(c)\right)\right] - \mathbb{E}\left[h\left(\frac{1}{2}\mathcal{Q}_C\right)\right]\right| \\ &\qquad \le C_\alpha[h'']_{C^{0,\alpha}} N^{-\alpha/2}\|c\|_{2+\alpha}^{2+\alpha} + \frac{\|h'\|_\infty \|c\|_2}{\sqrt{2N}}, \qquad N \ge 2.\end{aligned} \tag{5.2}$$

The constant depends only on α.

The proof in Section C.3.2 replaces one independent coordinate at a time, following the Lindeberg strategy of [Cha06, Section 2]. Moment matching cancels the first two Taylor terms. Only polynomial moments are used under the intermediate original/Gaussian product laws. The same lemma will give the absolute-moment estimate used for growing total variation.

Theorem 5.2 (Quantitative Laplace comparison)**.** *Fix $\beta > 0$ and $q > 1$, and suppose that*

$$M := \sup_{N\ge 2} \mathbb{E}_{\rho^{\otimes N}}\left[e^{-q\beta \mathsf{W}_N(c)/2}\right] < \infty. \tag{5.3}$$

Then every spectral value λ of C satisfies $1 + q\beta\lambda > 0$, and, for every $N \ge 2$,

$$\left|\mathbb{E}_{\rho^{\otimes N}}\left[e^{-\beta \mathsf{W}_N(c)/2}\right] - \det{}_2(\mathrm{Id} + \beta C)^{-1/2}\right|$$

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$$\leq C_{\alpha,\beta,q}\left(M + \|c\|_{2+\alpha}^{2+\alpha} + \|c\|_2\right) N^{-\alpha(q-1)/(2q)}. \tag{5.4}$$

The determinant is positive. The constant is independent of the rank, probability space, and N.

Proof. **The limiting exponential and its determinant.** Applying Lemma 5.1 to smooth bounded tests gives $\mathsf{W}_N(c)/2 \Longrightarrow \mathcal{Q}_C/2$. The Portmanteau theorem and (5.3) imply

$$\mathbb{E}\left[e^{-q\beta\mathcal{Q}_C/2}\right] \leq M. \tag{5.5}$$

The one-mode argument in the proof of Lemma 4.5 now gives $1 + q\beta\lambda > 0$ for every eigenvalue, including strictness at the boundary: the corresponding one-dimensional Gaussian integral would otherwise be infinite, while the independent complementary exponential is positive. The zero spectral value is harmless. Hence,

$$\mathrm{Id} + \beta C \geq (1 - q^{-1})\mathrm{Id}. \tag{5.6}$$

Let $\mathcal{Q}_C^{(m)}$ be the spectral partial sum through the first m Gaussian modes. It is the conditional expectation of $\mathcal{Q}_C$ given those modes. Conditional Jensen gives

$$\mathbb{E}\left[\left(e^{-\beta\mathcal{Q}_C^{(m)}/2}\right)^q\right] \leq \mathbb{E}\left[e^{-q\beta\mathcal{Q}_C/2}\right] \leq M. \tag{5.7}$$

The exponentials are therefore uniformly integrable. Their convergence in probability and finite-dimensional Gaussian integration yield

$$\mathbb{E}\left[e^{-\beta\mathcal{Q}_C/2}\right] = \prod_j e^{\beta\lambda_j(C)/2}(1 + \beta\lambda_j(C))^{-1/2} = \det{}_2(\mathrm{Id} + \beta C)^{-1/2}. \tag{5.8}$$

The logarithms of the factors are $O(\lambda_j(C)^2)$ near zero; (5.6) controls the negative spectrum. Thus, the product is finite and positive without a trace-class assumption.

Truncating the scalar exponential. Choose a fixed smooth χ with $0 \leq \chi \leq 1$, equal to zero on $(-\infty, -1]$ and to one on $[0, \infty)$. For $L \geq 0$, set

$$h_L(x) := e^{-\beta x}\chi(x + L). \tag{5.9}$$

Since $h_L(x) = e^{\beta L}h_0(x + L)$ and h_0 has bounded derivatives through order three,

$$\|h_L'\|_\infty + [h_L'']_{C^{0,\alpha}} \leq C_{\alpha,\beta}e^{\beta L}. \tag{5.10}$$

The error is bounded pointwise by

$$0 \leq e^{-\beta x} - h_L(x) \leq e^{-\beta x}\mathbf{1}_{\{x<-L\}} \leq e^{-(q-1)\beta L}e^{-q\beta x}. \tag{5.11}$$

Use Lemma 5.1 for h_L and (5.3), (5.5) for the two tails. The total error is at most

$$C_{\alpha,\beta}e^{\beta L}\left(N^{-\alpha/2}\|c\|_{2+\alpha}^{2+\alpha} + N^{-1/2}\|c\|_2\right) + 2Me^{-(q-1)\beta L}. \tag{5.12}$$

Taking $L = \alpha \log N/(2q\beta)$ proves (5.4); the finite-Gaussian error is no larger since $\alpha \leq 1$. The exponential moment has been used only at the original and limiting laws, not during coordinate replacement. □

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5.2. **Common inputs for the quantitative proof.** From this subsection through Section 5.5, assume the hypotheses of Theorem 3.1, with its fixed exponents p, ϑ, α. We use the centered normalization (4.86)–(4.87), the density moments (4.89), and the resolvent bound (4.90). Define

$$v_{N,\beta}(x,y) \coloneqq f^{(2)}_{N,\beta}(x,y) - f^{(1)}_{N,\beta}(x) - f^{(1)}_{N,\beta}(y) + 1, \tag{5.13}$$

and abbreviate it by v_N. As before, $u_N = f^{(1)}_{N,\beta} - 1$. The first-component estimate is (4.96). Lemma 4.3 and the density moment at exponent two give

$$\sum_{r=1}^{N} \binom{N}{r} \|h_{[r]}\|^2_{L^2(\mu^{\otimes r})} = \|f_{N,\beta} - 1\|^2_{L^2(\mu^{\otimes N})} \le C, \tag{5.14}$$

where $h_{[1]} = u_N$ and $h_{[2]} = v_N$. The scalar estimates below have rate

$$\mathfrak{e}_N \coloneqq N^{-3\alpha/8}. \tag{5.15}$$

This rate is distinct from the marginal rate (3.7).

Lemma 5.3 (Insertion of a centered one-body term)**.** *Let A, V be real random variables with $V \le C_0$, $\mathbb{E}[V^2] \le C_1/N$, and $\mathbb{E}[e^{2A}] \le C_2$. Then*

$$\left|\mathbb{E}[e^{A+V}] - \mathbb{E}[e^A]\right| \le CN^{-1/2}, \tag{5.16}$$

where C depends only on C_0, C_1, C_2.

Proof. On the half-line $v \le C_0$, one has $|e^v - 1| \le C_{C_0}|v|$. Multiply this inequality by e^A and apply Cauchy–Schwarz. □

5.3. **Correlated replicas and noise stability.** Under the resampling coupling below, the squared norms of the Hoeffding components are the coefficients of a power series with nonnegative coefficients. The replica representation expresses this series through partition functions. A uniform estimate for the corresponding determinant will allow us to extract its degree-two coefficient.

For $0 \le t < 1$, define the one-site coupling

$$\nu_t(\mathrm{d}x\,\mathrm{d}y) \coloneqq (1-t)\mu(\mathrm{d}x)\mu(\mathrm{d}y) + t\mu(\mathrm{d}x)\delta_x(\mathrm{d}y). \tag{5.17}$$

Both marginals are μ. Conditional on $X = x$, the variable Y is retained with probability t and otherwise resampled independently from μ.

On $E = \mathbb{R}^{\mathsf{d}} \times \mathbb{R}^{\mathsf{d}}$, define

$$K_t((x,y),(x',y')) \coloneqq \widetilde{W}(x,x') + \widetilde{W}(y,y'), \tag{5.18}$$

$$\psi_t(x,y) \coloneqq \psi_\mu(x) + \psi_\mu(y). \tag{5.19}$$

The kernel K_t is symmetric, canonical with respect to ν_t, and belongs to $L^2(\nu_t^{\otimes 2})$. For any canonical $r \in L^2(\mu^{\otimes 2})$ and independent sites $(X,Y),(X',Y')$ with law ν_t,

$$\mathbb{E}[r(Y,Y') \mid X, X'] = t^2 r(X,X'), \tag{5.20}$$

$$\|r(x,x') + r(y,y')\|^2_{L^2(\nu_t^{\otimes 2})} = 2(1+t^2)\|r\|_2^2. \tag{5.21}$$

Every term with a resampled coordinate vanishes by canonicity. Thus, the coupled Hilbert–Schmidt estimate uses independence between sites, without requiring independence within a site.

Let $\mathcal{T}_t$ be the integral operator with kernel K_t on $L^2(\nu_t)$.

Lemma 5.4 (Paired spectrum)**.** *If (λ_j, e_j) are the nonzero spectral data of T_μ, then the nonzero eigenvalues of $\mathcal{T}_t$ are*

$$(1+t)\lambda_j \quad \text{and} \quad (1-t)\lambda_j, \qquad j \ge 1. \tag{5.22}$$

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Proof. For centered $f, g \in L^2(\mu)$,

$$\int f(x)g(y)\,\mathrm{d}\nu_t(x,y) = t\int fg\,\mathrm{d}\mu. \tag{5.23}$$

A direct conditional-expectation calculation gives

$$\mathcal{T}_t[e_j(x)] = \lambda_j e_j(x) + t\lambda_j e_j(y), \tag{5.24}$$

$$\mathcal{T}_t[e_j(y)] = t\lambda_j e_j(x) + \lambda_j e_j(y). \tag{5.25}$$

Thus, $e_j(x)+e_j(y)$ and $e_j(x)-e_j(y)$ have the eigenvalues in (5.22). Their squared norms are $2(1+t)$ and $2(1-t)$, respectively.

For centered f, g, the additive map $J_t(f,g) = f(x) + g(y)$ satisfies

$$(1-t)(\|f\|_2^2 + \|g\|_2^2) \le \|J_t(f,g)\|^2_{L^2(\nu_t)} \le (1+t)(\|f\|_2^2 + \|g\|_2^2). \tag{5.26}$$

Its range is therefore closed for $t < 1$. The range of $\mathcal{T}_t$ is contained in this additive centered subspace, so every nonzero eigenfunction is additive. The orthogonal complement is annihilated by self-adjointness, and the remaining additive directions arise from the kernel of T_μ and are also annihilated. On the additive centered subspace, $\mathcal{T}_t$ is represented by

$$(f,g) \longmapsto (T_\mu(f+tg), T_\mu(tf+g)). \tag{5.27}$$

Decomposing f, g into the spectral subspaces of T_μ shows that the displayed symmetric and antisymmetric eigenfunctions exhaust the nonzero spectrum. □

Define

$$\mathsf{K}^{\mathrm{c},[2]}_{N,\beta}(t) \coloneqq \mathbb{E}_{\nu_t^{\otimes N}}\left[\exp\left\{-\frac{\beta}{2N}\sum_{i\neq j} K_t(Z_i,Z_j) + \frac{\beta}{N}\sum_{i=1}^N \psi_t(Z_i)\right\}\right]. \tag{5.28}$$

Let T_t be the product-resampling operator on $L^2(\mu^{\otimes N})$. On a Hoeffding component indexed by $A \subset [N]$,

$$\mathsf{T}_t h_A = t^{|A|} h_A. \tag{5.29}$$

Thus, the overlap and its coefficients satisfy

$$A_N(t) \coloneqq \langle f_{N,\beta}, \mathsf{T}_t f_{N,\beta}\rangle - 1 = \sum_{r=1}^N c_{N,r} t^r, \qquad c_{N,r} \coloneqq \binom{N}{r}\|h_{[r]}\|_2^2 \ge 0, \tag{5.30}$$

$$1 + A_N(t) = \frac{\mathsf{K}^{\mathrm{c},[2]}_{N,\beta}(t)}{(\mathsf{K}^{\mathrm{c}}_{N,\beta})^2}. \tag{5.31}$$

The coefficient sum is uniformly bounded by (5.14). The eigenvalues of $B_{\beta,\mu}$ are $-\beta\lambda/(1+\beta\lambda)$. By (4.90), a negative λ gives a value in $[0, 1/11)$, while a nonnegative λ gives a value in $(-1, 0]$. Since T_μ is bounded, these inequalities imply $\|B_{\beta,\mu}\|_{\mathrm{op}} < 1$. Put

$$D(t) \coloneqq \det(\mathrm{Id} - t^2 B^2_{\beta,\mu})^{-1/2}. \tag{5.32}$$

Only the square of the Hilbert–Schmidt operator occurs in this ordinary Fredholm determinant.

Proposition 5.5 (Quantitative correlated-replica estimate)**.** *Fix $0 < t_0 < 1$. Uniformly for $N \ge 2$,*

$$\sup_{0\le t\le t_0}\left|\mathsf{K}^{\mathrm{c},[2]}_{N,\beta}(t) - [\det_2(\mathrm{Id}+\beta(1+t)T_\mu)\det_2(\mathrm{Id}+\beta(1-t)T_\mu)]^{-1/2}\right| \le C\mathfrak{e}_N, \tag{5.33}$$

$$\sup_{0\le t\le t_0}|A_N(t) - (D(t)-1)| \le C\mathfrak{e}_N. \tag{5.34}$$

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Proof. **The paired Laplace transform.** By the marginal laws of ν_t and the triangle inequality,

$$\|K_t\|_{L^{2+\alpha}(\nu_t^{\otimes 2})} \le 2\|\widetilde{W}\|_{L^{2+\alpha}(\mu^{\otimes 2})}. \tag{5.35}$$

For the fourth exponential moment, Cauchy–Schwarz under $\nu_t^{\otimes N}$ gives

$$\mathbb{E}_{\nu_t^{\otimes N}}\left[e^{-2\beta \mathsf{W}_N(K_t)}\right] \le \mathbb{E}_{\mu^{\otimes N}}\left[e^{-4\beta \mathsf{W}_N(\widetilde{W})}\right] \le C. \tag{5.36}$$

The last estimate uses (3.6) at inverse temperature $8\beta \le \Gamma$. It uses independence between paired sites, not independence of the coordinates within a site. Apply Theorem 5.2 with $q = 4$ on (E, ν_t). Its constant is uniform for $0 \le t \le t_0$ by (5.35)–(5.36). The resulting error is $C\mathfrak{e}_N$.

To restore the source in (5.28), use Lemma 5.3 with $V = \beta N^{-1}\sum_i \psi_t(Z_i)$. This variable is bounded above by $2\beta C_+$, and (5.23) gives

$$\mathbb{E}_{\nu_t^{\otimes N}}[V^2] = \frac{2\beta^2(1+t)}{N}\|\psi_\mu\|_2^2 \le C/N. \tag{5.37}$$

The required second exponential moment follows from (5.36). Since $N^{-1/2} \le \mathfrak{e}_N$, source insertion does not change the rate. Lemma 5.4 identifies the limiting determinant and proves (5.33). Its additive-space argument is uniform on the fixed interval because $1 - t \ge 1 - t_0 > 0$.

The overlap ratio. Denote the displayed paired determinant by $\mathsf{K}^{\mathrm{c},[2]}_{\infty,\beta}(t)$ and put $\mathsf{K}^{\mathrm{c}}_{\infty,\beta} = \det_2(\mathrm{Id} + \beta T_\mu)^{-1/2}$. At $t = 0$ the two coordinates are independent, so

$$\mathsf{K}^{\mathrm{c},[2]}_{N,\beta}(0) = (\mathsf{K}^{\mathrm{c}}_{N,\beta})^2, \qquad \mathsf{K}^{\mathrm{c},[2]}_{\infty,\beta}(0) = (\mathsf{K}^{\mathrm{c}}_{\infty,\beta})^2. \tag{5.38}$$

Thus, the first estimate also controls the squared denominator with error $C\mathfrak{e}_N$. Both denominators are at least one, by Jensen for the centered finite-particle and Gaussian exponents. The paired limiting numerator is uniformly bounded by the Gaussian fourth-moment transfer in Theorem 5.2. Consequently,

$$\sup_{0\le t\le t_0}\left|1 + A_N(t) - \frac{\mathsf{K}^{\mathrm{c},[2]}_{\infty,\beta}(t)}{(\mathsf{K}^{\mathrm{c}}_{\infty,\beta})^2}\right| \le C\mathfrak{e}_N. \tag{5.39}$$

For every spectral value λ of T_μ, the regularization factors cancel and the remaining ratio is

$$\frac{1+\beta\lambda}{\sqrt{(1+\beta(1+t)\lambda)(1+\beta(1-t)\lambda)}} = \left(1 - t^2\left(\frac{\beta\lambda}{1+\beta\lambda}\right)^2\right)^{-1/2}. \tag{5.40}$$

All square roots are positive by spectral stability. Since $B^2_{\beta,\mu}$ is trace class and $\|B_{\beta,\mu}\|_{\mathrm{op}} < 1$, the product converges uniformly on this interval. It gives

$$\frac{\mathsf{K}^{\mathrm{c},[2]}_{\infty,\beta}(t)}{(\mathsf{K}^{\mathrm{c}}_{\infty,\beta})^2} = \det(\mathrm{Id} - t^2 B^2_{\beta,\mu})^{-1/2} = D(t). \tag{5.41}$$

Together with (5.39), this proves (5.34) for every fixed $t_0 < 1$. □

5.4. **Quantitative screened response.** To estimate the pairing with $b_{\beta,\mu}$, perturb the canonical pair interaction in that direction and leave the one-body source fixed. Set

$$A_\theta := \beta T_\mu + \theta B_{\beta,\mu}, \qquad K_\theta(x,y) := \beta\widetilde{W}(x,y) + \theta b_{\beta,\mu}(x,y), \tag{5.42}$$

$$\mathsf{K}^{\mathrm{c}}_{N,\beta}(\theta) := \mathbb{E}_{\mu^{\otimes N}}\left[\exp\left\{-\frac{1}{2N}\sum_{i\ne j}K_\theta(x_i,x_j) + \frac{\beta}{N}\sum_{i=1}^N \psi_\mu(x_i)\right\}\right], \tag{5.43}$$

$$\Lambda_N(\theta) := \log \mathsf{K}^{\mathrm{c}}_{N,\beta}(\theta), \qquad \Lambda(\theta) := -\frac{1}{2}\log\det_2(\mathrm{Id} + A_\theta). \tag{5.44}$$

In particular, $\mathsf{K}^{\mathrm{c}}_{N,\beta}(0) = \mathsf{K}^{\mathrm{c}}_{N,\beta}$.

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Proposition 5.6 (Quantitative screened response)**.** *There is $\theta_0 > 0$, depending only on the fixed hypothesis data, such that*

$$\sup_{|\theta|\le\theta_0} |\Lambda_N(\theta) - \Lambda(\theta)| \le C\mathfrak{e}_N, \qquad N \ge 2. \tag{5.45}$$

Both functions have uniformly bounded third derivatives on $[-\theta_0/2, \theta_0/2]$.

Proof. **The bounded correction and exponential moments.** Write $T = T_\mu$ and $R = R_{\beta,\mu}$ inside operator identities. By (4.90), $0 < R \le (12/11)\mathrm{Id}$. Therefore,

$$\mathcal{C} := \beta^2 TRT \ge 0, \qquad c_{\mathrm{corr}}(x,y) := \beta^2\langle w_x, Rw_y\rangle \tag{5.46}$$

has the centered feature $\beta R^{1/2} w_x$. With $M_2 = \operatorname{ess\,sup}_x \|w_x\|_2 \le M_p$, its squared norm is bounded by

$$K := \beta^2 \|R\|_{\mathrm{op}} M_2^2, \qquad |c_{\mathrm{corr}}(x,y)| \le K. \tag{5.47}$$

Strong measurability and centering follow as in Lemma 4.10. The resolvent identity gives

$$B_{\beta,\mu} = -\beta T + \mathcal{C}, \qquad K_\theta = \beta(1-\theta)\widetilde{W} + \theta c_{\mathrm{corr}}. \tag{5.48}$$

Choose

$$\theta_0 := \frac{1}{64(1+K)}. \tag{5.49}$$

Cauchy–Schwarz gives

$$\mathbb{E}\left[e^{-2\mathsf{W}_N(K_\theta)}\right] \le \mathbb{E}\left[e^{-4\beta(1-\theta)\mathsf{W}_N(\widetilde{W})}\right]^{1/2} \mathbb{E}\left[e^{-4\theta\mathsf{W}_N(c_{\mathrm{corr}})}\right]^{1/2}. \tag{5.50}$$

The first factor is controlled at inverse temperature $8\beta(1-\theta) \le 12\beta \le \Gamma$. For $\theta \ge 0$, the feature identity gives $\mathsf{W}_N(c_{\mathrm{corr}}) \ge -K$, so the second factor is bounded. For $\theta < 0$, apply Lemma C.1 with $a = 8|\theta|$:

$$\mathbb{E}\left[e^{4|\theta|\mathsf{W}_N(c_{\mathrm{corr}})}\right] \le (1 - 32|\theta|K)^{-1/2} \le \sqrt{2}. \tag{5.51}$$

It follows that

$$\sup_{N\ge 2}\sup_{|\theta|\le\theta_0} \mathbb{E}\left[e^{-2\mathsf{W}_N(K_\theta)}\right] < \infty. \tag{5.52}$$

The same conclusion at every exponent between zero and four follows by Hölder; convex interpolations of the kernels also preserve the bound. Positivity has been used only for $\mathcal{C}$, not for T.

The signed Laplace estimate and source insertion. The kernels K_θ are canonical and uniformly bounded in $L^{2+\alpha}(\mu^{\otimes 2})$ by (5.48) and (5.47). Apply Theorem 5.2 with target inverse temperature one and $q = 4$. Equations (5.52) and (5.4) give a uniform error $C\mathfrak{e}_N$, and also

$$\mathrm{Id} + A_\theta \ge \tfrac{3}{4}\mathrm{Id}, \qquad |\theta| \le \theta_0. \tag{5.53}$$

The one-body average $\beta N^{-1}\sum_i \psi_\mu(x_i)$ is bounded above, is centered, and has variance $\beta^2\|\psi_\mu\|_2^2/N$. Lemma 5.3 restores it at cost $CN^{-1/2}$. The finite-particle partition function and limiting Gaussian expectation are at least one by Jensen. Since the logarithm is 1-Lipschitz on $[1,\infty)$, this proves (5.45).

Third derivatives. Write the exponent as $H_N + \theta Y$, where $Y = -\mathsf{W}_N(b_{\beta,\mu})/2$ and H_N is the exponent at $\theta = 0$. For every fixed integer $j \ge 0$,

$$|y|^j e^{\theta y} \le C_{j,\theta_0}(e^{\theta_0 y} + e^{-\theta_0 y}), \qquad |\theta| \le \theta_0/2. \tag{5.54}$$

After multiplication by e^{H_N}, the right-hand side has uniformly bounded integral by the endpoint estimates above and the source upper bound. Dominated convergence justifies differentiation. If $\mathbb{E}_{N,\theta}$ denotes expectation under this normalized tilt, then

$$\Lambda_N'''(\theta) = \mathbb{E}_{N,\theta}\left[(Y - \mathbb{E}_{N,\theta}[Y])^3\right], \qquad |\Lambda_N'''(\theta)| \le 8\mathbb{E}_{N,\theta}[|Y|^3] \le C. \tag{5.55}$$

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Here, the denominator is at least one. This argument does not require $b_{\beta,\mu} \in L^3(\mu^{\otimes 2})$. Spectral calculus, commutation of A_θ and $B_{\beta,\mu}$, and (5.53) give

$$\Lambda'''(\theta) = -\operatorname{Tr}\left[((\mathrm{Id}+A_\theta)^{-1}B_{\beta,\mu})^3\right], \quad |\Lambda'''(\theta)| \le (4/3)^3 \|B_{\beta,\mu}\|_{\mathrm{op}} \|B_{\beta,\mu}\|_{\mathrm{HS}}^2. \tag{5.56}$$

The trace is absolutely convergent. This proves the derivative assertions. □

5.5. **Strong convergence of the pair kernel.**

Proposition 5.7 (Quantitative strong pair limit)**.** *Uniformly for $N \ge 2$,*

$$\|Nv_N - b_{\beta,\mu}\|_2 \le CN^{-\alpha/8}, \tag{5.57}$$

$$\left| \|Nv_N\|_2^2 - \|b_{\beta,\mu}\|_2^2 \right| \le CN^{-\alpha/4}, \tag{5.58}$$

$$\left| \langle Nv_N, b_{\beta,\mu}\rangle - \|b_{\beta,\mu}\|_2^2 \right| \le CN^{-\alpha/4}. \tag{5.59}$$

Proof. **Squared norm from correlated replicas.** The trace expansion near zero gives

$$\log D(t) = \frac{1}{2}\sum_{m\ge 1} \frac{t^{2m}}{m} \operatorname{Tr}(B_{\beta,\mu}^{2m}), \qquad D(t) = 1 + \frac{t^2}{2}\|B_{\beta,\mu}\|_{\mathrm{HS}}^2 + O(t^4). \tag{5.60}$$

It converges uniformly near $[0, 1/2]$ because $B_{\beta,\mu}$ is Hilbert–Schmidt and $\|B_{\beta,\mu}\|_{\mathrm{op}} < 1$. In particular, the sixth derivative of D is bounded on that interval. The nonnegative coefficients in (5.30) and (5.14) also give

$$\sup_{0\le t\le 1/2} |A_N^{(6)}(t)| \le C \sup_{r\ge 6} r(r-1)\cdots(r-5)2^{6-r} < \infty, \tag{5.61}$$

uniformly in N. For $5h \le 1/2$, set

$$\mathcal{D}_h F := \frac{45F(0) - 154F(h) + 214F(2h) - 156F(3h) + 61F(4h) - 10F(5h)}{24h^2}. \tag{5.62}$$

Direct substitution shows that this expression equals one on t^2 and zero on $1, t, t^3, t^4, t^5$. Taylor's formula through degree five therefore gives an error Ch^4 in extracting the quadratic coefficient. Applying this to A_N and $D - 1$, and using (5.34) at $t_0 = 1/2$, gives

$$\left| c_{N,2} - \tfrac{1}{2}\|B_{\beta,\mu}\|_{\mathrm{HS}}^2 \right| \le C\left(h^4 + \mathfrak{e}_N/h^2\right). \tag{5.63}$$

Choose $h = \mathfrak{e}_N^{1/6}$ for all sufficiently large N. The exact normalization is

$$\|Nv_N\|_2^2 = \frac{2N}{N-1} c_{N,2}. \tag{5.64}$$

Thus, its error is $C(\mathfrak{e}_N^{2/3} + N^{-1}) \le CN^{-\alpha/4}$, proving (5.58).

Cross term from the screened response. Differentiation is justified by (5.54). Canonicity gives

$$\Lambda_N'(0) = -\frac{1}{2N}\mathbb{E}_{\mathbb{Q}_{N,\beta}(\mu)}\left[\sum_{i\ne j} b_{\beta,\mu}(x_i, x_j)\right] = -\frac{N-1}{2}\langle v_N, b_{\beta,\mu}\rangle. \tag{5.65}$$

The product and singleton terms vanish. Spectral differentiation gives

$$\Lambda'(0) = -\frac{1}{2}\operatorname{Tr}[(R_{\beta,\mu} - \mathrm{Id})B_{\beta,\mu}] = -\frac{1}{2}\|b_{\beta,\mu}\|_2^2. \tag{5.66}$$

The third-derivative bounds in Proposition 5.6 imply, for $F = \Lambda_N$ and $F = \Lambda$,

$$\left| \frac{F(h) - F(-h)}{2h} - F'(0) \right| \le Ch^2, \qquad 0 < h \le \theta_0/2. \tag{5.67}$$

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Together with (5.45), this bounds the difference of derivatives by $C(h^2 + \mathfrak{e}_N/h)$. Choose $h = \mathfrak{e}_N^{1/3}$ for large N. Converting from $(N-1)v_N$ to Nv_N in (5.65) costs $O(N^{-1})$, so (5.59) follows. The finite initial range of N in both arguments is covered by (5.14), with an enlarged constant.

Finally, polarization gives

$$\|Nv_N - b_{\beta,\mu}\|_2^2 = \|Nv_N\|_2^2 + \|b_{\beta,\mu}\|_2^2 - 2\langle Nv_N, b_{\beta,\mu}\rangle \le CN^{-\alpha/4}. \tag{5.68}$$

Taking the square root proves (5.57). □

6. Singleton profiles and statistical divergences

Assume the hypotheses of Theorem 3.1 and use u_N, v_N from Section 5.2. The common second-order identity and cavity variance expansion below, together with Lemma 4.10, identify the singleton profile. We then assemble the two kernels and prove the marginal and divergence conclusions of the abstract theorem. The model hypotheses are verified afterward in Section 7.4.

6.1. **The second-order normalized-exponential identity.** To identify the singleton profile, we expand the normalized one-particle conditional density to second order. Throughout this cavity argument, expectations without a subscript are under $\mathbb{Q}_{N,\beta}(\mu)$. We use B_N, π_N, r_N, ϵ, and δ_B from (3.19), (4.45), (4.63), (4.55), and (4.56). Each cavity field is centered, so $\delta_B \ge 0$.

For a fixed reservoir, abbreviate $B = B_N(\cdot; \hat{\Xi}_1)$ and set

$$m_2(B) := \int B^2 \,\mathrm{d}\mu, \tag{6.1}$$

$$\rho(t) := e^t - 1 - t - \frac{1}{2}t^2, \tag{6.2}$$

$$\bar{\rho}(B) := \int \rho(B) \,\mathrm{d}\mu. \tag{6.3}$$

Then

$$\delta_B = \frac{1}{2}m_2(B) + \bar{\rho}(B). \tag{6.4}$$

Substituting this identity into (4.61) gives

$$\pi_N(x \mid \hat{\Xi}_1) - 1 - B(x) - \frac{1}{2}\left(B(x)^2 - m_2(B)\right) = \frac{\rho(B(x)) - \bar{\rho}(B) - B(x)\delta_B - \frac{1}{2}\delta_B\left(B(x)^2 - m_2(B)\right)}{1 + \delta_B}. \tag{6.5}$$

No asymptotic approximation has yet been made.

Define

$$q_N(x) := \mathbb{E}\left[B_N(x; \hat{\Xi}_1)^2\right], \qquad \bar{q}_N := \int q_N \,\mathrm{d}\mu. \tag{6.6}$$

6.2. **The common cavity variance identity.** The quantitative row assumption gives $M_2 = \operatorname{ess\,sup}_x \|w_x\|_2 \le M_p < \infty$. The density moments in (4.89) and Lemma 4.10 justify the expectations below for μ-almost every x. In particular, the row-square term is integrable by Hölder at the available exponent $p/(p-2)$, and the distinct-pair term by the L^2 row and density bounds.

Define the bounded linear map

$$(\mathcal{M}g)(x) := \iint w_x(y)w_x(z)g(y,z) \,\mathrm{d}\mu(y)\mathrm{d}\mu(z), \qquad g \in L^2(\mu^{\otimes 2}). \tag{6.7}$$

By the uniform row bound,

$$\|\mathcal{M}g\|_{L^2(\mu)} \le M_2^2 \|g\|_{L^2(\mu^{\otimes 2})}. \tag{6.8}$$

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Let $S_x = \sum_{j=2}^N w_x(\xi_j)$. Expanding (3.19),

$$Nq_N(x) = \frac{\beta^2}{N}\left[\psi_\mu(x)^2 - 2\psi_\mu(x)\mathbb{E}[S_x] + \mathbb{E}[S_x^2]\right]. \tag{6.9}$$

Exchangeability and canonicity give

$$\mathbb{E}[S_x] = (N-1)T_\mu u_N(x). \tag{6.10}$$

For the second moment,

$$\begin{aligned}\mathbb{E}[S_x^2] = (N-1)\int w_x(y)^2(1+u_N(y))\,\mathrm{d}\mu(y)\\ + (N-1)(N-2)\iint w_x(y)w_x(z)v_N(y,z)\,\mathrm{d}\mu(y)\mathrm{d}\mu(z).\end{aligned} \tag{6.11}$$

Indeed, in the distinct-label contribution, the product term and the two first-Hoeffding terms vanish because w_x is centered.

Substituting (6.10)–(6.11) into (6.9),

$$\begin{aligned}Nq_N = \beta^2\Bigg[\frac{\psi_\mu^2}{N} - 2\frac{N-1}{N}\psi_\mu T_\mu u_N + \frac{N-1}{N}\int w_\cdot(y)^2\,\mathrm{d}\mu(y)\\ + \frac{N-1}{N}\int w_\cdot(y)^2 u_N(y)\,\mathrm{d}\mu(y) + \frac{(N-1)(N-2)}{N^2}\mathcal{M}(Nv_N)\Bigg].\end{aligned} \tag{6.12}$$

6.3. **The second-order remainder and singleton profile.**

Lemma 6.1 (Second-order cavity remainder)**.** *Under the hypotheses of Theorem 3.1,*

$$N\|r_N - \tfrac{1}{2}(q_N - \bar{q}_N)\|_{L^2(\mu)} \le CN^{-\vartheta/2}. \tag{6.13}$$

Proof. Use the exact identity (6.5). On the upper-bounded range of B_N, $|\rho(B_N)| \le C|B_N|^{2+\vartheta}$. The pointwise moment estimate (4.91) and $\psi_\mu \in L^{4+2\vartheta}(\mu)$ give

$$\|\mathbb{E}[|\rho(B_N(\cdot))|]\|_2 + \mathbb{E}[|\bar{\rho}(B_N)|] \le CN^{-1-\vartheta/2}. \tag{6.14}$$

The source assumption supplies this moment because $\vartheta < 1$. By Minkowski and (4.93),

$$\|\mathbb{E}[|B_N|\delta_B]\|_2 \le C\mathbb{E}[\|B_N\|_2^3] \le CN^{-3/2}. \tag{6.15}$$

For the remaining mixed term, take the conjugate exponents $q = (2+\vartheta)/\vartheta$ and $q' = (2+\vartheta)/2$. Then

$$\mathbb{E}[\delta_B B_N(x)^2] \le \mathbb{E}[\delta_B^q]^{1/q}\mathbb{E}[|B_N(x)|^{2+\vartheta}]^{1/q'} \le C\left(N^{-2} + N^{-3}|\psi_\mu(x)|^2\right). \tag{6.16}$$

This is $O(N^{-2})$ in $L^2(\mu)$, and $\mathbb{E}[\delta_B m_2(B_N)] \le C\mathbb{E}[\|B_N\|_2^4] \le CN^{-2}$. Taking absolute values in (6.5), then expectation and the spatial norm, and using $1 + \delta_B \ge 1$, proves (6.13). No fourth pointwise moment of the interaction is used. □

Proposition 6.2 (Quantitative singleton profile)**.** *Under the hypotheses of Theorem 3.1,*

$$\|Nu_N - a_{\beta,\mu}\|_{L^2(\mu)} \le C\mathfrak{r}_N, \qquad N \ge 2, \tag{6.17}$$

with $\mathfrak{r}_N$ defined in (3.7).

Proof. We evaluate the common variance identity (6.12). The term ψ_μ^2/N is $O(N^{-1})$ in L^2 because $\psi_\mu \in L^4$. The next term has the same bound, since $\|T_\mu u_N\|_\infty \le M_2\|u_N\|_2 \le C/N$ and $\psi_\mu \in L^2$. For the square-row term, take the same q, q' as in (6.16). Then $2q' = 2 + \vartheta < p$ and $2q \le r_*$. Hölder and interpolation yield

$$\operatorname*{ess\,sup}_x \left|\int w_x(y)^2 u_N(y)\,\mathrm{d}\mu(y)\right| \le M_p^2\|u_N\|_q$$

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$$\leq M_p^2 \|u_N\|_2^{1/(q-1)} \|u_N\|_{2q}^{1-1/(q-1)} \leq C N^{-\vartheta/2}. \tag{6.18}$$

Here, (4.89) bounds the higher marginal norm, (4.96) gives the L^2 bound, and $1/(q-1) = \vartheta/2$. The pair term is controlled by (6.8) and (5.57). Its limit satisfies

$$\|w_x\|_2^2 + (\mathcal{M} b_{\beta,\mu})(x) = \langle w_x, (\mathrm{Id} + B_{\beta,\mu}) w_x \rangle = \langle w_x, R_{\beta,\mu} w_x \rangle. \tag{6.19}$$

All finite-particle coefficients differ from their limits by $O(N^{-1})$, including the coefficient zero at $N = 2$ for distinct reservoir labels. Thus,

$$\|N q_N - \omega_{\beta,\mu}\|_2 \leq C \left(N^{-\alpha/8} + N^{-\vartheta/2} \right) = C \mathfrak{r}_N. \tag{6.20}$$

Centering is an L^2 contraction, so Lemma 6.1 identifies $N r_N$ with $\omega^\circ_{\beta,\mu}/2$ with this error. The resolvent identity and (4.90) give

$$\left\| \left(\mathrm{Id} + \beta \frac{N-1}{N} T_\mu \right)^{-1} - R_{\beta,\mu} \right\|_{\mathrm{op}} \leq \frac{\beta}{N} \left(\frac{r_*}{r_* - 1} \right)^2 \|T_\mu\|_{\mathrm{op}}. \tag{6.21}$$

Multiply (4.80) by N, apply the bounded inverse, and compare with (2.20). Equations (6.20) and (6.21) prove (6.17). □

6.4. **Uniform marginal assembly and divergence estimates.** Propositions 5.7 and 6.2 give

$$\|N u_N - a_{\beta,\mu}\|_{L^2(\mu)} + \|N v_N - b_{\beta,\mu}\|_{L^2(\mu^{\otimes 2})} \leq C \mathfrak{r}_N. \tag{6.22}$$

We also retain the sharper squared-norm estimate (5.58). For every $1 \leq k \leq N$, marginalization of (4.14) gives

$$f_{N,\beta}^{(k)} - 1 = \sum_{i=1}^k u_N(x_i) + \sum_{1 \leq i < j \leq k} v_N(x_i, x_j) + \mathcal{R}_{N,k}^{(\geq 3)}, \tag{6.23}$$

where

$$\mathcal{R}_{N,k}^{(\geq 3)} := \sum_{r=3}^k \sum_{\substack{A \subset [k] \\ |A| = r}} h_A. \tag{6.24}$$

An empty sum is zero. For $3 \leq r \leq k \leq N$,

$$\frac{\binom{k}{r}}{\binom{N}{r}} = \prod_{j=0}^{r-1} \frac{k-j}{N-j} \leq \left(\frac{k}{N} \right)^r \leq \left(\frac{k}{N} \right)^3. \tag{6.25}$$

Consequently, orthogonality and the total mass in (5.14) give

$$\|\mathcal{R}_{N,k}^{(\geq 3)}\|_2^2 = \sum_{r=3}^k \binom{k}{r} \|h_{[r]}\|_2^2 \leq \left(\frac{k}{N} \right)^3 \sum_{r=3}^N \binom{N}{r} \|h_{[r]}\|_2^2 \leq C \left(\frac{k}{N} \right)^3, \tag{6.26}$$

$$N^2 \|\mathcal{R}_{N,k}^{(\geq 3)}\|_2^2 \leq C \frac{k^3}{N}. \tag{6.27}$$

This estimate uses the total higher-order Hoeffding mass, so it remains valid for $k = N$; summing a separate bound for each level is unnecessary.

The profile norm is given by (2.22).

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6.4.1. *Moments of the comparison profile.* To pass from the density profile to relative entropy, we need a higher moment of $G_{k,\beta,\mu}/k$ that is uniform in k. The canonical pair sum is a martingale sum under the product law. Applying Lemma C.2 to it gives the required moment bound without the extra factor of k from the triangle inequality.

Lemma 6.3 (Canonical profile moments)**.** *Let $a \in L^p(\mu)$ have mean zero and let $b \in L^p(\mu^{\otimes 2})$ be symmetric and canonical, where $2 \le p < \infty$. For independent $X_1, \dots, X_k$ with law μ,*

$$\left\| \sum_{i=1}^{k} a(X_i) \right\|_p \le \sqrt{(p-1)k}\|a\|_p, \tag{6.28}$$

$$\left\| \sum_{i<j} b(X_i, X_j) \right\|_p \le (p-1)\sqrt{\binom{k}{2}}\|b\|_p. \tag{6.29}$$

Consequently, for $G_k = \sum_i a(X_i) + \sum_{i<j} b(X_i, X_j)$,

$$\left\| \frac{G_k}{k} \right\|_p \le \sqrt{\frac{p-1}{k}}\|a\|_p + (p-1)\sqrt{\frac{k-1}{2k}}\|b\|_p. \tag{6.30}$$

For the profiles in Theorem 3.1, the normalized moment is uniformly finite at every exponent $2 \le q \le \min\{p, 6\}$, where p is the row exponent of that theorem.

Proof. The first estimate follows immediately from Lemma C.2. For the second, put $D_j = \sum_{i<j} b(X_i, X_j)$. The sequence is a martingale-difference sequence for $\sigma(X_1, \dots, X_j)$ by canonicity in the second variable. To bound each D_j, reveal X_j first and then $X_1, \dots, X_{j-1}$. The summands $b(X_i, X_j)$ are martingale differences for this filtration by canonicity in the first variable. Thus, two applications of (C.5) give

$$\left\| \sum_{j=2}^{k} D_j \right\|_p^2 \le (p-1)\sum_{j=2}^{k}\|D_j\|_p^2 \le (p-1)^2\sum_{j=2}^{k}(j-1)\|b\|_p^2. \tag{6.31}$$

This proves (6.29), and Minkowski gives (6.30).

For the abstract profiles, write $R = R_{\beta,\mu}$ and $T = T_\mu$. The resolvent identity and the bounded correction from (5.46) give

$$b(x, y) = -\beta\widetilde{W}(x, y) + \beta^2\langle w_x, Rw_y\rangle. \tag{6.32}$$

Thus, $b \in L^p(\mu^{\otimes 2})$ and $\omega_{\beta,\mu} \in L^\infty(\mu)$. For $F = \beta\psi_\mu + \omega^\circ_{\beta,\mu}/2$, one has $F \in L^6(\mu)$ and

$$a = RF = F - \beta TRF, \qquad \|TRF\|_\infty \le M_p\|R\|_{\mathrm{op}}\|F\|_2. \tag{6.33}$$

Hence, $a \in L^6(\mu)$, proving the abstract profile assertion. For the exact models, the row and source bounds in Lemmas 7.1 and 7.3 give the stronger conclusions: $a \in L^\infty$ and $b \in L^q$ for every $2 \le q < \mathsf{d}/\mathsf{s}$ in the Riesz case, and $a, b \in L^q$ for every finite $q \ge 2$ in the logarithmic case. Thus, the corresponding normalized profile moments remain uniform in k throughout these model ranges. □

6.4.2. *Uniform relative-entropy comparison.* The next comparison is applied to the marginal density at scale k/N. Its error is quadratic in the density-profile approximation, and its moment above order two is required only for the explicit comparison profile.

Lemma 6.4 (Entropy comparison at an arbitrary scale)**.** *Let ν be a probability measure, $t > 0$, $H \in L^2(\nu)$ have mean zero, and $f = 1 + tH \ge 0$. Let $P \in L^{2+\alpha}(\nu)$ for some $0 < \alpha \le 1$. Then*

$$\left| t^{-2}\mathsf{H}(f\nu \mid \nu) - \frac{1}{2}\|H\|_2^2 \right| \le 4\|H - P\|_2^2 + 2^{2+\alpha}t^\alpha\|P\|_{2+\alpha}^{2+\alpha}. \tag{6.34}$$

No $L^{2+\alpha}$ moment of H is required.

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Proof. For $z \geq -1$, use

$$\Psi(z) = \begin{cases} ((1+z)\log(1+z) - z)/z^2, & z \neq 0, \\ 1/2, & z = 0, \end{cases} \tag{6.35}$$

with $0 \log 0 = 0$. Twice integrating the second derivative of the numerator gives

$$\Psi(z) = \int_0^1 \frac{1-r}{1+rz}\,\mathrm{d}r \qquad (z > -1). \tag{6.36}$$

For $z \geq 0$, this is between zero and $1/2$. For $-1 < z < 0$, the denominator is at least $1 - r$, so it is between zero and one; the same holds at $z = -1$ by continuity. For $|z| \leq 1/2$, subtracting $1/2$ in (6.36) gives $|\Psi(z) - 1/2| \leq |z|/3$. For $|z| \geq 1/2$, the bound $0 \leq \Psi \leq 1$ gives

$$|\Psi(z) - 1/2| \leq \min\{|z|, 1\} \qquad (z \geq -1). \tag{6.37}$$

Since $\int H \,\mathrm{d}\nu = 0$, and the entropy integrand after subtracting the linear term is at most $t^2 H^2$,

$$t^{-2}\mathsf{H}(f\nu \mid \nu) = \int H^2 \Psi(tH)\,\mathrm{d}\nu. \tag{6.38}$$

All quantities are finite, including the value at $f = 0$.

On the set $|H| > 2|P|$, one has $H^2 \leq 4|H - P|^2$. On the complementary set, $H^2 \leq 4P^2$ and $\min\{t|H|, 1\} \leq \min\{2t|P|, 1\} \leq (2t|P|)^\alpha$. Therefore, pointwise,

$$H^2 \min\{t|H|, 1\} \leq 4|H - P|^2 + 2^{2+\alpha} t^\alpha |P|^{2+\alpha}. \tag{6.39}$$

Integrating (6.39) and using (6.37)–(6.38) proves (6.34). □

6.4.3. *Proof of the quantitative profile theorem.*

Proof of Theorem 3.1: marginal estimates. **The density profile.** Subtract (2.21) from N times (6.23). Hoeffding orthogonality gives the exact identity

$$\begin{aligned} &\left\| N(f_{N,\beta}^{(k)} - 1) - G_{k,\beta,\mu} \right\|_2^2 \\ &\quad = k\|Nu_N - a_{\beta,\mu}\|_2^2 + \binom{k}{2}\|Nv_N - b_{\beta,\mu}\|_2^2 + N^2\|\mathcal{R}_{N,k}^{(\geq 3)}\|_2^2. \end{aligned} \tag{6.40}$$

Equations (6.22) and (6.27) bound this by $C(k^2\mathfrak{r}_N^2 + k^3/N)$. Consequently,

$$\|N(f_{N,\beta}^{(k)} - 1) - G_{k,\beta,\mu}\|_2 \leq C\left(k\mathfrak{r}_N + \frac{k^{3/2}}{\sqrt{N}}\right). \tag{6.41}$$

Division by k proves (2.29).

Chi-square divergence. The exact squared norm is

$$N^2\chi^2\left(\mathbb{Q}_{N,\beta}^{(k)}(\mu) \mid \mu^{\otimes k}\right) = k\|Nu_N\|_2^2 + \binom{k}{2}\|Nv_N\|_2^2 + N^2\|\mathcal{R}_{N,k}^{(\geq 3)}\|_2^2. \tag{6.42}$$

The singleton norms are uniformly bounded, so their difference of squares is $O(\mathfrak{r}_N)$ by (6.22). For the pair term, use the sharper bound (5.58), not only the strong pair estimate. Subtracting (2.22) from (6.42) gives

$$\left|N^2\chi^2 - \|G_{k,\beta,\mu}\|_2^2\right| \leq C\left(k\mathfrak{r}_N + k^2\mathfrak{r}_N^2 + \frac{k^2}{N} + \frac{k^3}{N}\right), \tag{6.43}$$

which implies (2.30).

Total variation. The normalization of total variation and the reverse triangle inequality give

$$\left|N\,\mathrm{TV}\left(\mathbb{Q}_{N,\beta}^{(k)}(\mu), \mu^{\otimes k}\right) - \frac{1}{2}\|G_{k,\beta,\mu}\|_1\right|$$

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$$\leq \frac{1}{2}\left\|N(f_{N,\beta}^{(k)}-1)-G_{k,\beta,\mu}\right\|_1 \leq \frac{1}{2}\left\|N(f_{N,\beta}^{(k)}-1)-G_{k,\beta,\mu}\right\|_2. \tag{6.44}$$

This proves (2.32).

Relative entropy. Set $t = k/N$, $H = (N/k)(f_{N,\beta}^{(k)}-1)$, and $P = G_{k,\beta,\mu}/k$. Then $\int H\,\mathrm{d}\mu^{\otimes k} = 0$, $1+tH \geq 0$, and Lemma 6.3 gives a uniform $L^{2+\alpha}$ bound for P. Applying Lemma 6.4 and (2.29),

$$\left|\frac{N^2}{k^2}\mathsf{H}\Big(\mathbb{Q}_{N,\beta}^{(k)}(\mu)\mid\mu^{\otimes k}\Big)-\frac{1}{2}\frac{N^2}{k^2}\chi^2\Big(\mathbb{Q}_{N,\beta}^{(k)}(\mu)\mid\mu^{\otimes k}\Big)\right|$$
$$\leq C_\alpha\left(\mathfrak{r}_N^2+\frac{k}{N}+(k/N)^\alpha\right)\leq C_\alpha\left(\mathfrak{r}_N^2+(k/N)^\alpha\right). \tag{6.45}$$

Here, $k/N \leq (k/N)^\alpha$. Multiply by k^2 and combine with (2.30); the term k^3/N is absorbed by $k^2(k/N)^\alpha$. This proves (2.31). The comparison uses a higher moment only for the explicit profile, not for H.

The growing-marginal conclusions are proved below. □

For any integer sequence $1 \leq k_N \leq N$ with $k_N/N \to 0$, (2.29) gives

$$\sup_{1\leq k\leq k_N}\left\|\frac{N}{k}(f_{N,\beta}^{(k)}-1)-\frac{1}{k}G_{k,\beta,\mu}\right\|_{L^2(\mu^{\otimes k})}\leq C\left(\mathfrak{r}_N+\sqrt{\frac{k_N}{N}}\right)\longrightarrow 0 \qquad (N\to\infty). \tag{6.46}$$

Thus, the normalized error vanishes throughout every sublinear marginal range in either model, without a condition on $k_N\mathfrak{r}_N$.

6.5. **Divergence limits for growing marginals.** The chi-square and relative-entropy limits use the exact squared norm of the profile and the estimates just proved. The total-variation limit also requires the following approximation of the absolute first moment of the canonical pair statistic. For a self-adjoint Hilbert–Schmidt operator C, let $\mathcal{Q}_C$ be the centered Gaussian quadratic form defined in (2.33), with C in place of $B_{\beta,\mu}$. Section C.3 gives its joint construction and proves the approximation.

Proposition 6.5 (Absolute-moment rate)**.** *Let $0<\alpha\leq 1$ and let $c\in L^{2+\alpha}(\mu^{\otimes 2})$ be real, symmetric, and canonical. For independent $X_1,\dots,X_k$ with law μ, and the associated Gaussian quadratic form $\mathcal{Q}_C$,*

$$\left|\mathbb{E}\left[\left|\frac{1}{k}\sum_{i<j}c(X_i,X_j)\right|\right]-\frac{1}{2}\mathbb{E}[|\mathcal{Q}_C|]\right|\leq C_\alpha\|c\|_{2+\alpha}\,k^{-\alpha/(2(2+\alpha))} \qquad (k\geq 1). \tag{6.47}$$

The constant depends only on α, and in particular is independent of the rank of the kernel.

Completion of the proof of Theorem 3.1. Divide (2.22) by k^2:

$$\left\|\frac{1}{k}G_{k,\beta,\mu}\right\|_2^2=\frac{1}{k}\|a_{\beta,\mu}\|_2^2+\frac{k-1}{2k}\|B_{\beta,\mu}\|_{\mathrm{HS}}^2. \tag{6.48}$$

Equations (2.30) and (6.48) give (2.35), since $\mathfrak{r}_N/k \leq 2/k$. The entropy estimate (2.36) then follows from (6.45).

For total variation, (2.32) gives

$$\left|\frac{N}{k}\,\mathrm{TV}-\frac{1}{2}\left\|\frac{1}{k}G_{k,\beta,\mu}\right\|_1\right|\leq C\left(\mathfrak{r}_N+\sqrt{k/N}\right). \tag{6.49}$$

The singleton part satisfies

$$\mathbb{E}_{\mu^{\otimes k}}\left[\left|\frac{1}{k}\sum_{i=1}^k a_{\beta,\mu}(X_i)\right|\right]\leq k^{-1/2}\|a_{\beta,\mu}\|_2. \tag{6.50}$$

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By Lemma 6.3, $b_{\beta,\mu} \in L^{2+\alpha}(\mu^{\otimes 2})$. Proposition 6.5 gives

$$\left| \mathbb{E}_{\mu^{\otimes k}} \left[\left| \frac{1}{k} \sum_{i<j} b_{\beta,\mu}(X_i, X_j) \right| \right] - \frac{1}{2} \mathbb{E}[|\mathcal{Q}_{\beta,\mu}|] \right| \leq C_\alpha k^{-\alpha/(2(2+\alpha))}. \tag{6.51}$$

Combining (6.49)–(6.51) proves (2.37). The factor $1/4$ is the product of the $1/2$ in total variation and the $1/2$ in the canonical quadratic U-statistic limit. Only the absolute first moment of that statistic is compared. If $k_N \to \infty$ and $k_N/N \to 0$, then

$$\mathfrak{r}_N^2 + \frac{1}{k_N} + \left(\frac{k_N}{N}\right)^\alpha \longrightarrow 0, \tag{6.52}$$

$$\mathfrak{r}_N + \sqrt{\frac{k_N}{N}} + k_N^{-\alpha/(2(2+\alpha))} \longrightarrow 0 \qquad (N \to \infty). \tag{6.53}$$

Since $k_N/N \leq (k_N/N)^\alpha$, the three estimates therefore imply the asserted divergence limits. In the logarithmic case, $\alpha = 1$, giving the stated simplification. □

6.6. **Sufficient windows for undivided errors.**

Corollary 6.6 (Sufficient windows for the undivided errors)**.** *Under the model hypotheses of Theorem 2.1, the undivided density error in (6.41) and the three divergence errors in (2.30)–(2.32) tend to zero along every integer sequence $1 \leq k_N \leq N$ satisfying*

$$k_N = \begin{cases} o(N^{\alpha/8}), & 0 < \mathsf{s} < \mathsf{d}/2, \\ o(N^{1/8}), & \mathsf{s} = 0. \end{cases} \tag{6.54}$$

Here, α satisfies (2.25) in the Riesz case. These windows are sufficient and are not claimed optimal.

Proof. Take $\alpha = 1$ for the logarithmic model. In both cases, $k_N \mathfrak{r}_N \to 0$ and $k_N^2 \mathfrak{r}_N^2 \to 0$, and, as $N \to \infty$,

$$k_N^3/N = o(N^{3\alpha/8-1}) \longrightarrow 0, \tag{6.55}$$

$$k_N^{2+\alpha}/N^\alpha = o(N^{\alpha(\alpha-6)/8}) \longrightarrow 0. \tag{6.56}$$

Both exponents are negative because $0 < \alpha \leq 1$. The density and total-variation finite-size error is the square root of k_N^3/N. Apply Theorem 2.1 and (6.41). □

7. Verification for logarithmic and Riesz interactions

We collect the potential and row estimates once, then apply Theorem 3.1 to prove the model approximation and divergence theorems. The final subsection verifies the separate order criterion, retaining its weaker logarithmic hypothesis and weighted alternative. Assumptions made for the quantitative logarithmic inputs are confined to that subsection and its application; they do not govern the order verification.

7.1. **Riesz row and potential estimates.**

Lemma 7.1 (Uniform row integrability)**.** *Under the Riesz hypotheses in Theorem 2.1, for every $1 \leq q < \mathsf{d}/\mathsf{s}$, there is a finite constant C_q, depending only on $\mathsf{d}, \mathsf{s}, q$, and $\|\mu\|_{L^\infty}$, such that*

$$\sup_{x \in \mathbb{R}^{\mathsf{d}}} \|\widetilde{W}(x, \cdot)\|_{L^q(\mu)} \leq C_q. \tag{7.1}$$

Moreover,

$$\|h_\mu\|_{L^\infty} + \|\psi_\mu\|_{L^\infty} < \infty, \qquad \operatorname*{ess\,inf}_{x,y} \widetilde{W}(x, y) > -\infty. \tag{7.2}$$

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In particular,

$$M_2 := \sup_x \|w_x\|_{L^2(\mu)} < \infty. \tag{7.3}$$

Proof. Split the Riesz potential at unit distance. Uniformly in x,

$$\int W(x,y)\,\mathrm{d}\mu(y) \le \frac{\|\mu\|_{L^\infty}}{\mathsf{s}} \int_{|z|\le 1} |z|^{-\mathsf{s}}\,\mathrm{d}z + \frac{1}{\mathsf{s}}, \tag{7.4}$$

which is finite because $\mathsf{s} < \mathsf{d}$. This proves the first bound in (7.2); the remaining bounded one-body terms follow from the definitions.

For $q < \mathsf{d}/\mathsf{s}$, the same decomposition gives

$$\sup_x \int W(x,y)^q\,\mathrm{d}\mu(y) \le \frac{\|\mu\|_{L^\infty}}{\mathsf{s}^q} \int_{|z|\le 1} |z|^{-q\mathsf{s}}\,\mathrm{d}z + \frac{1}{\mathsf{s}^q} < \infty. \tag{7.5}$$

Adding the bounded centering terms in (2.3) proves (7.1). Since $W \ge 0$ and h_μ, I_μ are bounded, the lower bound in (7.2) follows. Finally, $2 < \mathsf{d}/\mathsf{s}$ by hypothesis, so (7.3) is the case $q = 2$. □

7.2. **Logarithmic potential estimates.** For the elementary estimates below, assume only that μ has bounded density and a finite polynomial moment of some positive order. Set

$$\ell(z) := (-\log|z|)_+, \qquad L(x) := \log(e + |x|). \tag{7.6}$$

The elementary bounds

$$(\log|x-y|)_+ \le L(x) + L(y), \qquad |\log|x-y|| \le \ell(x-y) + L(x) + L(y) \tag{7.7}$$

hold for all x, y. Since $\mu \in L^\infty$, for every finite $m \ge 1$,

$$\sup_{x\in\mathbb{R}^{\mathsf{d}}} \int_{\mathbb{R}^{\mathsf{d}}} \ell(x-y)^m\,\mathrm{d}\mu(y) \le \|\mu\|_{L^\infty} \int_{|z|<1} (-\log|z|)^m\,\mathrm{d}z < \infty. \tag{7.8}$$

Moreover, the positive polynomial moment implies that $L \in L^m(\mu)$ for every finite m. It follows from (7.7)–(7.8) that the absolute logarithmic energy is finite.

The logarithmic potential satisfies

$$-L(x) - \int L\,\mathrm{d}\mu \le h_\mu(x) \le \sup_{z\in\mathbb{R}^{\mathsf{d}}} \int \ell(z-y)\,\mathrm{d}\mu(y) < \infty. \tag{7.9}$$

Hence, $|h_\mu(x) - 2I_\mu| \le C_\mu(1 + L(x))$. Also,

$$|\widetilde{W}(x,y)| \le C_\mu + \ell(x-y) + L(x) + L(y). \tag{7.10}$$

In particular, the finiteness assertions in (3.16) follow. The same estimates give

$$(\widetilde{W})_-(x,y) \le C_\mu + L(x) + L(y), \tag{7.11}$$

$$(h_\mu - 2I_\mu)_+(x) \le C_\mu, \tag{7.12}$$

which verifies (3.21) for suitable finite constants.

7.3. **Logarithmic inputs for marginal approximation.** We now assume the logarithmic hypotheses of Theorem 2.1. The following potential and centered-row estimates verify the source and kernel assumptions of the quantitative theorem. The weighted hypotheses of the order theorem above are unchanged.

Lemma 7.2 (Logarithmic analytic input)**.** *Under the logarithmic hypotheses in Theorem 2.1, including both* (2.27) *and* (2.28)*, the following assertions hold.*

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(i) *The absolute logarithmic energy is finite, h_μ is finite μ-almost everywhere, and*

$$-L(x) - \int L \,\mathrm{d}\mu \le h_\mu(x) \le C_{\mathrm{loc}} \quad \textit{for every } x \textit{ for which } h_\mu(x) \textit{ is defined.} \tag{7.13}$$

In particular, for a finite constant C,

$$|\psi_\mu(x)| \le C(1 + L(x)). \tag{7.14}$$

(ii) *For every finite $p \ge 1$,*

$$\widetilde{W} \in L^p(\mu^{\otimes 2}). \tag{7.15}$$

Moreover, with

$$a_0 \coloneqq 2C_{\mathrm{loc}} + 2|I_\mu|, \qquad a_1 \coloneqq 1, \qquad a_h \coloneqq C_{\mathrm{loc}} + 2|I_\mu|, \tag{7.16}$$

one has

$$(\widetilde{W})_-(x,y) \le a_0 + a_1(L(x) + L(y)), \tag{7.17}$$

$$(\psi_\mu)_+(x) \le a_h(1 + L(x)). \tag{7.18}$$

Thus, $\mathsf{c}_{\log,\beta,\mu} = \beta(a_0 + a_1 + a_h)$.

(iii) *The logarithmic determinant and fluctuation inputs of* [DGR26] *satisfy*

$$\int e^{-4\beta h_\mu} \,\mathrm{d}\mu < \infty, \qquad \widetilde{W} \in L^2(\mu^{\otimes 2}). \tag{7.19}$$

(iv) *The operator T_μ is self-adjoint, Hilbert–Schmidt, and nonnegative on $L^2(\mu)$.*

Proof. The estimates (7.7)–(7.10) prove finite absolute logarithmic energy, (7.13), and (7.14). They also give (7.15), since every finite power of L is integrable under a positive polynomial moment. Using $h_\mu \le C_{\mathrm{loc}}$ in the centering formula gives the precise envelopes (7.17)–(7.18) with the constants in (7.16).

The lower bound in (7.13) yields

$$e^{-4\beta h_\mu(x)} \le e^{4\beta \int L \,\mathrm{d}\mu}(e + |x|)^{4\beta}. \tag{7.20}$$

The first part of (2.28) proves the first assertion in (7.19); the second follows from (7.15).

Only nonnegativity of T_μ remains. Let $f \in L^2(\mu)$ satisfy $\int f \,\mathrm{d}\mu = 0$. Since $\widetilde{W} \in L^2(\mu^{\otimes 2})$, the quadratic form

$$f \longmapsto \iint \widetilde{W}(x,y) f(x) f(y) \,\mathrm{d}\mu(x)\mathrm{d}\mu(y) \tag{7.21}$$

is continuous on $L^2(\mu)$. Approximate f in $L^2(\mu)$ by bounded, compactly supported, centered functions f_n. Indeed, truncate f in value and space and subtract its mean times a fixed bounded compactly supported function having μ-mean one. Put $\nu_n = f_n \mu$.

For $0 < s < \mathsf{d}$, the Riesz Fourier representation gives

$$\frac{1}{s} \iint |x - y|^{-s} \,\mathrm{d}\nu_n(x)\mathrm{d}\nu_n(y) = c_{\mathsf{d},s} \int_{\mathbb{R}^{\mathsf{d}}} |\widehat{\nu_n}(\xi)|^2 |\xi|^{s-\mathsf{d}} \,\mathrm{d}\xi \ge 0, \tag{7.22}$$

where $c_{\mathsf{d},s} > 0$. Since $\nu_n(\mathbb{R}^{\mathsf{d}}) = 0$, the left-hand side is unchanged if $|x - y|^{-s}$ is replaced by $|x - y|^{-s} - 1$. As $s \downarrow 0$,

$$\frac{|x - y|^{-s} - 1}{s} \longrightarrow -\log|x - y|. \tag{7.23}$$

The measures ν_n have bounded compactly supported densities. Taking $0 < s \le s_0 < \mathsf{d}$, the integrands in (7.23) are dominated near the diagonal by an integrable multiple of $|x-y|^{-s_0}(1+|\log|x-y||)$, and they are uniformly bounded away from the diagonal on the common compact support. Dominated convergence in (7.22) therefore gives

$$\iint -\log|x - y| \,\mathrm{d}\nu_n(x)\mathrm{d}\nu_n(y) \ge 0. \tag{7.24}$$

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The one-variable centering terms vanish against ν_n, so the left-hand side equals the quadratic form of $\widetilde{W}$. Letting $n \to \infty$ in (7.21) proves nonnegativity on the centered subspace. Since $T_\mu \mathbf{1} = 0$, decomposing an arbitrary $L^2(\mu)$ function into its mean and centered part proves nonnegativity on all of $L^2(\mu)$. Self-adjointness and the Hilbert–Schmidt property follow from symmetry and (7.15) with $p = 2$.

□

Lemma 7.3 (Uniform logarithmic rows)**.** *Assume the logarithmic hypotheses in Theorem 2.1. Then $h_\mu(x) + L(x)$ is uniformly bounded, $\widetilde{W}$ has a uniform lower bound, and, for every finite $p \geq 1$,*

$$\sup_x \|w_x\|_{L^p(\mu)} < \infty. \tag{7.25}$$

All statements are understood outside null sets where a chosen kernel representative is undefined.

Proof. Let $\ell(z) = (-\log|z|)_+$ and $C_0 = \log(1+e)$. The triangle inequality implies

$$-L(y) \leq -\log|x-y| + L(x) \leq L(y) + C_0 + \ell(x-y). \tag{7.26}$$

For the lower bound, use $|x-y| \leq (e+|x|)(e+|y|)$. For the upper bound, use $e + |x| \leq (e+|x-y|)(e+|y|)$ and $\log(1+e/|z|) \leq C_0 + \ell(z)$.

The local logarithmic moments are bounded uniformly by (7.8). Moreover, (2.28) implies $L \in L^p(\mu)$ for every finite p. Integrating (7.26), with $m_L = \int L \, \mathrm{d}\mu$, gives

$$-m_L \leq h_\mu(x) + L(x) \leq m_L + C_0 + C_{\mathrm{loc}}. \tag{7.27}$$

Rewrite the centered kernel as

$$\begin{aligned} \widetilde{W}(x,y) = {} & \big[-\log|x-y| + L(x) + L(y)\big] \\ & - \big[h_\mu(x) + L(x)\big] - \big[h_\mu(y) + L(y)\big] + 2I_\mu. \end{aligned} \tag{7.28}$$

The first bracket is nonnegative, and (7.27) proves a uniform lower bound for $\widetilde{W}$. Its upper bound, by (7.26) and (7.27), is $C + 2L(y) + \ell(x-y)$. Therefore,

$$|\widetilde{W}(x,y)| \leq C + 2L(y) + \ell(x-y). \tag{7.29}$$

Combining (7.29) with (7.8) proves (7.25). The displayed one-sided envelope does not assert that the kernel itself is bounded; its collision singularity remains. □

7.4. **Verification of the quantitative hypotheses.**

Proof of Theorems 2.1 and 2.2. It suffices to verify the hypotheses of Theorem 3.1. The estimates used here concern the interaction and the reference measure; no marginal estimate enters this verification.

For Riesz interactions, fix α as in (2.25), choose $2 + \alpha < p < \mathsf{d}/\mathsf{s}$, and take $\vartheta = \alpha/2$. Lemma 7.1 gives (3.2)–(3.3). Proposition 5.1, equation (5.4), of [DGR26], applied at zero cutoff and at the fixed inverse temperature Γ in (3.5), gives (3.6). Its ordered statistic has the normalization (3.4). There is no additional spatial-moment assumption in this case, and

$$N^{-\alpha/8} + N^{-\vartheta/2} = N^{-\alpha/8} + N^{-\alpha/4} \leq 2N^{-\alpha/8}. \tag{7.30}$$

For the logarithmic model, take $p = 4$, $\vartheta = 1/2$, and $\alpha = 1$. Then $r_* = 12$ and $\Gamma = 12\beta$. The potential bounds in Lemma 7.2 and the centered row estimates in Lemma 7.3 give (3.2)–(3.3). In particular, ψ_μ is bounded above, and belongs to every finite $L^q(\mu)$, although it need not be bounded below. The logarithmic centered-family bound in [DGR26, Proposition 5.1] requires finite absolute logarithmic energy and $\int e^{-2\Gamma h_\mu} \, \mathrm{d}\mu < \infty$. The former was proved above. Since $\mathsf{c}_{\log,\beta,\mu} \geq \beta$, (2.28) implies $q > 24\beta$, and (7.9) gives

$$\int e^{-24\beta h_\mu(x)} \, \mathrm{d}\mu(x) \leq e^{24\beta \int L \, \mathrm{d}\mu} \int (e+|x|)^{24\beta} \, \mathrm{d}\mu(x) < \infty. \tag{7.31}$$

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Thus, the zero-cutoff conclusion of that proposition supplies (3.6), and

$$N^{-\alpha/8} + N^{-\vartheta/2} = N^{-1/8} + N^{-1/4} \le 2N^{-1/8}. \tag{7.32}$$

Applying Theorem 3.1 proves all the asserted estimates and limits. The product-law case $\beta = 0$ is immediate. □

7.5. **Verification of the order criterion.** The following applications of Theorem 3.4 give the sharp scales recorded in the introduction. They do not use the quantitative marginal theorem. In particular, the logarithmic criterion below retains its original, weaker polynomial-moment requirement and the more general weighted formulation.

7.5.1. *Riesz order bounds.* For every fixed $\beta \ge 0$, $0 < \mathsf{s} < \mathsf{d}/2$, and bounded probability density μ, the hypotheses of Theorem 3.4 hold for $W(x,y) = \mathsf{g}_{\mathsf{s}}(x-y)$. Thus, its upper bounds hold for every $1 \le k \le N$; for $\beta > 0$, its lower bounds hold for every $2 \le k \le N$ once $N \ge N_0$, with N_0 independent of k.

Riesz order bounds. Choose $2 < p < \mathsf{d}/\mathsf{s}$. Lemma 7.1 supplies (3.16) and the bounded lower-tail condition (3.20). To verify the separate nontriviality condition (3.17), let $\nu = f\mu$, where f is a nonzero bounded centered function. The standard Fourier representation gives, whenever the energy is finite,

$$\int_{(\mathbb{R}^{\mathsf{d}})^2} W(x,y)\,\mathrm{d}\nu(x)\mathrm{d}\nu(y) = c_{\mathsf{d},\mathsf{s}} \int_{\mathbb{R}^{\mathsf{d}}} |\widehat{\nu}(\xi)|^2 |\xi|^{\mathsf{s}-\mathsf{d}}\,\mathrm{d}\xi, \tag{7.33}$$

with $c_{\mathsf{d},\mathsf{s}} > 0$; see, for example, [Ser24, Chapter 4, Section 4.1]. The right-hand side is strictly positive. Since the one-variable centering terms vanish against a zero-mass measure, the same quadratic form is generated by $\widetilde{W}$; in particular, $\widetilde{W}$ cannot vanish in $L^2(\mu^{\otimes 2})$. Applying the companion uniform-bound theorem [DGR26, Theorem 2.2(i)] at the two fixed inverse temperatures $r\beta$ and 2β supplies (3.27). The claim follows from Theorem 3.4. □

7.5.2. *Logarithmic order bounds.* Use the exponents p, r from (3.15) and the logarithmic lower-tail data of (3.21). These are the order-theory parameters, independent of the choices in the preceding quantitative verification.

The following lemma gives an explicit, nonoptimal sufficient condition for the weighted estimates (3.26). Its auxiliary notation and the stronger pointwise estimate used in the proof are deferred to Appendix A.

Lemma 7.4 (Weighted logarithmic cavity estimates)**.** *Assume the logarithmic lower-tail condition* (3.21) *and* $\mathfrak{P}_{r\beta}(\mu) < \infty$*, and write* $c := \mathsf{c}_{\log,\beta,\mu}$*. Suppose that*

$$\int_{\mathbb{R}^{\mathsf{d}}} (e + |x|)^{p^2 c}\,\mathrm{d}\mu(x) < \infty, \tag{7.34a}$$

$$\left\| (e + |\cdot|)^{2c} \left(|h_\mu - 2I_\mu|^2 + \left[\int_{\mathbb{R}^{\mathsf{d}}} |\widetilde{W}(\cdot,y)|^p (e+|y|)^{p^2 c}\,\mathrm{d}\mu(y) \right]^{2/p} \right) \right\|_{L^2(\mu)} < \infty. \tag{7.34b}$$

Then the weighted cavity estimates (3.26) *hold.*

The proof of Lemma 7.4 is given in Appendix A.

Lemma 7.5 (Polynomial moments imply the logarithmic cavity hypotheses)**.** *Fix* $\beta \ge 0$ *and* $p > 2$*, let* $W(x,y) = -\log|x-y|$*, and let* $\mu \in L^\infty \cap \mathcal{P}(\mathbb{R}^{\mathsf{d}})$*. Suppose that*

$$\int_{\mathbb{R}^{\mathsf{d}}} (e + |x|)^q\,\mathrm{d}\mu(x) < \infty \tag{7.35}$$

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for some $q > 0$. Then the logarithmic lower-tail condition (3.21) holds for suitable finite constants a_0, a_1, a_h. Fix one such choice and let $c = \mathsf{c}_{\log,\beta,\mu}$ be as in (3.23). If

$$q > \max\left\{p^2 c,\, 4(p+1)c\right\}, \tag{7.36}$$

then the finiteness requirements in (3.16), finite absolute logarithmic energy, and the assumptions (7.34) all hold. Consequently, if $\mathfrak{P}_{r\beta}(\mu) < \infty$, then the weighted cavity estimates (3.26) hold.

Proof. The elementary estimates above give finite absolute logarithmic energy, (3.16), and the logarithmic lower-tail condition. We verify the remaining weighted requirements.

The first condition follows directly from (7.35) and $q > p^2 c$. By (7.10) and the inequality $(e+|y|)^{p^2 c} \le C_{p,c}(e+|x|)^{p^2 c}$ when $|x-y| < 1$,

$$\begin{aligned}\left[\int_{\mathbb{R}^{\mathsf{d}}} |\widetilde{W}(x,y)|^p (e+|y|)^{p^2 c}\, \mathrm{d}\mu(y)\right]^{1/p} \\ \le C_{\beta,p,\mu}\left(1 + L(x) + (e+|x|)^{pc}\right).\end{aligned} \tag{7.37}$$

Together with (7.9), this gives

$$\begin{aligned}(e+|x|)^{2c}\left(|h_\mu(x) - 2I_\mu|^2 + \left[\int_{\mathbb{R}^{\mathsf{d}}} |\widetilde{W}(x,y)|^p (e+|y|)^{p^2 c}\, \mathrm{d}\mu(y)\right]^{2/p}\right) \\ \le C_{\beta,p,\mu}\left[(e+|x|)^{2c}(1+L(x))^2 + (e+|x|)^{2(p+1)c}\right].\end{aligned} \tag{7.38}$$

After squaring, the right-hand side is integrable under μ whenever $q > 4(p+1)c$; logarithmic factors are absorbed by the strict inequality. This proves the second condition in (7.34). The final assertion follows from Lemma 7.4.

□

For $W(x,y) = -\log|x-y|$ and a bounded probability density satisfying the criterion of Lemma 7.5 for some $p > 2$, the conclusions of Theorem 3.4 hold. Its constants and onset are independent of the marginal size. This statement uses the order criterion, not the stronger logarithmic hypothesis of Theorem 2.1.

Logarithmic order bounds. Choose $p > 2$ and q as in Lemma 7.5. That lemma supplies the quantitative integrability, logarithmic lower-tail, and weighted cavity hypotheses. It also gives finite absolute logarithmic energy, which is precisely the additional logarithmic hypothesis in the companion uniform-bound theorem [DGR26, Theorem 2.2(i)]. Applying that theorem at $r\beta$ and 2β supplies (3.27). If $\nu = f\mu$ has zero total mass, the logarithmic energy admits the Fourier representation

$$-\int_{(\mathbb{R}^{\mathsf{d}})^2} \log|x-y|\, \mathrm{d}\nu(x)\mathrm{d}\nu(y) = c_{\mathsf{d}} \int_{\mathbb{R}^{\mathsf{d}}} |\widehat{\nu}(\xi)|^2 |\xi|^{-\mathsf{d}}\, \mathrm{d}\xi, \tag{7.39}$$

whenever finite, with $c_{\mathsf{d}} > 0$; see, for example, [Ser24, Chapter 4, Section 4.1]. Choose any nonzero bounded centered f and set $\nu = f\mu$. The right-hand side is strictly positive, and the same quadratic form is generated by the Hoeffding-centered kernel because the centering terms vanish against ν. Thus, (3.17) holds, and Theorem 3.4 applies. □

Remark 7.6 (Compactly supported logarithmic backgrounds). If μ has compact support, then (7.35) holds for every $q > 0$, so one may choose q above the threshold (7.36). Thus, compact support satisfies the logarithmic order criterion, but is not required. In this special case, one may alternatively use the simpler bounded lower-tail condition (3.20).

Corollary 7.7 (General weighted logarithmic backgrounds)**.** *Let $\beta \ge 0$, let $W(x,y) = -\log|x-y|$, and let $\mu \in L^\infty \cap \mathcal{P}(\mathbb{R}^{\mathsf{d}})$ satisfy*

$$\int_{(\mathbb{R}^{\mathsf{d}})^2} |\log|x-y||\, \mathrm{d}\mu(x)\mathrm{d}\mu(y) < \infty. \tag{7.40}$$

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Suppose that, for some $p > 2$, the quantitative integrability assumptions (3.16), the logarithmic lower-tail condition (3.21), and the weighted cavity estimates (3.26) hold. Then the conclusions of Theorem 3.4 hold. The conditions of Lemma 7.4 are sufficient for the weighted cavity estimates.

Proof. The zero-mass Fourier representation used above verifies the nontriviality condition (3.17) under the stated integrability assumptions. The remaining external input is (3.27), which follows by applying the logarithmic uniform-bound theorem [DGR26, Theorem 2.2(i)] at $r\beta$ and 2β; the finite absolute logarithmic energy required there is assumed in (7.40). The weighted estimates are assumed directly or supplied by Lemma 7.4; hence, Theorem 3.4 applies. □

8. Fixed marginals under a subquadratic moment

We first estimate canonical U-statistics under the product law and then pair them with the full Gibbs density. The degree-two estimate preserves the N^{-1} scale; the higher-order terms are negligible at that scale for each fixed marginal.

Let $(E, \mathcal{E}, \mu)$ be a probability space. For integers $1 \le j \le N$, write

$$(N)_j := N(N-1)\cdots(N-j+1). \tag{8.1}$$

A measurable kernel $h : E^j \to \mathbb{R}$ is *canonical* if its integral in each variable vanishes for almost every choice of the remaining variables. For independent $X_1, \dots, X_N \sim \mu$, define the ordered canonical U-statistic

$$\mathcal{U}_{N,j} h := \frac{1}{(N)_j} \sum_{\substack{1 \le i_1,\dots,i_j \le N \\ i_1,\dots,i_j \text{ all distinct}}} h(X_{i_1}, \dots, X_{i_j}). \tag{8.2}$$

Lemma 8.1 (Canonical U-statistics of arbitrary fixed order)**.** *For every integer $j \ge 2$ and every $q \ge 2$, there is a finite constant $\mathsf{A}_{j,q}$, depending only on j and q, such that, for every $N \ge j$ and every canonical $h \in L^q(\mu^{\otimes j})$,*

$$\|\mathcal{U}_{N,j} h\|_{L^q(\mu^{\otimes N})} \le \frac{\mathsf{A}_{j,q}}{\sqrt{(N)_j}} \|h\|_{L^q(\mu^{\otimes j})}. \tag{8.3}$$

If C_j is the constant in the upper-tail comparison of de la Peña and Montgomery-Smith [dlPnMS95, Theorem 1], and K_q is an admissible scalar Khintchine constant, one may take

$$\mathsf{D}_{j,q} := \mathsf{C}_j^{1+1/q}, \qquad \mathsf{A}_{j,q} := \mathsf{D}_{j,q} 2^j \mathsf{K}_q^j. \tag{8.4}$$

Proof. Let S^{cpl} denote the unnormalized sum in (8.2), and let

$$S^{\mathrm{dec}} := \sum_{\substack{1 \le i_1,\dots,i_j \le N \\ i_1,\dots,i_j \text{ all distinct}}} h(X_{i_1}^{(1)}, \dots, X_{i_j}^{(j)}), \tag{8.5}$$

where the j coordinate families are independent copies. The upper direction of the cited tail-decoupling theorem requires no symmetry of h and gives

$$\mathbb{P}(|S^{\mathrm{cpl}}| \ge t) \le \mathsf{C}_j \mathbb{P}(\mathsf{C}_j |S^{\mathrm{dec}}| \ge t). \tag{8.6}$$

Layer-cake integration yields

$$\begin{aligned} \mathbb{E}\left[|S^{\mathrm{cpl}}|^q\right] &\le \mathsf{C}_j q \int_0^\infty t^{q-1} \mathbb{P}(|S^{\mathrm{dec}}| \ge t/\mathsf{C}_j)\, \mathrm{d}t \\ &= \mathsf{C}_j^{q+1} \mathbb{E}\left[|S^{\mathrm{dec}}|^q\right]. \end{aligned} \tag{8.7}$$

Thus, $\|S^{\mathrm{cpl}}\|_q \le \mathsf{D}_{j,q} \|S^{\mathrm{dec}}\|_q$, which is the tail-to-moment conversion in (8.4).

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Complete degeneracy permits successive symmetrization in the j independent coordinate families:

$$\|S^{\mathrm{dec}}\|_{L^q} \le 2^j \left\| \sum_{\substack{1\le i_1,\dots,i_j\le N \\ i_1,\dots,i_j \text{ all distinct}}} \varepsilon_{i_1}^{(1)}\cdots\varepsilon_{i_j}^{(j)} h(X_{i_1}^{(1)},\dots,X_{i_j}^{(j)}) \right\|_{L^q}. \tag{8.8}$$

Conditionally on the X-variables, repeated Khintchine inequality gives

$$\begin{aligned} &\left\| \sum_{\substack{1\le i_1,\dots,i_j\le N \\ i_1,\dots,i_j \text{ all distinct}}} \varepsilon_{i_1}^{(1)}\cdots\varepsilon_{i_j}^{(j)} h(X_{i_1}^{(1)},\dots,X_{i_j}^{(j)}) \right\|_{L^q(\varepsilon)} \\ &\qquad \le \mathsf{K}_q^j \left(\sum_{\substack{1\le i_1,\dots,i_j\le N \\ i_1,\dots,i_j \text{ all distinct}}} \left| h(X_{i_1}^{(1)},\dots,X_{i_j}^{(j)}) \right|^2 \right)^{1/2}. \end{aligned} \tag{8.9}$$

Since $q/2 \ge 1$, Minkowski's inequality implies

$$\begin{aligned} &\left\| \left(\sum_{\substack{1\le i_1,\dots,i_j\le N \\ i_1,\dots,i_j \text{ all distinct}}} \left| h(X_{i_1}^{(1)},\dots,X_{i_j}^{(j)}) \right|^2 \right)^{1/2} \right\|_{L^q(X)} \\ &\qquad \le \sqrt{(N)_j}\, \|h\|_{L^q(\mu^{\otimes j})}. \end{aligned} \tag{8.10}$$

Combining these estimates and dividing by $(N)_j$ proves (8.3). □

Lemma 8.2 (Refined canonical pair moment)**.** *For every $q > 2$, there is a finite constant B_q, depending only on q, such that, for every $N \ge 2$ and every canonical $h \in L^q(\mu^{\otimes 2})$,*

$$\|\mathcal{U}_{N,2} h\|_{L^q(\mu^{\otimes N})} \le \mathsf{B}_q \left(\frac{\|h\|_{L^2(\mu^{\otimes 2})}}{N-1} + \frac{N^{1/q-1/2}}{N-1} \|h\|_{L^q(\mu^{\otimes 2})} \right). \tag{8.11}$$

Proof. Let $(X_i)_{i\le N}$ and $(Y_j)_{j\le N}$ be independent samples with law μ. The order-two case of the moment decoupling obtained from [dlPnMS95, Theorem 1] reduces the coupled ordered sum, up to a constant depending only on q, to

$$S := \sum_{i\ne j} h(X_i, Y_j). \tag{8.12}$$

Condition on $Y = (Y_1,\dots,Y_N)$ and set

$$Z_i := \sum_{j\ne i} h(X_i, Y_j). \tag{8.13}$$

The variables Z_i are conditionally independent and centered. Conditional scalar Rosenthal inequality [Ros70] gives

$$\|S\|_{L^q} \le C_q \left\{ \left\| \left(\sum_i \mathbb{E}[Z_i^2 \mid Y] \right)^{1/2} \right\|_{L^q(Y)} + \left(\sum_i \mathbb{E}\left[|Z_i|^q\right] \right)^{1/q} \right\}. \tag{8.14}$$

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For the first term, set

$$A_i(Y) \coloneqq \int_E \left| \sum_{j \neq i} h(x, Y_j) \right|^2 \mathrm{d}\mu(x). \tag{8.15}$$

The $L^2(\mu)$-valued variables $h(\cdot, Y_j)$ are independent and centered. A Hilbert-space-valued Rosenthal inequality, for example the two-smooth Banach-space estimate in [Pin15], yields

$$\|A_i^{1/2}\|_{L^q(Y)} \leq C_q \left(\sqrt{N} \|h\|_{L^2(\mu^{\otimes 2})} + N^{1/q} \|h\|_{L^q(\mu^{\otimes 2})} \right). \tag{8.16}$$

Here, the L^q-norm of $y \mapsto \|h(\cdot, y)\|_{L^2(\mu)}$ is bounded by $\|h\|_{L^q(\mu^{\otimes 2})}$. Minkowski's inequality in $L^{q/2}$ therefore gives

$$\left\| \left(\sum_i A_i \right)^{1/2} \right\|_{L^q(Y)} \leq C_q \left(N \|h\|_{L^2(\mu^{\otimes 2})} + N^{1/2+1/q} \|h\|_{L^q(\mu^{\otimes 2})} \right). \tag{8.17}$$

For the second term in (8.14), condition on X_i and apply scalar Rosenthal inequality to the centered sum over j. Since

$$\int_E \left(\int_E |h(x,y)|^2 \, \mathrm{d}\mu(y) \right)^{q/2} \mathrm{d}\mu(x) \leq \|h\|^q_{L^q(\mu^{\otimes 2})}, \tag{8.18}$$

we obtain

$$\left(\sum_i \mathbb{E}\left[|Z_i|^q \right] \right)^{1/q} \leq C_q N^{1/2+1/q} \|h\|_{L^q(\mu^{\otimes 2})}. \tag{8.19}$$

Combining the preceding estimates, restoring the decoupling constant, and dividing by $N(N-1)$ proves (8.11). □

Proposition 8.3 (Exchangeable fixed-marginal L^τ estimate)**.** *Fix $1 < \tau < 2$ and set $q = \tau/(\tau - 1)$. For each N, let F_N be an exchangeable probability density with respect to $\mu^{\otimes N}$, with labeled marginal density $F_N^{(k)}$. Suppose that*

$$\sup_{N \geq 1} \|F_N\|_{L^\tau(\mu^{\otimes N})} \leq M_\tau, \qquad \|F_N^{(1)} - 1\|_{L^2(\mu)} \leq \frac{U}{N}, \quad N \geq 2. \tag{8.20}$$

Then, for every $N \geq 2$ and $2 \leq k \leq N$,

$$\begin{aligned} \|F_N^{(k)} - 1\|_{L^\tau(\mu^{\otimes k})} \leq \frac{U\sqrt{k}}{N} + (M_\tau + 1)\mathsf{B}_q \left[\frac{\sqrt{\binom{k}{2}}}{N-1} + \frac{4\binom{k}{2} N^{1/q - 1/2}}{N-1} \right] \\ + (M_\tau + 1) \sum_{j=3}^{k} \binom{k}{j} \frac{2^j \mathsf{A}_{j,q}}{\sqrt{(N)_j}}. \end{aligned} \tag{8.21}$$

The empty sum is zero. For $k = 1$,

$$\|F_N^{(1)} - 1\|_{L^\tau(\mu)} \leq \frac{U}{N}. \tag{8.22}$$

For every fixed $k \geq 2$,

$$\limsup_{N \to \infty} N \|F_N^{(k)} - 1\|_{L^\tau(\mu^{\otimes k})} \leq U\sqrt{k} + (M_\tau + 1)\mathsf{B}_q \sqrt{\binom{k}{2}}. \tag{8.23}$$

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In particular, there is a finite constant $C = C(\tau, M_\tau, U)$, independent of k, such that, for every fixed k, there is an onset $N_0 = N_0(k, \tau, M_\tau, U)$ for which

$$\|F_N^{(k)} - 1\|_{L^\tau(\mu^{\otimes k})} \le C\frac{k}{N}, \qquad N \ge N_0. \tag{8.24}$$

Proof. By L^τ–L^q duality, pair $F_N^{(k)} - 1$ against $\varphi \in L^q(\mu^{\otimes k})$ with $\|\varphi\|_q \le 1$. Permutation averaging allows us to assume that φ is symmetric. Its Hoeffding decomposition has the form

$$\varphi = \mathbb{E}_{\mu^{\otimes k}}[\varphi] + \sum_{j=1}^{k} \sum_{\substack{A \subset \{1,\dots,k\} \\ |A|=j}} \varphi_j(x_A), \tag{8.25}$$

with symmetric canonical kernels φ_j. Inclusion–exclusion and contractivity of conditional expectation give

$$\|\varphi_j\|_{L^q(\mu^{\otimes j})} \le 2^j. \tag{8.26}$$

Since $\|\varphi\|_{L^2} \le 1$, orthogonality also gives

$$\|\varphi_1\|_{L^2(\mu)} \le k^{-1/2}, \qquad \|\varphi_2\|_{L^2(\mu^{\otimes 2})} \le \binom{k}{2}^{-1/2}. \tag{8.27}$$

Marginal consistency and exchangeability yield

$$\int_{E^k} \varphi(F_N^{(k)} - 1)\,\mathrm{d}\mu^{\otimes k} = \sum_{j=1}^{k} \binom{k}{j} \mathbb{E}_{F_N\mu^{\otimes N}}[\varphi_j(X_1, \dots, X_j)]. \tag{8.28}$$

The first level is bounded by $U\sqrt{k}/N$. For $j \ge 2$, exchangeability and canonicity give

$$\mathbb{E}_{F_N\mu^{\otimes N}}[\varphi_j(X_1, \dots, X_j)] = \int_{E^N} (F_N - 1)\mathcal{U}_{N,j}\varphi_j\,\mathrm{d}\mu^{\otimes N}, \tag{8.29}$$

and hence

$$\left|\mathbb{E}_{F_N\mu^{\otimes N}}[\varphi_j(X_1, \dots, X_j)]\right| \le (M_\tau + 1)\|\mathcal{U}_{N,j}\varphi_j\|_{L^q(\mu^{\otimes N})}. \tag{8.30}$$

For $j = 2$, apply Lemma 8.2 together with (8.26)–(8.27). For $j \ge 3$, apply Lemma 8.1 and (8.26). Taking the supremum over φ proves (8.21).

For fixed k, the second pair term is $o(N^{-1})$ because $1/q - 1/2 < 0$, and every term of degree $j \ge 3$ is $O_{k,\tau}(N^{-j/2})$. This proves (8.23). The last assertion follows from $\sqrt{k} \le k$ and $\sqrt{\binom{k}{2}} \le k/\sqrt{2}$; the onset may depend on k. □

Proof of Theorem 3.6. By Lemma 4.4 with exponent τ,

$$\sup_{N \ge 1} \|f_{N,\beta}\|_{L^\tau(\mu^{\otimes N})} \le \mathsf{M}_{\tau,\beta,\mu}. \tag{8.31}$$

Moreover, Lemma 3.3 with $s = r$ and $t = \tau$ gives

$$\mathfrak{P}_{r\beta}(\mu)^{1/r} \le \mathfrak{P}_{\tau\beta}(\mu)^{1/\tau}. \tag{8.32}$$

Thus, the bounded cavity proof and the sufficient weighted logarithmic proof have exactly the required full-density L^r input, and the literal hypothesis $\mathfrak{P}_{r\beta}(\mu) < \infty$ in Lemma 7.4 is satisfied. No partition-function moment at 2β is used. Applying the one-point argument in Section 4 with $\theta = \tau$ gives

$$\|f_{N,\beta}^{(1)} - 1\|_{L^2(\mu)} \le \frac{\mathsf{U}_{\tau,\beta,\mu}}{N}. \tag{8.33}$$

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Now apply Proposition 8.3 with $M_\tau = \mathsf{M}_{\tau,\beta,\mu}$ and $U = \mathsf{U}_{\tau,\beta,\mu}$. This gives (3.50)–(3.52) and the L^τ line of (3.53). Since $\mu^{\otimes k}$ is a probability measure,

$$2\,\mathrm{TV}(\mathbb{Q}^{(k)}_{N,\beta}(\mu), \mu^{\otimes k}) = \|f^{(k)}_{N,\beta} - 1\|_{L^1(\mu^{\otimes k})} \le \|f^{(k)}_{N,\beta} - 1\|_{L^\tau(\mu^{\otimes k})}, \tag{8.34}$$

which proves the total-variation line.

The first line of (3.55) is the τ-th power of the L^τ estimate. Finally, there is a finite constant C_τ such that

$$t\log t - t + 1 \le C_\tau |t-1|^\tau, \qquad t \ge 0. \tag{8.35}$$

Indeed, the quotient of the left-hand side by $|t-1|^\tau$ extends continuously at $t = 1$ with value zero, is finite at $t = 0$, and tends to zero as $t \to \infty$. Integrating (8.35) proves the second line of (3.55). □

9. Matching lower bounds

We prove the lower estimates in Theorem 3.4 under that theorem's own hypotheses, without using the quantitative marginal theorem. On the common hypotheses with a nontrivial interaction and $\beta > 0$, the quantitative estimates give another deduction: the two-particle limit covers finitely many small marginal sizes, direct subtraction gives a lower bound up to a sufficiently small fixed fraction of N, and contraction under marginalization covers every larger size. The resulting lower bound holds for every $2 \le k \le N$ once $N \ge N_0$, with N_0 independent of k. This deduction would, however, impose the quantitative theorem's row, source, and amplified centered-pair moment assumptions. The direct argument below preserves the distinct order criterion, including its weighted logarithmic lower-tail alternative. In Section 9.1, a strict Jensen gap and convexity give a lower bound for the derivative of the product-law log-Laplace transform. Section 9.2 uses this derivative to estimate the pair response and tests with the pair sum. Section 9.3 obtains total variation by a bounded truncation, and then relative entropy by Pinsker's inequality.

9.1. Strict product-law Jensen gap.

Recall the centered energy $\mathcal{X}_N$ from (4.18). For $X_N \sim \mu^{\otimes N}$, one has $\mathbb{E}[\mathcal{X}_N] = 0$, and

$$\mathsf{K}_{N,\beta}(\mu) = e^{\beta I_\mu}\mathbb{E}_{\mu^{\otimes N}}\left[e^{-\beta\mathcal{X}_N}\right]. \tag{9.1}$$

Denote the log-Laplace transform of $\mathcal{X}_N$, equivalently the cumulant generating function of $-\mathcal{X}_N$, by

$$G_N(\alpha) := \log \mathbb{E}_{\mu^{\otimes N}}\left[e^{-\alpha\mathcal{X}_N}\right], \qquad \alpha \ge 0. \tag{9.2}$$

Lemma 9.1 (Finiteness and differentiability of the log partition function)**.** *Fix $\beta > 0$ and assume $\mathsf{K}_{N,2\beta}(\mu) < \infty$. Then G_N is finite on $[0, 2\beta]$, is continuously differentiable on $(0, 2\beta)$, and*

$$G_N'(\alpha) = -\frac{\mathbb{E}_{\mu^{\otimes N}}\left[\mathcal{X}_N e^{-\alpha\mathcal{X}_N}\right]}{\mathbb{E}_{\mu^{\otimes N}}\left[e^{-\alpha\mathcal{X}_N}\right]} = -\mathbb{E}_{\mathbb{Q}_{N,\alpha}(\mu)}[\mathcal{X}_N], \qquad 0 < \alpha < 2\beta. \tag{9.3}$$

Proof. The identity (9.1) at 2β gives exponential integrability at the right endpoint. Hölder's inequality and $G_N(0) = 0$ then imply finiteness throughout $[0, 2\beta]$.

Write $J = [a, b] \Subset (0, 2\beta)$, and choose $\epsilon > 0$ so that $a - \epsilon > 0$ and $b + \epsilon < 2\beta$. For every $\alpha \in J$,

$$|x|e^{-\alpha x} \le C_\epsilon\left(e^{-(a-\epsilon)x} + e^{-(b+\epsilon)x}\right), \qquad x \in \mathbb{R}. \tag{9.4}$$

After increasing C_ϵ if necessary, the right-hand side of (9.4) is integrable by finiteness of G_N on $[0, 2\beta]$, and it is independent of $\alpha \in J$. Differentiation under the integral sign, as well as continuity of the derivative on J, now follows by dominated convergence.

The preceding gives the first identity. Since

$$e^{-\alpha N\mathsf{F}_N} = e^{\alpha I_\mu}e^{-\alpha\mathcal{X}_N}, \tag{9.5}$$

the second identity follows from the definition of $\mathbb{Q}_{N,\alpha}(\mu)$. □

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We use Lemma 8.1 with $j = 2$ to control the pair part of $\mathcal{X}_N$. The resulting moment estimate serves only to make the Jensen gap uniform in N: the variance gives a fixed nonzero L^2 mass, while the L^p bound prevents that mass from escaping to rare large values.

Lemma 9.2 (Uniform product-law nondegeneracy)**.** *Let $p > 2$ be fixed by (3.15) under the integrability assumptions (3.16), and set*

$$\mathsf{M}_{p,\mu} := (\frac{\mathsf{A}_{2,p}}{2}\|\widetilde{W}\|_{L^p(\mu^{\otimes 2})} + \|h_\mu - 2I_\mu\|_{L^p(\mu)})^p. \tag{9.6}$$

For all $N \geq 2$,

$$\mathbb{E}_{X_N\sim\mu^{\otimes N}}\left[\mathcal{X}_N^2\right] \geq \frac{1}{4}\|\widetilde{W}\|^2_{L^2(\mu^{\otimes 2})}, \tag{9.7}$$

and

$$\mathbb{E}_{X_N\sim\mu^{\otimes N}}\left[|\mathcal{X}_N|^p\right] \leq \mathsf{M}_{p,\mu}. \tag{9.8}$$

Proof. Write

$$A_N := \frac{1}{N}\sum_{1\leq i<j\leq N}\widetilde{W}(x_i, x_j), \tag{9.9}$$

$$B_N := \frac{1}{N}\sum_{i=1}^N (h_\mu(x_i) - 2I_\mu). \tag{9.10}$$

so that $\mathcal{X}_N = A_N - B_N$. Since $\widetilde{W}$ is centered in each variable,

$$\mathbb{E}\left[A_N\right] = 0, \qquad \mathbb{E}\left[A_N^2\right] = \frac{\binom{N}{2}}{N^2}\|\widetilde{W}\|^2_{L^2(\mu^{\otimes 2})} = \frac{N-1}{2N}\|\widetilde{W}\|^2_{L^2(\mu^{\otimes 2})}. \tag{9.11}$$

Moreover, $\mathbb{E}\left[A_N B_N\right] = 0$, again by the centering of $\widetilde{W}$. Therefore,

$$\mathbb{E}\left[\mathcal{X}_N^2\right] = \mathbb{E}\left[A_N^2\right] + \mathbb{E}\left[B_N^2\right] \geq \frac{1}{4}\|\widetilde{W}\|^2_{L^2(\mu^{\otimes 2})} > 0, \qquad \forall N \geq 2, \tag{9.12}$$

where positivity is exactly the standing nondegeneracy input $\|\widetilde{W}\|_{L^2(\mu^{\otimes 2})} > 0$.

Recall that $p > 2$ was fixed in (3.15) so that the integrability assumptions in (3.16) hold. Since $\widetilde{W}$ is symmetric and centered in each variable,

$$A_N = \frac{N-1}{2}\mathcal{U}_{N,2}\widetilde{W}. \tag{9.13}$$

Therefore, Lemma 8.1 and Minkowski's inequality give

$$\|A_N\|_{L^p(\mu^{\otimes N})} \leq \frac{\mathsf{A}_{2,p}}{2}\sqrt{\frac{N-1}{N}}\|\widetilde{W}\|_{L^p(\mu^{\otimes 2})} \leq \frac{\mathsf{A}_{2,p}}{2}\|\widetilde{W}\|_{L^p(\mu^{\otimes 2})}, \tag{9.14}$$

$$\|B_N\|_{L^p(\mu^{\otimes N})} \leq \|h_\mu - 2I_\mu\|_{L^p(\mu)}. \tag{9.15}$$

Thus, by the triangle inequality,

$$\|\mathcal{X}_N\|_{L^p(\mu^{\otimes N})} \leq \frac{\mathsf{A}_{2,p}}{2}\|\widetilde{W}\|_{L^p(\mu^{\otimes 2})} + \|h_\mu - 2I_\mu\|_{L^p(\mu)}, \tag{9.16}$$

which is (9.8). □

The next lemma quantifies the gap in Jensen's inequality for the product law $\mu^{\otimes N}$ in terms of the second moment of $\mathcal{X}_N$ and the local curvature of the exponential function. For $\beta > 0$, define

$$\mathsf{R}_{p,\mu} := \max\left\{1, \left(\frac{8\mathsf{M}_{p,\mu}}{\|\widetilde{W}\|^2_{L^2(\mu^{\otimes 2})}}\right)^{1/(p-2)}\right\}, \tag{9.17}$$

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$$\mathsf{a}_{\beta,\mu} \coloneqq \inf_{0<|t|\le \mathsf{R}_{p,\mu}} \frac{e^{-\beta t}-1+\beta t}{t^2}, \tag{9.18}$$

$$\mathsf{J}_{\beta,\mu} \coloneqq \log\left(1+\frac{1}{8}\mathsf{a}_{\beta,\mu}\|\widetilde{W}\|^2_{L^2(\mu^{\otimes 2})}\right). \tag{9.19}$$

Lemma 9.3 (Strict Jensen gap)**.** *For every $\beta>0$,*

$$G_N(\beta) \ge \mathsf{J}_{\beta,\mu}, \qquad N\ge 2. \tag{9.20}$$

Consequently,

$$G_N'(\beta) \ge \frac{\mathsf{J}_{\beta,\mu}}{\beta}, \qquad N\ge 2. \tag{9.21}$$

Proof. By the L^p estimate (9.8) in Lemma 9.2 and the definition (9.17) of $\mathsf{R}_{p,\mu}$,

$$\mathbb{E}\left[\mathcal{X}_N^2\mathbf{1}_{\{|\mathcal{X}_N|>\mathsf{R}_{p,\mu}\}}\right] \le \mathsf{R}_{p,\mu}^{2-p}\mathbb{E}\left[|\mathcal{X}_N|^p\right] \le \frac{1}{8}\|\widetilde{W}\|^2_{L^2(\mu^{\otimes 2})}, \qquad N\ge 2. \tag{9.22}$$

Combining this with the variance lower bound (9.7) from Lemma 9.2,

$$\mathbb{E}\left[\mathcal{X}_N^2\mathbf{1}_{\{|\mathcal{X}_N|\le \mathsf{R}_{p,\mu}\}}\right] \ge \frac{1}{8}\|\widetilde{W}\|^2_{L^2(\mu^{\otimes 2})}. \tag{9.23}$$

Since

$$\varphi_\beta(t) \coloneqq e^{-\beta t}-1+\beta t \ge 0, \qquad \forall t\in\mathbb{R}, \tag{9.24}$$

the definition (9.18) of $\mathsf{a}_{\beta,\mu}$, the truncated variance bound (9.23), and $\mathbb{E}[\mathcal{X}_N]=0$ give

$$\mathbb{E}\left[e^{-\beta\mathcal{X}_N}\right]-1 = \mathbb{E}\left[\varphi_\beta(\mathcal{X}_N)\right] \ge \frac{1}{8}\mathsf{a}_{\beta,\mu}\|\widetilde{W}\|^2_{L^2(\mu^{\otimes 2})}. \tag{9.25}$$

Therefore,

$$G_N(\beta) = \log\mathbb{E}\left[e^{-\beta\mathcal{X}_N}\right] \ge \log(1+\frac{1}{8}\mathsf{a}_{\beta,\mu}\|\widetilde{W}\|^2_{L^2(\mu^{\otimes 2})}) = \mathsf{J}_{\beta,\mu}. \tag{9.26}$$

By Lemma 9.1, G_N is differentiable at β. Since it is convex and $G_N(0)=0$,

$$G_N'(\beta) \ge \frac{G_N(\beta)-G_N(0)}{\beta} \ge \frac{\mathsf{J}_{\beta,\mu}}{\beta}, \tag{9.27}$$

which proves (9.21). □

9.2. **Pair statistic and the chi-square lower bound.** We now convert the Jensen gap into a marginal lower bound. Set

$$m_N \coloneqq \mathbb{E}_{\Xi_N\sim\mathbb{Q}_{N,\beta}(\mu)}\left[\widetilde{W}(\xi_1,\xi_2)\right], \qquad \forall N\ge 2. \tag{9.28}$$

Thus, m_N records the response of one Hoeffding-centered pair interaction to the Gibbs tilt. To detect this response at the level of k observed labels, set

$$A_k(x_1,\dots,x_k) \coloneqq \sum_{1\le i<j\le k}\widetilde{W}(x_i,x_j). \tag{9.29}$$

Because $\widetilde{W}$ is centered in each variable, distinct pair summands are orthogonal under $\mu^{\otimes k}$, including when the two pairs share one index. Consequently,

$$\|A_k\|^2_{L^2(\mu^{\otimes k})} = \binom{k}{2}\|\widetilde{W}\|^2_{L^2(\mu^{\otimes 2})}. \tag{9.30}$$

By exchangeability,

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$$\int A_k(f_{N,\beta}^{(k)} - 1)\,\mathrm{d}\mu^{\otimes k} = \mathbb{E}_{\mathbb{Q}_{N,\beta}^{(k)}(\mu)}[A_k] = \binom{k}{2} m_N. \tag{9.31}$$

Cauchy–Schwarz inequality gives

$$\begin{aligned}\chi^2(\mathbb{Q}_{N,\beta}^{(k)}(\mu) \mid \mu^{\otimes k}) &= \|f_{N,\beta}^{(k)} - 1\|^2_{L^2(\mu^{\otimes k})} \\ &\geq \frac{\left(\int A_k(f_{N,\beta}^{(k)} - 1)\,\mathrm{d}\mu^{\otimes k}\right)^2}{\|A_k\|^2_{L^2(\mu^{\otimes k})}}.\end{aligned} \tag{9.32}$$

Substituting (9.30) and (9.31), we obtain

$$\chi^2(\mathbb{Q}_{N,\beta}^{(k)}(\mu) \mid \mu^{\otimes k}) \geq \binom{k}{2} \frac{m_N^2}{\|\widetilde{W}\|^2_{L^2(\mu^{\otimes 2})}}. \tag{9.33}$$

Thus, the chi-square lower bound of order k^2/N^2 follows once we prove that the Gibbs expectation m_N is negative and has magnitude of order N^{-1}. The next lemma derives exactly this estimate from the strict product-law Jensen gap. Define the threshold

$$\mathsf{N}_{\beta,\mu} \coloneqq \left\lceil \max\left\{2, \frac{2\beta\|h_\mu - 2I_\mu\|_{L^2(\mu)}\mathsf{U}_{\beta,\mu}}{\mathsf{J}_{\beta,\mu}}\right\}\right\rceil. \tag{9.34}$$

Lemma 9.4 (Negative pair response)**.** *For every $\beta > 0$,*

$$m_N \leq -\frac{\mathsf{J}_{\beta,\mu}}{\beta N}, \qquad \forall N \geq \mathsf{N}_{\beta,\mu}. \tag{9.35}$$

Proof. By (9.1),

$$\log \mathsf{K}_{N,\beta}(\mu) = \beta I_\mu + G_N(\beta). \tag{9.36}$$

By Lemma 9.1, differentiating in β is justified and gives

$$G_N'(\beta) = -\mathbb{E}_{\mathbb{Q}_{N,\beta}(\mu)}\left[N\mathsf{F}_N(X_N,\mu) + I_\mu\right]. \tag{9.37}$$

Using the definition of F_N and exchangeability,

$$\mathbb{E}_{\mathbb{Q}_{N,\beta}(\mu)}[N\mathsf{F}_N(X_N,\mu)] = \frac{N-1}{2} m_N - \mathbb{E}_{\mathbb{Q}_{N,\beta}^{(1)}(\mu)}[h_\mu] + I_\mu. \tag{9.38}$$

Combining (9.37) and (9.38),

$$G_N'(\beta) = -\frac{N-1}{2} m_N + (\mathbb{E}_{\mathbb{Q}_{N,\beta}^{(1)}(\mu)}[h_\mu] - 2I_\mu). \tag{9.39}$$

Cauchy–Schwarz inequality and the one-point upper bound (4.82) give

$$\begin{aligned}\left|\mathbb{E}_{\mathbb{Q}_{N,\beta}^{(1)}(\mu)}[h_\mu] - 2I_\mu\right| &= \left|\int (h_\mu - 2I_\mu)(f_{N,\beta}^{(1)} - 1)\,\mathrm{d}\mu\right| \\ &\leq \|h_\mu - 2I_\mu\|_{L^2(\mu)} \frac{\mathsf{U}_{\beta,\mu}}{N}.\end{aligned} \tag{9.40}$$

By Lemma 9.3,

$$\begin{aligned}-\frac{N-1}{2} m_N &= G_N'(\beta) - (\mathbb{E}_{\mathbb{Q}_{N,\beta}^{(1)}(\mu)}[h_\mu] - 2I_\mu) \\ &\geq \frac{\mathsf{J}_{\beta,\mu}}{\beta} - \frac{\|h_\mu - 2I_\mu\|_{L^2(\mu)}\mathsf{U}_{\beta,\mu}}{N}.\end{aligned} \tag{9.41}$$

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If $N \geq \mathsf{N}_{\beta,\mu}$, the right-hand side is at least $\mathsf{J}_{\beta,\mu}/(2\beta)$. Therefore,

$$m_N \leq -\frac{\mathsf{J}_{\beta,\mu}}{\beta(N-1)} \leq -\frac{\mathsf{J}_{\beta,\mu}}{\beta N}, \tag{9.42}$$

which proves (9.35). □

Combining (9.33) with Lemma 9.4, we obtain

$$\chi^2(\mathbb{Q}^{(k)}_{N,\beta}(\mu) \mid \mu^{\otimes k}) \geq \frac{\mathsf{J}^2_{\beta,\mu}}{\beta^2 \|\widetilde{W}\|^2_{L^2(\mu^{\otimes 2})}} \frac{\binom{k}{2}}{N^2}, \tag{9.43}$$

for every $N \geq \mathsf{N}_{\beta,\mu}$ and $2 \leq k \leq N$. The total-variation estimate proved below would also recover a chi-square lower bound of the same order. We present the direct argument because it identifies the L^2 pair-response mechanism without the additional losses introduced by truncation and by the chi-square upper bound used to control the discarded tail.

9.3. **Bounded two-body test and the total-variation lower bound.** The preceding lower bound tested the marginal against A_k, which need not be bounded. This is harmless for χ^2, where the Cauchy–Schwarz inequality uses the L^2-norm of A_k. For total variation, however, the test function must be bounded. Although A_k contains $\binom{k}{2}$ summands, Hoeffding centering yields an $L^p(\mu^{\otimes k})$ norm of order k, rather than the order k^2 given by the triangle inequality. Hence, truncation at height Rk discards only a controlled tail.

Lemma 9.5 (Moment bound for the pair statistic)**.** *With p as in (3.15),*

$$\|A_k\|_{L^p(\mu^{\otimes k})} \leq \frac{\mathsf{A}_{2,p}}{2} \|\widetilde{W}\|_{L^p(\mu^{\otimes 2})} k, \qquad \forall k \geq 2. \tag{9.44}$$

Consequently, for every $R \geq 1$,

$$\|A_k \mathbf{1}_{\{|A_k| > Rk\}}\|_{L^2(\mu^{\otimes k})} \leq (\frac{\mathsf{A}_{2,p}}{2} \|\widetilde{W}\|_{L^p(\mu^{\otimes 2})})^{p/2} R^{1-p/2} k. \tag{9.45}$$

Proof. By the standing integrability assumptions in (3.16), $\widetilde{W} \in L^p(\mu^{\otimes 2})$. By (9.29) and the symmetry of $\widetilde{W}$,

$$A_k = \frac{k(k-1)}{2} \mathcal{U}_{k,2} \widetilde{W}. \tag{9.46}$$

Applying Lemma 8.1 gives (9.44). Finally,

$$\|A_k \mathbf{1}_{\{|A_k| > Rk\}}\|^2_{L^2} \leq (Rk)^{2-p} \|A_k\|^p_{L^p} \leq (\frac{\mathsf{A}_{2,p}}{2} \|\widetilde{W}\|_{L^p(\mu^{\otimes 2})})^p R^{2-p} k^2, \tag{9.47}$$

and taking square roots gives (9.45). □

To convert this moment estimate into a total-variation lower bound, we truncate A_k at height Rk. The preceding lemma controls the discarded tail, while the truncated statistic is bounded and may therefore be used in the dual characterization of total variation. We now fix the constants used in this argument:

$$\mathsf{S}_{\beta,\mu} := \frac{\mathsf{J}_{\beta,\mu}}{4\beta}, \tag{9.48}$$

$$\mathsf{R}_{\mathrm{TV},\beta,\mu} := \max\left\{1, \left(\frac{2(\frac{\mathsf{A}_{2,p}}{2} \|\widetilde{W}\|_{L^p(\mu^{\otimes 2})})^{p/2} \sqrt{\mathsf{C}^{\mathrm{up}}_{\chi,\beta,\mu}}}{\mathsf{S}_{\beta,\mu}}\right)^{2/(p-2)}\right\}, \tag{9.49}$$

$$\mathsf{c}^{\mathrm{low}}_{\mathrm{TV},\beta,\mu} := \frac{\mathsf{S}_{\beta,\mu}}{4\mathsf{R}_{\mathrm{TV},\beta,\mu}}. \tag{9.50}$$

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Proposition 9.6 (Total-variation lower bound)**.** *Let $\beta > 0$. For all $N \geq \mathsf{N}_{\beta,\mu}$ and all $2 \leq k \leq N$,*

$$\mathrm{TV}(\mathbb{Q}^{(k)}_{N,\beta}(\mu), \mu^{\otimes k}) \geq \mathsf{c}^{\mathrm{low}}_{\mathrm{TV},\beta,\mu} \frac{k}{N}. \tag{9.51}$$

Proof. Recall that $f^{(k)}_{N,\beta} = \mathrm{d}\mathbb{Q}^{(k)}_{N,\beta}(\mu)/\mathrm{d}\mu^{\otimes k}$. By the chi-square upper bound in (3.28), proved in Section 4.6, with the admissible constant $\mathsf{C}^{\mathrm{up}}_{\chi,\beta,\mu}$ fixed in (4.101),

$$\|f^{(k)}_{N,\beta} - 1\|_{L^2(\mu^{\otimes k})} \leq \sqrt{\mathsf{C}^{\mathrm{up}}_{\chi,\beta,\mu}} \frac{k}{N}. \tag{9.52}$$

By (9.31) and Lemma 9.4,

$$\left| \int A_k (f^{(k)}_{N,\beta} - 1) \, \mathrm{d}\mu^{\otimes k} \right| = \binom{k}{2} |m_N| \geq \mathsf{S}_{\beta,\mu} \frac{k^2}{N}, \qquad 2 \leq k \leq N, \tag{9.53}$$

where we used $\binom{k}{2} \geq k^2/4$. By Lemma 9.5 and (9.52),

$$\begin{aligned} \left| \int A_k \mathbf{1}_{\{|A_k| > Rk\}} (f^{(k)}_{N,\beta} - 1) \, \mathrm{d}\mu^{\otimes k} \right| &\leq \|A_k \mathbf{1}_{\{|A_k| > Rk\}}\|_{L^2} \|f^{(k)}_{N,\beta} - 1\|_{L^2} \\ &\leq \Big(\frac{\mathsf{A}_{2,p}}{2} \|\widetilde{W}\|_{L^p(\mu^{\otimes 2})}\Big)^{p/2} \sqrt{\mathsf{C}^{\mathrm{up}}_{\chi,\beta,\mu}} R^{1-p/2} \frac{k^2}{N}. \end{aligned} \tag{9.54}$$

With $R = \mathsf{R}_{\mathrm{TV},\beta,\mu}$, the right side of (9.54) is at most one half of the right side of (9.53). Set

$$A_{k,R} := A_k \mathbf{1}_{\{|A_k| \leq Rk\}}. \tag{9.55}$$

Then

$$\left| \int A_{k,R} (f^{(k)}_{N,\beta} - 1) \, \mathrm{d}\mu^{\otimes k} \right| \geq \frac{1}{2} \mathsf{S}_{\beta,\mu} \frac{k^2}{N}. \tag{9.56}$$

On the other hand, $\|A_{k,R}\|_{L^\infty} \leq Rk$ and $\|f^{(k)}_{N,\beta} - 1\|_{L^1(\mu^{\otimes k})} = 2\,\mathrm{TV}(\mathbb{Q}^{(k)}_{N,\beta}(\mu), \mu^{\otimes k})$. Therefore, the dual formulation of total variation gives

$$\left| \int A_{k,R} (f^{(k)}_{N,\beta} - 1) \, \mathrm{d}\mu^{\otimes k} \right| \leq 2\|A_{k,R}\|_{L^\infty} \mathrm{TV}(\mathbb{Q}^{(k)}_{N,\beta}(\mu), \mu^{\otimes k}) \leq 2Rk \ \mathrm{TV}(\mathbb{Q}^{(k)}_{N,\beta}(\mu), \mu^{\otimes k}). \tag{9.57}$$

Combining this with (9.56) and the definition of $\mathsf{c}^{\mathrm{low}}_{\mathrm{TV},\beta,\mu}$ proves (9.51). □

We now finish the proof of Theorem 3.4.

Proof of Theorem 3.4. The chi-square upper bound was proved in Section 4.6. The remaining upper bounds follow from the standard inequalities

$$\mathsf{H}(\nu \mid \pi) \leq \chi^2(\nu \mid \pi), \tag{9.58}$$

$$\mathrm{TV}(\nu, \pi) \leq \frac{1}{2} \chi^2(\nu \mid \pi)^{1/2}. \tag{9.59}$$

Thus, admissible upper constants are

$$\mathsf{C}^{\mathrm{up}}_{\mathrm{TV},\beta,\mu} := \frac{1}{2} \sqrt{\mathsf{C}^{\mathrm{up}}_{\chi,\beta,\mu}}, \tag{9.60}$$

$$\mathsf{C}^{\mathrm{up}}_{\mathsf{H},\beta,\mu} := \mathsf{C}^{\mathrm{up}}_{\chi,\beta,\mu}. \tag{9.61}$$

Assume now $\beta > 0$. For the lower bounds, (9.43) and $\binom{k}{2} \geq k^2/4$, for $k \geq 2$, give the chi-square estimate in (3.33) with

$$\mathsf{c}^{\mathrm{low}}_{\chi,\beta,\mu} := \frac{\mathsf{J}^2_{\beta,\mu}}{4\beta^2 \|\widetilde{W}\|^2_{L^2(\mu^{\otimes 2})}}. \tag{9.62}$$

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The total-variation lower bound is Proposition 9.6. Finally, Pinsker's inequality,

$$\mathsf{H}(\nu \mid \pi) \geq 2\,\mathrm{TV}(\nu,\pi)^2, \tag{9.63}$$

gives the entropy lower bound with

$$\mathsf{c}^{\mathrm{low}}_{\mathsf{H},\beta,\mu} \coloneqq 2(\mathsf{c}^{\mathrm{low}}_{\mathrm{TV},\beta,\mu})^2. \tag{9.64}$$

This proves all assertions of the theorem. □

10. Ultraviolet obstructions

We now record whole-space Riesz examples showing that $\widetilde{W} \in L^2(\mu^{\otimes 2})$ is the exact ultraviolet threshold for the classical fixed-particle chi-square and entropy scales, and that the uniform k/N total-variation bound fails throughout $\mathsf{d}/2 \leq \mathsf{s} < \mathsf{d}$. After introducing the common Riesz estimates, the two-point density representation, and the ultraviolet scale in Section 10.1, we prove the all-temperature two-point chi-square lower bounds in Section 10.2. Section 10.3 transfers these estimates to every fixed number of labels and to relative entropy and total variation, thereby proving Theorem 2.3(i). Finally, Section 10.4 proves the two mesoscopic assertions in Theorem 2.3(ii): the strictly supercritical argument uses a union of depleted close-pair events, whereas the critical argument uses sequential Hellinger accumulation across transition annuli.

10.1. Common setup and the two-point density.

Throughout this section,

$$\frac{\mathsf{d}}{2} \leq \mathsf{s} < \mathsf{d}, \qquad W(x,y) = \mathsf{g}_{\mathsf{s}}(x-y) = \frac{1}{\mathsf{s}}|x-y|^{-\mathsf{s}}, \tag{10.1}$$

and μ is a probability measure on $\mathbb{R}^{\mathsf{d}}$ with bounded Lebesgue density.

The Riesz potential is bounded without any compact-support assumption. Indeed, splitting at an arbitrary radius and optimizing that radius gives

$$H_\mu \coloneqq \|h_\mu\|_{L^\infty} \leq \frac{1}{\mathsf{s}} \inf_{R>0} \left\{ \frac{|\mathbb{S}^{\mathsf{d}-1}|}{\mathsf{d}-\mathsf{s}} \|\mu\|_{L^\infty} R^{\mathsf{d}-\mathsf{s}} + R^{-\mathsf{s}} \right\} \leq C_{\mathsf{d},\mathsf{s}} \|\mu\|_{L^\infty}^{\mathsf{s}/\mathsf{d}}. \tag{10.2}$$

Since $W \geq 0$, it follows that

$$0 \leq 2I_\mu = \int h_\mu \,\mathrm{d}\mu \leq H_\mu, \tag{10.3}$$

$$\|h_\mu - 2I_\mu\|_{L^\infty} \leq H_\mu, \tag{10.4}$$

$$\widetilde{W}(x,y) \geq -2H_\mu \quad \text{for } \mu^{\otimes 2}\text{-a.e. } (x,y). \tag{10.5}$$

The same bounds make the failure of the Hilbert–Schmidt condition immediate. Near the diagonal,

$$\widetilde{W}(x,y) = \frac{1}{\mathsf{s}}|x-y|^{-\mathsf{s}} + O_\mu(1). \tag{10.6}$$

Choose a bounded Borel set $A \subset \mathbb{R}^{\mathsf{d}}$ of positive Lebesgue measure and $a > 0$ such that $\mu \geq a$ almost everywhere on A. Since $|A \cap (A-z)| \to |A|$ as $z \to 0$, for a sufficiently small $r_0 > 0$,

$$\int_{(\mathbb{R}^{\mathsf{d}})^2} |\widetilde{W}(x,y)|^2 \,\mathrm{d}\mu(x)\mathrm{d}\mu(y) \geq c_{\mu,\mathsf{s}} \int_{|z|<r_0} |z|^{-2\mathsf{s}} |A \cap (A-z)| \,\mathrm{d}z = +\infty, \tag{10.7}$$

where the last equality holds exactly because $2\mathsf{s} \geq \mathsf{d}$.

We next specialize the marked representation of Section 4.1 to two observed particles. For $N \geq 3$, define the residual two-point prefactor by

$$\mathcal{R}_{N,\beta}(x,y) \coloneqq \frac{\exp\left\{\frac{\beta}{N}\left(h_\mu(x) + h_\mu(y) - 2I_\mu\right)\right\} \mathsf{M}_{N,2}(x,y)}{\displaystyle\int_{(\mathbb{R}^{\mathsf{d}})^2} \Psi_{N,2} \,\mathrm{d}\mu^{\otimes 2}}. \tag{10.8}$$

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Then Lemma 4.2 gives the exact factorization

$$f^{(2)}_{N,\beta}(x,y) = \mathcal{R}_{N,\beta}(x,y)\exp\left\{-\frac{\beta}{N}\widetilde{W}(x,y)\right\}. \tag{10.9}$$

For every fixed configuration of the remaining $N-2$ particles, Hoeffding centering and Jensen's inequality give

$$\int_{(\mathbb{R}^{\mathsf{d}})^2} \Psi_{N,2}\,\mathrm{d}\mu^{\otimes 2} \geq \exp\left(\frac{2\beta I_\mu}{N}\right) \geq 1. \tag{10.10}$$

Moreover, (10.5) and the definition of $\mathsf{M}_{N,2}$ imply

$$0 \leq \mathsf{M}_{N,2}(x,y) \leq e^{4\beta H_\mu}. \tag{10.11}$$

Combining (10.2)–(10.11), we obtain

$$0 \leq \mathcal{R}_{N,\beta}(x,y) \leq e^{5\beta H_\mu} \tag{10.12}$$

for $\mu^{\otimes 2}$-a.e. (x,y), and consequently

$$0 \leq f^{(2)}_{N,\beta}(x,y) \leq e^{6\beta H_\mu}. \tag{10.13}$$

Recall the ultraviolet scale $\mathfrak{a}_N$ from (2.38). All constants below may depend on $(\beta,\mu,\mathsf{d},\mathsf{s})$, but never on N or on the number of observed particles.

10.2. **All-temperature fixed-particle χ^2 failure.**

10.2.1. *The strictly supercritical case.* The mutual Boltzmann factor in (10.9) suppresses pairs separated by a distance of order $N^{-1/\mathsf{s}}$. Comparing the probability of such pairs under the Gibbs and product laws gives the supercritical lower bound.

Proposition 10.1 (Strictly supercritical two-point obstruction)**.** *Let $\mathsf{d}/2 < \mathsf{s} < \mathsf{d}$ and $\beta > 0$. Then there are constants $c > 0$ and $N_0 < \infty$ such that, for every $N \geq N_0$,*

$$\chi^2(\mathbb{Q}^{(2)}_{N,\beta}(\mu) \mid \mu^{\otimes 2}) \geq cN^{-\mathsf{d}/\mathsf{s}}. \tag{10.14}$$

In particular,

$$N^2\chi^2(\mathbb{Q}^{(2)}_{N,\beta}(\mu) \mid \mu^{\otimes 2}) \longrightarrow +\infty. \tag{10.15}$$

Proof. Retain the bounded set A, lower-density constant a, and radius r_0 chosen above. After decreasing r_0, if necessary, (10.6) gives a constant $c_0 > 0$ such that

$$\widetilde{W}(x,y) \geq c_0|x-y|^{-\mathsf{s}} \qquad \text{for } (x,y) \in A\times A \text{ with } |x-y| \leq r_0. \tag{10.16}$$

Choose $\rho_* > 0$ sufficiently small that

$$\theta := \exp\left\{5\beta H_\mu - \beta c_0 \rho_*^{-\mathsf{s}}\right\} < \frac{1}{2}. \tag{10.17}$$

For all sufficiently large N, set

$$r_N := \rho_* N^{-1/\mathsf{s}}, \tag{10.18}$$

$$E_N := \{(x,y) \in A\times A : |x-y| \leq r_N\}. \tag{10.19}$$

By (10.9), (10.12), and (10.16),

$$f^{(2)}_{N,\beta}(x,y) \leq \theta \qquad \text{for } (x,y) \in E_N. \tag{10.20}$$

Hence,

$$\mathbb{Q}^{(2)}_{N,\beta}(\mu)(E_N) \leq \theta\mu^{\otimes 2}(E_N). \tag{10.21}$$

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Boundedness of the density gives $\mu^{\otimes 2}(E_N) \le C r_N^{\mathsf{d}}$. For the reverse bound, use $\mu \ge a$ on A, the change of variables $y = x + z$, and continuity of translations of $\mathbf{1}_A$ in $L^1(\mathbb{R}^{\mathsf{d}})$. We obtain

$$cN^{-\mathsf{d}/\mathsf{s}} \le a^2 \int_{|z|\le r_N} |A \cap (A - z)| \, \mathrm{d}z \le \mu^{\otimes 2}(E_N) \le CN^{-\mathsf{d}/\mathsf{s}}. \tag{10.22}$$

Therefore,

$$\chi^2(\mathbb{Q}^{(2)}_{N,\beta}(\mu) \mid \mu^{\otimes 2}) \ge (1-\theta)^2 \mu^{\otimes 2}(E_N) \ge cN^{-\mathsf{d}/\mathsf{s}}. \tag{10.23}$$

Since $\mathsf{d}/\mathsf{s} < 2$, (10.15) follows. □

We retain the close-pair data $A, a, \rho_*, \theta, r_N, E_N$ from the preceding proof. The estimates (10.21) and (10.22) will be used again in Section 10.4.1.

10.2.2. *The critical case.* At the critical exponent $\mathsf{s} = \mathsf{d}/2$, the set $|x - y| \lesssim N^{-1/\mathsf{s}}$ has product probability only of order N^{-2}. The additional logarithm comes from the range of transition annuli on which the mutual Boltzmann factor differs perturbatively from one. The factorization (10.9) gives the pointwise inequality

$$(1 - f^{(2)}_{N,\beta})^2 \ge \frac{1}{2}\left(1 - e^{-\beta \widetilde{W}/N}\right)^2 - (\mathcal{R}_{N,\beta} - 1)^2 e^{-2\beta \widetilde{W}/N}. \tag{10.24}$$

The first term has integral of order $N^{-2} \log N$ across the transition annuli. Thus, an error of order $o(N^{-2} \log N)$ in the second term could not cancel the leading contribution. For the actual residual prefactor, however, the natural estimate approximates $\mathcal{R}_{N,\beta}(x, y)$ by a function of one variable, rather than by the constant one. We first isolate the scalar transition-annulus calculation used both here and in the mesoscopic critical argument.

Lemma 10.2 (Scalar transition-annulus estimate)**.** *For every fixed $\lambda > 0$,*

$$\inf_{v \in \mathbb{R}} \int_\epsilon^1 \left(1 - v e^{-\lambda t}\right)^2 t^{-3} \, \mathrm{d}t = \lambda^2 \log \frac{1}{\epsilon} + O_\lambda(1) \tag{10.25}$$

as $\epsilon \downarrow 0$.

Proof. Set

$$A_\epsilon := \int_\epsilon^1 t^{-3} \, \mathrm{d}t, \tag{10.26}$$

$$B_{\epsilon,\lambda} := \int_\epsilon^1 e^{-\lambda t} t^{-3} \, \mathrm{d}t, \tag{10.27}$$

$$C_{\epsilon,\lambda} := \int_\epsilon^1 e^{-2\lambda t} t^{-3} \, \mathrm{d}t. \tag{10.28}$$

The minimum in (10.25) equals $A_\epsilon - B^2_{\epsilon,\lambda}/C_{\epsilon,\lambda}$. Taylor expansion at zero gives

$$A_\epsilon = \frac{1}{2\epsilon^2} + O(1), \tag{10.29}$$

$$B_{\epsilon,\lambda} = \frac{1}{2\epsilon^2} - \frac{\lambda}{\epsilon} + \frac{\lambda^2}{2} \log \frac{1}{\epsilon} + O_\lambda(1), \tag{10.30}$$

$$C_{\epsilon,\lambda} = \frac{1}{2\epsilon^2} - \frac{2\lambda}{\epsilon} + 2\lambda^2 \log \frac{1}{\epsilon} + O_\lambda(1). \tag{10.31}$$

Dividing these expansions yields (10.25). □

The following lemma records the corresponding one-variable prefactor criterion.

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Lemma 10.3 (One-variable prefactor criterion)**.** *Let* $\mathsf{s} = \mathsf{d}/2$*, let* $\beta > 0$*, and suppose that* $\mu \geq m_B > 0$ *on a ball* $B \subset \mathbb{R}^{\mathsf{d}}$*. Let* $B' \Subset B$*, fix* $0 < \alpha < 1/\mathsf{s} = 2/\mathsf{d}$*, and let* $0 < c_- < c_+ < \infty$*. Define*

$$\mathcal{T}_N := \left\{(x,y) \in B' \times B : c_- N^{-1/\mathsf{s}} \leq |x-y| \leq c_+ N^{-\alpha}\right\}. \tag{10.32}$$

Assume that

$$\inf_{v \in L^2(B',\mu)} \int_{\mathcal{T}_N} |\mathcal{R}_{N,\beta}(x,y) - v(x)|^2 e^{-2\beta \widetilde{W}(x,y)/N} \, \mathrm{d}\mu(x)\mathrm{d}\mu(y) = o\left(N^{-2}\log N\right). \tag{10.33}$$

Then there are $c > 0$ *and* $N_0 < \infty$ *such that*

$$\chi^2(\mathbb{Q}^{(2)}_{N,\beta}(\mu) \mid \mu^{\otimes 2}) \geq cN^{-2}\log N \tag{10.34}$$

for every $N \geq N_0$*.*

Proof. For each N, choose $v_N \in L^2(B',\mu)$ such that the integral in (10.33) is at most the displayed infimum plus N^{-3}. By the hypothesis, this integral is $o(N^{-2}\log N)$. The pointwise identity

$$1 - f^{(2)}_{N,\beta}(x,y) = \left(1 - v_N(x) e^{-\beta\widetilde{W}(x,y)/N}\right) - \left(\mathcal{R}_{N,\beta}(x,y) - v_N(x)\right) e^{-\beta\widetilde{W}(x,y)/N} \tag{10.35}$$

holds independently of the domain of integration. Therefore,

$$\chi^2(\mathbb{Q}^{(2)}_{N,\beta}(\mu) \mid \mu^{\otimes 2}) \geq \frac{1}{2}\int_{\mathcal{T}_N} \left(1 - v_N(x) e^{-\beta\widetilde{W}(x,y)/N}\right)^2 \mathrm{d}\mu(x)\mathrm{d}\mu(y) - o\left(N^{-2}\log N\right). \tag{10.36}$$

It remains to lower bound the displayed integral.

To this end, choose

$$\bar{\alpha} \in \left(\max\left\{\alpha, \frac{1}{2\mathsf{s}}\right\}, \frac{1}{\mathsf{s}}\right). \tag{10.37}$$

Fix $c_{\mathrm{in}} > c_-$ and then choose $c_{\mathrm{out}} > 0$ so small that $x + r\omega \in B$ whenever $x \in B'$, $\omega \in \mathbb{S}^{\mathsf{d}-1}$, and $0 \leq r \leq c_{\mathrm{out}}$. For all sufficiently large N,

$$\overline{\mathcal{T}}_N := \left\{(x,y) \in B' \times B : c_{\mathrm{in}} N^{-1/\mathsf{s}} \leq |x-y| \leq c_{\mathrm{out}} N^{-\bar{\alpha}}\right\} \subset \mathcal{T}_N. \tag{10.38}$$

By (10.6), write

$$\widetilde{W}(x,y) = \frac{1}{\mathsf{s}}|x-y|^{-\mathsf{s}} + R_\mu(x,y), \qquad |R_\mu(x,y)| \leq 2H_\mu. \tag{10.39}$$

Set

$$\lambda := \frac{\beta}{\mathsf{s}c_{\mathrm{in}}^{\mathsf{s}}}, \tag{10.40}$$

$$\epsilon_N := \left(\frac{c_{\mathrm{in}}}{c_{\mathrm{out}}}\right)^{\mathsf{s}} N^{\bar{\alpha}\mathsf{s}-1}. \tag{10.41}$$

Under the change of variables

$$t = \left(\frac{c_{\mathrm{in}} N^{-1/\mathsf{s}}}{r}\right)^{\mathsf{s}}, \qquad r = c_{\mathrm{in}} N^{-1/\mathsf{s}} t^{-1/\mathsf{s}}, \tag{10.42}$$

the radial interval in $\overline{\mathcal{T}}_N$ becomes $\epsilon_N \leq t \leq 1$, and criticality $\mathsf{d}/\mathsf{s} = 2$ gives

$$r^{\mathsf{d}-1}\,\mathrm{d}r = \frac{c_{\mathrm{in}}^{\mathsf{d}}}{\mathsf{s}} N^{-2} t^{-3}\,\mathrm{d}t. \tag{10.43}$$

Define

$$\vartheta_N(x,\omega,t) := \exp\left\{-\frac{\beta}{N} R_\mu\left(x, x + c_{\mathrm{in}} N^{-1/\mathsf{s}} t^{-1/\mathsf{s}}\omega\right)\right\}. \tag{10.44}$$

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Then

$$\|\vartheta_N - 1\|_{L^\infty} \le \frac{C_{\beta,\mu}}{N}, \tag{10.45}$$

and the mutual Boltzmann factor is $e^{-\lambda t}\vartheta_N(x,\omega,t)$.

For $0 < \epsilon < 1$, define the perturbed coefficients

$$B^{\vartheta_N}_{\epsilon,\lambda}(x) \coloneqq \frac{1}{|\mathbb{S}^{\mathsf{d}-1}|}\int_{\mathbb{S}^{\mathsf{d}-1}}\int_\epsilon^1 e^{-\lambda t}\vartheta_N(x,\omega,t)t^{-3}\,\mathrm{d}t\,\mathrm{d}\omega, \tag{10.46}$$

$$C^{\vartheta_N}_{\epsilon,\lambda}(x) \coloneqq \frac{1}{|\mathbb{S}^{\mathsf{d}-1}|}\int_{\mathbb{S}^{\mathsf{d}-1}}\int_\epsilon^1 e^{-2\lambda t}\vartheta_N(x,\omega,t)^2t^{-3}\,\mathrm{d}t\,\mathrm{d}\omega. \tag{10.47}$$

Since $\mu \ge m_B$ on B, polar coordinates and (10.43) yield

$$\begin{aligned}\int_{\overline{\mathcal{T}}_N}&\left(1 - v_N(x)e^{-\beta\widetilde{W}(x,y)/N}\right)^2 \mathrm{d}\mu(x)\mathrm{d}\mu(y)\\ &\ge cN^{-2}\int_{B'}\left[A_{\epsilon_N} - 2v_N(x)B^{\vartheta_N}_{\epsilon_N,\lambda}(x) + v_N(x)^2C^{\vartheta_N}_{\epsilon_N,\lambda}(x)\right]\,\mathrm{d}x.\end{aligned} \tag{10.48}$$

For fixed x, the bracket is a quadratic polynomial in the scalar $v_N(x)$. Completing the square shows that its minimum over $v \in \mathbb{R}$ is

$$A_{\epsilon_N} - \frac{B^{\vartheta_N}_{\epsilon_N,\lambda}(x)^2}{C^{\vartheta_N}_{\epsilon_N,\lambda}(x)}. \tag{10.49}$$

For the unperturbed factor $\vartheta_N \equiv 1$, use $A_\epsilon, B_{\epsilon,\lambda}, C_{\epsilon,\lambda}$ from (10.26)–(10.28). By Lemma 10.2,

$$A_\epsilon - \frac{B^2_{\epsilon,\lambda}}{C_{\epsilon,\lambda}} = \lambda^2\log\frac{1}{\epsilon} + O_\lambda(1). \tag{10.50}$$

The bound (10.45) implies, uniformly in $x \in B'$,

$$|B^{\vartheta_N}_{\epsilon_N,\lambda}(x) - B_{\epsilon_N,\lambda}| + |C^{\vartheta_N}_{\epsilon_N,\lambda}(x) - C_{\epsilon_N,\lambda}| \le C_{\beta,\mu}N^{-1}\epsilon_N^{-2}. \tag{10.51}$$

Moreover,

$$C^{\vartheta_N}_{\epsilon_N,\lambda}(x) \asymp C_{\epsilon_N,\lambda} \asymp \epsilon_N^{-2}, \qquad |B^{\vartheta_N}_{\epsilon_N,\lambda}(x)| + B_{\epsilon_N,\lambda} \lesssim \epsilon_N^{-2}. \tag{10.52}$$

Consequently,

$$\left|\frac{B^{\vartheta_N}_{\epsilon_N,\lambda}(x)^2}{C^{\vartheta_N}_{\epsilon_N,\lambda}(x)} - \frac{B^2_{\epsilon_N,\lambda}}{C_{\epsilon_N,\lambda}}\right| \le C_{\beta,\mu}N^{-1}\epsilon_N^{-2}. \tag{10.53}$$

Because $\bar\alpha\mathsf{s} > 1/2$,

$$N^{-1}\epsilon_N^{-2} = CN^{1-2\bar\alpha\mathsf{s}} = o\left(\log\frac{1}{\epsilon_N}\right). \tag{10.54}$$

Also, $\log(1/\epsilon_N) = (1-\bar\alpha\mathsf{s})\log N + O(1)$. Therefore, (10.48)–(10.54) give

$$\int_{\mathcal{T}_N}\left(1 - v_N(x)e^{-\beta\widetilde{W}(x,y)/N}\right)^2 \mathrm{d}\mu(x)\mathrm{d}\mu(y) \ge cN^{-2}\log N. \tag{10.55}$$

Combining this lower bound with the negligible residual-prefactor error in (10.36) proves (10.34). □

The next two lemmas provide the inputs needed to verify the one-variable prefactor criterion for $\mathcal{R}_{N,\beta}$.

Lemma 10.4 (Uniform one-point density bound)**.** *For every $\beta_* < \infty$,*

$$\sup_{M\ge1}\sup_{0\le\beta'\le\beta_*}\|f^{(1)}_{M,\beta'}\|_{L^\infty(\mu)} \le e^{3\beta_*H_\mu}. \tag{10.56}$$

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Proof. Assume first that $M \geq 2$. The one-mark identity of Lemma 4.2 gives

$$f^{(1)}_{M,\beta'}(x) = \frac{e^{\beta'(h_\mu(x)-I_\mu)/M}\mathsf{M}_{M,1}(x)}{\int \Psi_{M,1}\,\mathrm{d}\mu}. \tag{10.57}$$

By (10.5),

$$e^{\beta'(h_\mu(x)-I_\mu)/M}\mathsf{M}_{M,1}(x) \leq e^{3\beta_* H_\mu}. \tag{10.58}$$

For each fixed configuration of the remaining particles, Jensen's inequality and Hoeffding centering imply

$$\int \Psi_{M,1}\,\mathrm{d}\mu \geq 1. \tag{10.59}$$

This proves the estimate for $M \geq 2$. For $M = 1$, the same argument without the remaining particles gives

$$f^{(1)}_{1,\beta'}(x) = \frac{e^{\beta'(h_\mu(x)-I_\mu)}}{\int e^{\beta'(h_\mu-I_\mu)}\,\mathrm{d}\mu} \leq e^{\beta_* H_\mu}, \tag{10.60}$$

which completes the proof. □

Lemma 10.5 (Critical Riesz translation modulus)**.** *Let $K \subset \mathbb{R}^{\mathsf{d}}$ be compact. For $0 < r < 1/2$, define the raw Riesz translation modulus*

$$\omega_K(r) \coloneqq \sup_{\substack{x,y\in K\\ |x-y|\leq r}} \int_{\mathbb{R}^{\mathsf{d}}} |W(x,z) - W(y,z)|\,\mathrm{d}\mu(z). \tag{10.61}$$

At the critical exponent $\mathsf{s} = \mathsf{d}/2$,

$$\omega_K(r) \leq C_K \begin{cases} r^{1/2}, & \mathsf{d} = 1,\\ r\log(e/r), & \mathsf{d} = 2,\\ r, & \mathsf{d} \geq 3. \end{cases} \tag{10.62}$$

Moreover, the centering identity gives

$$\sup_{\substack{x,y\in K\\ |x-y|\leq r}} \int_{\mathbb{R}^{\mathsf{d}}} |\widetilde{W}(x,z) - \widetilde{W}(y,z)|\,\mathrm{d}\mu(z) \leq 2\omega_K(r). \tag{10.63}$$

Proof. Set $\delta = |x-y| < 1/2$. On $\{|x-z| \leq 2\delta\} \cup \{|y-z| \leq 2\delta\}$, boundedness of the density gives a contribution of order $\delta^{\mathsf{d}-\mathsf{s}}$. On the region where both distances exceed 2δ and $|x-z| \leq 1$, the mean-value theorem gives

$$|W(x,z) - W(y,z)| \leq C\delta|x-z|^{-\mathsf{s}-1}. \tag{10.64}$$

Its integral is bounded by

$$C\delta \int_{2\delta}^{1} r^{\mathsf{d}-\mathsf{s}-2}\,\mathrm{d}r. \tag{10.65}$$

On the remaining region, the same mean-value estimate is bounded by $C\delta$, and integration uses only that μ has mass one. Thus, the raw translation modulus is of order $\delta^{\mathsf{d}-\mathsf{s}}$ if $\mathsf{d}-\mathsf{s} < 1$, $\delta\log(e/\delta)$ if $\mathsf{d}-\mathsf{s} = 1$, and δ if $\mathsf{d}-\mathsf{s} > 1$. Substituting $\mathsf{s} = \mathsf{d}/2$ proves (10.62). Finally,

$$\widetilde{W}(x,z) - \widetilde{W}(y,z) = W(x,z) - W(y,z) - h_\mu(x) + h_\mu(y), \tag{10.66}$$

and

$$|h_\mu(x) - h_\mu(y)| \leq \int |W(x,z) - W(y,z)|\,\mathrm{d}\mu(z), \tag{10.67}$$

which gives (10.63). □

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Lemma 10.6 (One-variable approximation of the residual prefactor)**.** *Let* $\mathsf{s} = \mathsf{d}/2$*, let* $\beta > 0$*, and suppose that* $\mu \geq m_B > 0$ *on a ball* B*. Let* $B' \Subset B$*, and choose*

$$\alpha \in \begin{cases} (1,2), & \mathsf{d} = 1, \\ (1/2,1), & \mathsf{d} = 2, \\ (2/(\mathsf{d}+2), 2/\mathsf{d}), & \mathsf{d} \geq 3. \end{cases} \tag{10.68}$$

Then there are constants $0 < c_- < c_+ < \infty$ *such that*

$$\inf_{v \in L^2(B',\mu)} \int_{\substack{(x,y)\in B'\times B \\ c_- N^{-1/\mathsf{s}} \leq |x-y| \leq c_+ N^{-\alpha}}} |\mathcal{R}_{N,\beta}(x,y) - v(x)|^2 e^{-2\beta \widetilde{W}(x,y)/N} \, \mathrm{d}\mu(x)\mathrm{d}\mu(y) = o\left(N^{-2}\log N\right). \tag{10.69}$$

Proof. Let $n = N-2$, $\gamma = \beta n/N$. Define

$$\mathsf{M}_N^{\Delta}(x) := \mathbb{E}_{\mathbb{Q}_{n,\gamma}(\mu)}\left[\exp\left\{-\frac{2\beta}{N}\sum_{j=1}^{n}\widetilde{W}(x,Y_j)\right\}\right] \tag{10.70}$$

and

$$v_N(x) := \left(\int_{(\mathbb{R}^{\mathsf{d}})^2} \Psi_{N,2}\, \mathrm{d}\mu^{\otimes 2}\right)^{-1} \exp\left\{\frac{2\beta}{N}\big(h_\mu(x) - I_\mu\big)\right\} \mathsf{M}_N^{\Delta}(x). \tag{10.71}$$

The one-body factors in (10.8) and (10.71) differ by at most $C_{\beta,\mu}/N$, uniformly for $(x,y) \in B \times B$. For the reservoir factors, the mean-value theorem and (10.5) give

$$|\mathsf{M}_{N,2}(x,y) - \mathsf{M}_N^{\Delta}(x)| \leq \frac{C_{\beta,\mu}}{N}\mathbb{E}_{\mathbb{Q}_{n,\gamma}(\mu)}\left[\sum_{j=1}^{n}|\widetilde{W}(y,Y_j) - \widetilde{W}(x,Y_j)|\right]. \tag{10.72}$$

By exchangeability, Lemma 10.4 with $\beta_* = \beta$, and Lemma 10.5,

$$|\mathsf{M}_{N,2}(x,y) - \mathsf{M}_N^{\Delta}(x)| \leq C_{\beta,\mu}\omega_B(|x-y|). \tag{10.73}$$

Since $\int \Psi_{N,2}\, \mathrm{d}\mu^{\otimes 2} \geq 1$ by (10.10) and all remaining factors are uniformly bounded, it follows that

$$|\mathcal{R}_{N,\beta}(x,y) - v_N(x)| \leq C_{\beta,\mu}\left(\omega_B(|x-y|) + N^{-1}\right) \qquad \text{for } (x,y) \in B \times B. \tag{10.74}$$

Also, (10.5) gives

$$e^{-2\beta\widetilde{W}(x,y)/N} \leq e^{4\beta H_\mu/N}. \tag{10.75}$$

Choose any fixed $0 < c_- < c_+ < \infty$. By (10.74)–(10.75), the integral in (10.69), evaluated at v_N, is bounded by

$$C_{\beta,\mu}\int_{\substack{(x,y)\in B'\times B \\ |x-y|\leq c_+N^{-\alpha}}} \left(\omega_B(|x-y|)^2 + N^{-2}\right) \mathrm{d}x\,\mathrm{d}y. \tag{10.76}$$

Using (10.62), the contribution of ω_B^2 is

$$\begin{cases} O(N^{-2\alpha}), & \mathsf{d} = 1, \\ O(N^{-4\alpha}(\log N)^2), & \mathsf{d} = 2, \\ O(N^{-\alpha(\mathsf{d}+2)}), & \mathsf{d} \geq 3, \end{cases} \tag{10.77}$$

whereas the constant term is $O(N^{-2-\alpha\mathsf{d}})$. The range (10.68) makes every contribution $o(N^{-2}\log N)$, which proves (10.69). □

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Proposition 10.7 (Critical two-point obstruction)**.** *Let* $\mathsf{s} = \mathsf{d}/2$*, let* $\beta > 0$*, and suppose that* $\mu \geq m_B > 0$ *on a ball. Then there are constants* $c > 0$ *and* $N_0 < \infty$ *such that, for every* $N \geq N_0$*,*

$$\chi^2(\mathbb{Q}^{(2)}_{N,\beta}(\mu) \mid \mu^{\otimes 2}) \geq cN^{-2}\log N. \tag{10.78}$$

In particular,

$$N^2\chi^2(\mathbb{Q}^{(2)}_{N,\beta}(\mu) \mid \mu^{\otimes 2}) \longrightarrow +\infty. \tag{10.79}$$

Proof. Let B be a ball on which $\mu \geq m_B > 0$, and choose $B' \Subset B$. Select α in the range (10.68). Lemma 10.6 verifies the hypothesis of Lemma 10.3, which gives (10.78). The divergence in (10.79) is immediate. □

10.3. **Fixed-particle consequences for all three divergences.** The preceding propositions give two-point chi-square estimates. Monotonicity under coordinate projection transfers them to every fixed number of observed particles, while the uniform upper bound (10.13) converts the two-point chi-square estimate into relative-entropy and total-variation lower bounds.

Corollary 10.8 (Fixed-particle ultraviolet lower bounds)**.** *Let* $\beta > 0$ *and fix an integer* $k \geq 2$*. If* $\mathsf{d}/2 < \mathsf{s} < \mathsf{d}$*, assume the standing Riesz hypotheses. If* $\mathsf{s} = \mathsf{d}/2$*, assume in addition that* $\mu \geq m_B > 0$ *on some ball. Then there are positive constants* $c_\chi, c_H, c_{\mathrm{TV}}$ *such that, for all sufficiently large* N*,*

$$\chi^2(\mathbb{Q}^{(k)}_{N,\beta}(\mu) \mid \mu^{\otimes k}) \geq c_\chi \mathfrak{a}_N, \tag{10.80}$$

$$\mathsf{H}(\mathbb{Q}^{(k)}_{N,\beta}(\mu) \mid \mu^{\otimes k}) \geq c_H \mathfrak{a}_N, \tag{10.81}$$

$$\mathrm{TV}(\mathbb{Q}^{(k)}_{N,\beta}(\mu), \mu^{\otimes k}) \geq c_{\mathrm{TV}} \mathfrak{a}_N. \tag{10.82}$$

Proof. Coordinate projection from k labels to two and monotonicity of f-divergences give

$$\chi^2(\mathbb{Q}^{(k)}_{N,\beta}(\mu) \mid \mu^{\otimes k}) \geq \chi^2(\mathbb{Q}^{(2)}_{N,\beta}(\mu) \mid \mu^{\otimes 2}). \tag{10.83}$$

Thus, (10.80) follows from Propositions 10.1 and 10.7.

By (10.13), $0 \leq f^{(2)}_{N,\beta} \leq e^{6\beta H_\mu}$. With the convention $0\log 0 = 0$, compactness of $[0, e^{6\beta H_\mu}]$ and the quadratic expansion at one give a constant $c_{\beta,\mu} > 0$ such that

$$t\log t - t + 1 \geq c_{\beta,\mu}(t-1)^2 \qquad \text{for every } 0 \leq t \leq e^{6\beta H_\mu}, \tag{10.84}$$

and

$$(t-1)^2 \leq (e^{6\beta H_\mu} + 1)|t-1| \qquad \text{for every } 0 \leq t \leq e^{6\beta H_\mu}. \tag{10.85}$$

Since $\int (f^{(2)}_{N,\beta} - 1)\,\mathrm{d}\mu^{\otimes 2} = 0$,

$$\mathsf{H}(\mathbb{Q}^{(2)}_{N,\beta}(\mu) \mid \mu^{\otimes 2}) \geq c_{\beta,\mu}\chi^2(\mathbb{Q}^{(2)}_{N,\beta}(\mu) \mid \mu^{\otimes 2}), \tag{10.86}$$

$$\mathrm{TV}(\mathbb{Q}^{(2)}_{N,\beta}(\mu), \mu^{\otimes 2}) \geq \frac{1}{2(e^{6\beta H_\mu}+1)}\chi^2(\mathbb{Q}^{(2)}_{N,\beta}(\mu) \mid \mu^{\otimes 2}). \tag{10.87}$$

Relative entropy and total variation are also monotone under coordinate projection. Applying Propositions 10.1 and 10.7 in (10.86)–(10.87), and then projecting from k labels to two, proves (10.81)–(10.82). □

10.4. **Mesoscopic total-variation obstructions.** Recall the common scale $\mathfrak{a}_N$ from (2.38). In both the critical and strictly supercritical regimes, we prove

$$\mathrm{TV}(\mathbb{Q}^{(k)}_{N,\beta}(\mu), \mu^{\otimes k}) \geq ck^2\mathfrak{a}_N \qquad \text{for } 2 \leq k \leq c_0\mathfrak{a}_N^{-1/2}. \tag{10.88}$$

The scale is the same, but the mechanisms are different. In the strictly supercritical case, we take a union of the depleted close-pair events constructed in Section 10.2.1. At criticality, we accumulate one-step conditional Hellinger deficits along the chain of labeled marginals.

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10.4.1. *The strictly supercritical case: close-pair accumulation.* The strictly supercritical two-point argument becomes stronger when the number of observed labels grows. At the birthday-paradox scale, the product law sees at least one close pair with probability of order one, whereas the Gibbs law suppresses each such pair by the fixed factor in (10.21).

Proposition 10.9 (Strictly supercritical mesoscopic obstruction)**.** *Let* $\mathsf{d}/2 < \mathsf{s} < \mathsf{d}$ *and* $\beta > 0$. *There are constants* $c_0, c_1 > 0$ *and* $N_0 < \infty$ *such that, for every* $N \geq N_0$ *and every integer*

$$2 \leq k \leq c_0 \mathfrak{a}_N^{-1/2}, \tag{10.89}$$

one has

$$\mathrm{TV}(\mathbb{Q}_{N,\beta}^{(k)}(\mu), \mu^{\otimes k}) \geq c_1 k^2 \mathfrak{a}_N. \tag{10.90}$$

Proof. Fix the set A, the radii r_N, and the events E_N from the proof of Proposition 10.1. The estimates (10.21) and (10.22) give

$$\mathbb{Q}_{N,\beta}^{(2)}(\mu)(E_N) \leq \theta \mu^{\otimes 2}(E_N), \tag{10.91}$$

$$cN^{-\mathsf{d}/\mathsf{s}} \leq \mu^{\otimes 2}(E_N) \leq CN^{-\mathsf{d}/\mathsf{s}}, \qquad 0 < \theta < 1. \tag{10.92}$$

For $1 \leq i < j \leq k$, define

$$E_{ij} \coloneqq \{(x_1, \dots, x_k) : (x_i, x_j) \in E_N\}, \tag{10.93}$$

$$E_{N,k} \coloneqq \bigcup_{1 \leq i < j \leq k} E_{ij}. \tag{10.94}$$

For each pair $i < j$,

$$\mu^{\otimes k}(E_{ij}) = \mu^{\otimes 2}(E_N). \tag{10.95}$$

If the index pairs $\{i, j\}$ and $\{\ell, m\}$ are disjoint, then the corresponding events are independent. If they share one vertex, then, for example,

$$\begin{aligned} \mu^{\otimes k}(E_{12} \cap E_{13}) &= \int_A \mu(A \cap B(x, r_N))^2 \, \mathrm{d}\mu(x) \\ &\leq C r_N^{\mathsf{d}} \mu^{\otimes 2}(E_N) \leq C \mu^{\otimes 2}(E_N)^2, \end{aligned} \tag{10.96}$$

where the last inequality uses the lower bound in (10.22). The same upper bound holds for disjoint pairs. Bonferroni's inequality therefore gives

$$\mu^{\otimes k}(E_{N,k}) \geq \binom{k}{2} \mu^{\otimes 2}(E_N) - C \binom{k}{2}^2 \mu^{\otimes 2}(E_N)^2. \tag{10.97}$$

On the other hand, the union bound, exchangeability, and (10.21) give

$$\mathbb{Q}_{N,\beta}^{(k)}(\mu)(E_{N,k}) \leq \theta \binom{k}{2} \mu^{\otimes 2}(E_N). \tag{10.98}$$

Hence,

$$\mu^{\otimes k}(E_{N,k}) - \mathbb{Q}_{N,\beta}^{(k)}(\mu)(E_{N,k}) \geq \binom{k}{2} \mu^{\otimes 2}(E_N) \left(1 - \theta - C \binom{k}{2} \mu^{\otimes 2}(E_N)\right). \tag{10.99}$$

Choose $c_0 > 0$ sufficiently small. For $k \leq c_0 \mathfrak{a}_N^{-1/2}$, the upper bound in (10.22) makes the final error at most $(1 - \theta)/2$. The lower bound in the same display then yields

$$\mu^{\otimes k}(E_{N,k}) - \mathbb{Q}_{N,\beta}^{(k)}(\mu)(E_{N,k}) \geq c k^2 \mathfrak{a}_N. \tag{10.100}$$

The event characterization of total variation proves (10.90). □

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10.4.2. *The critical case: sequential Hellinger accumulation.*

Proposition 10.10 (Critical mesoscopic total-variation obstruction)**.** *Let* $\mathsf{s} = \mathsf{d}/2$*, let* $\beta > 0$*, and let* μ *be a probability measure on* $\mathbb{R}^{\mathsf{d}}$ *with bounded Lebesgue density. Assume that* $\mu \geq m_B > 0$ *on some ball. Then there are constants* $c_0, c_1 > 0$ *and* $N_0 \geq 3$*, depending only on* (β, μ, d)*, such that, for every* $N \geq N_0$ *and every integer*

$$2 \leq k \leq c_0 \mathfrak{a}_N^{-1/2}, \tag{10.101}$$

one has

$$\mathrm{TV}(\mathbb{Q}_{N,\beta}^{(k)}(\mu), \mu^{\otimes k}) \geq c_1 k^2 \mathfrak{a}_N. \tag{10.102}$$

Fix a ball $B \subset \mathbb{R}^{\mathsf{d}}$ and $m_B > 0$ such that

$$\mu(x) \geq m_B \qquad \text{for Lebesgue-a.e. } x \in B. \tag{10.103}$$

Since $W \geq 0$, the common bound (10.2) also gives $0 \leq h_\mu \leq H_\mu$.

The proof has three parts. First, an abstract Hellinger recursion converts one-step conditional deficits into a lower bound for the total variation of the k-particle marginal. Second, the raw-kernel marked-reservoir factorization shows that each isolated observed particle with controlled far-field interaction contributes a conditional deficit of order $\mathfrak{a}_N$. Third, a product-law estimate shows that a positive fraction of the observed particles satisfy these conditions in expectation.

Hellinger affinity and accumulation of conditional deficits.

We use Hellinger affinity at the iterative stage because the square root is compatible with disintegration. Indeed, the conditional factorization in (10.105) below yields an exact multiplicative recursion for the affinity. Total variation has no comparable multiplicative conditional identity. After accumulating the resulting affinity deficits, we return to total variation through (10.109).

For $0 \leq j \leq N$, recall the marginal density ratio $f_{N,\beta}^{(j)}$ from (2.12) and define its Hellinger affinity with the product law, also called the Bhattacharyya coefficient, by

$$\mathcal{A}_{N,j} \coloneqq \int \sqrt{f_{N,\beta}^{(j)}}\, \mathrm{d}\mu^{\otimes j}. \tag{10.104}$$

With the normalization in which the squared Hellinger distance is $\frac{1}{2}\int \left(\sqrt{\frac{\mathrm{d}P}{\mathrm{d}\lambda}} - \sqrt{\frac{\mathrm{d}Q}{\mathrm{d}\lambda}}\right)^2 \mathrm{d}\lambda$, it equals $1 - \mathcal{A}(P, Q)$; see [Tsy09, Section 2.4, pp. 83–86]. We use the conventions $\mathcal{A}_{N,0} = 1$ and $f_{N,\beta}^{(0)} = 1$.

For $0 \leq j < N$, let $q_{N,j}(x \mid X_j)$ denote the conditional density, relative to μ, of the $(j+1)$-st coordinate under $\mathbb{Q}_{N,\beta}^{(j+1)}(\mu)$ given $X_j = (x_1, \ldots, x_j)$.[2] Thus,

$$f_{N,\beta}^{(j+1)}(X_j, x) = f_{N,\beta}^{(j)}(X_j) q_{N,j}(x \mid X_j) \tag{10.105}$$

for $\mu^{\otimes(j+1)}$-a.e. (X_j, x). Define

$$\delta_{N,j}(X_j) \coloneqq 1 - \int \sqrt{q_{N,j}(x \mid X_j)}\, \mathrm{d}\mu(x) = \frac{1}{2}\int (\sqrt{q_{N,j}(x \mid X_j)} - 1)^2\, \mathrm{d}\mu(x) \tag{10.106}$$

and the probability measure

$$\mathrm{d}\nu_{N,j}(X_j) \coloneqq \frac{\sqrt{f_{N,\beta}^{(j)}(X_j)}}{\mathcal{A}_{N,j}}\, \mathrm{d}\mu^{\otimes j}(X_j). \tag{10.107}$$

[2]For each pair (N, j), we take the explicit conditional-density version below on the full-measure set of pairwise-distinct marked configurations and extend it arbitrarily elsewhere. Since the family of pairs (N, j) is countable and every square-root bridge introduced below is absolutely continuous with respect to $\mu^{\otimes j}$, these versions may be chosen on one common full-measure set.

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We call $\nu_{N,j}$ the *square-root bridge* because its density is the normalized geometric mean $\sqrt{f^{(j)}_{N,\beta}\cdot 1}$ of the Gibbs marginal density and the product density. It is precisely the probability measure under which the one-step conditional affinity deficits are averaged in the recursion below.

Lemma 10.11 (Affinity recursion, bridge estimate, and accumulation criterion)**.** *For every $0 \le j < N$,*

$$\mathcal{A}_{N,j+1} = \mathcal{A}_{N,j}(1 - \mathbb{E}_{\nu_{N,j}}[\delta_{N,j}]). \tag{10.108}$$

Moreover,

$$\mathrm{TV}(\mathbb{Q}^{(j)}_{N,\beta}(\mu), \mu^{\otimes j}) \ge 1 - \mathcal{A}_{N,j}, \tag{10.109}$$

and

$$\mathrm{TV}(\nu_{N,j}, \mu^{\otimes j}) \le \frac{1}{2}\sqrt{\mathcal{A}_{N,j}^{-2} - 1}. \tag{10.110}$$

Suppose, for some $N \ge 3$ and $k \ge 2$, that measurable functions $G_{N,j}$, $1 \le j < k$, and constants $c_\delta, c_G > 0$ satisfy

$$0 \le G_{N,j} \le j, \tag{10.111}$$

$$\delta_{N,j} \ge c_\delta \mathfrak{a}_N G_{N,j}, \tag{10.112}$$

$$\mathbb{E}_{\mu^{\otimes j}}[G_{N,j}] \ge c_G j. \tag{10.113}$$

Then there are constants $c_0, c_1 > 0$, depending only on c_δ and c_G, such that

$$\mathrm{TV}(\mathbb{Q}^{(k)}_{N,\beta}(\mu), \mu^{\otimes k}) \ge c_1 k^2 \mathfrak{a}_N \qquad \text{whenever } 2 \le k \le c_0 \mathfrak{a}_N^{-1/2}. \tag{10.114}$$

Proof. Using (10.105) and then (10.107),

$$\begin{aligned}\mathcal{A}_{N,j+1} &= \int \sqrt{f^{(j)}_{N,\beta}(X_j)} \left(\int \sqrt{q_{N,j}(x \mid X_j)}\, \mathrm{d}\mu(x) \right) \mathrm{d}\mu^{\otimes j}(X_j) \\ &= \mathcal{A}_{N,j}(1 - \mathbb{E}_{\nu_{N,j}}[\delta_{N,j}]).\end{aligned} \tag{10.115}$$

Since $\min\{t, 1\} \le \sqrt{t}$ for $t \ge 0$,

$$\mathrm{TV}(\mathbb{Q}^{(j)}_{N,\beta}(\mu), \mu^{\otimes j}) = 1 - \int \min\{f^{(j)}_{N,\beta}, 1\}\, \mathrm{d}\mu^{\otimes j} \ge 1 - \mathcal{A}_{N,j}. \tag{10.116}$$

The density of $\nu_{N,j}$ relative to $\mu^{\otimes j}$ is $\sqrt{f^{(j)}_{N,\beta}}/\mathcal{A}_{N,j}$, and hence

$$\chi^2(\nu_{N,j} \mid \mu^{\otimes j}) = \mathcal{A}_{N,j}^{-2} - 1. \tag{10.117}$$

The standard bound $\mathrm{TV} \le \frac{1}{2}\sqrt{\chi^2}$ gives (10.110).

It remains to prove the accumulation assertion. Choose $a_* \in (0, 1)$ sufficiently close to one that

$$\frac{1}{2}\sqrt{a_*^{-2} - 1} \le \frac{c_G}{8}. \tag{10.118}$$

If $\mathcal{A}_{N,j} < a_*$ for some $1 \le j < k$, then (10.108) shows that the affinities are nonincreasing, and (10.109) gives

$$\mathrm{TV}(\mathbb{Q}^{(k)}_{N,\beta}(\mu), \mu^{\otimes k}) \ge 1 - a_*. \tag{10.119}$$

Suppose instead that $\mathcal{A}_{N,j} \ge a_*$ for every $1 \le j < k$. Since $0 \le G_{N,j} \le j$, (10.110), (10.118), and (10.113) give

$$\mathbb{E}_{\nu_{N,j}}[G_{N,j}] \ge \mathbb{E}_{\mu^{\otimes j}}[G_{N,j}] - 2j\,\mathrm{TV}(\nu_{N,j}, \mu^{\otimes j}) \ge \frac{c_G}{2} j. \tag{10.120}$$

Combining this estimate with (10.112),

$$\mathbb{E}_{\nu_{N,j}}[\delta_{N,j}] \ge c j \mathfrak{a}_N, \qquad 1 \le j < k. \tag{10.121}$$

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Iteration of (10.108) and the inequality $1-u \le e^{-u}$ yield

$$\mathcal{A}_{N,k} \le \exp\left\{-c\mathfrak{a}_N \sum_{j=1}^{k-1} j\right\} \le \exp\left\{-c'k^2\mathfrak{a}_N\right\}. \tag{10.122}$$

Consequently,

$$\mathrm{TV}(\mathbb{Q}^{(k)}_{N,\beta}(\mu), \mu^{\otimes k}) \ge 1 - e^{-c'k^2\mathfrak{a}_N}. \tag{10.123}$$

Choose $c_0 > 0$ so that $c'c_0^2 \le 1$, and then choose

$$c_1 \le \min\left\{\frac{c'}{2}, \frac{1-a_*}{c_0^2}\right\}. \tag{10.124}$$

For $k \le c_0\mathfrak{a}_N^{-1/2}$, the right-hand side of (10.123) is at least $(c'/2)k^2\mathfrak{a}_N$, while (10.119) gives the same conclusion in the alternative case. This proves (10.114). □

Marked-reservoir factorization and uniform bounds.

The centered Gibbs exponent can be expanded back into the nonnegative raw Riesz interaction. Up to a normalizing constant independent of the configuration,

$$f_{N,\beta}(X_N) \propto \exp\left\{\beta\sum_{i=1}^N h_\mu(x_i) - \frac{\beta}{N}\sum_{1\le i<\ell\le N} W(x_i,x_\ell)\right\}. \tag{10.125}$$

Indeed, this follows by inserting $\widetilde{W}(x,y) = W(x,y) - h_\mu(x) - h_\mu(y) + 2I_\mu$ in the modulated Gibbs exponent. The raw form (10.125) is advantageous because every interaction term in the exponential is nonpositive.

Fix $1 \le j < N$, $X_j = (x_1,\dots,x_j)$, and set $n = N-j-1$. When $n \ge 1$, let $\mathsf{R}^{X_j}_{N,j}$ be the probability measure on $(\mathbb{R}^{\mathsf{d}})^n$ whose density relative to $\mu^{\otimes n}$ is proportional to

$$\exp\left\{\beta\sum_{\ell=1}^n h_\mu(y_\ell) - \frac{\beta}{N}\sum_{1\le\ell<m\le n} W(y_\ell,y_m) - \frac{\beta}{N}\sum_{a=1}^j\sum_{\ell=1}^n W(x_a,y_\ell)\right\}. \tag{10.126}$$

For $n=0$, this is the point mass on the empty configuration. Set

$$H_{N,j}(x;X_j) := e^{\beta h_\mu(x)}\mathbb{E}_{\mathsf{R}^{X_j}_{N,j}}\left[\exp\left\{-\frac{\beta}{N}\sum_{\ell=1}^n W(x,Y_\ell)\right\}\right] \tag{10.127}$$

and

$$\mathcal{N}_{N,j}(X_j) := \int H_{N,j}(x;X_j)\exp\left\{-\frac{\beta}{N}\sum_{a=1}^j W(x_a,x)\right\}\mathrm{d}\mu(x). \tag{10.128}$$

A direct splitting of (10.125) gives the exact conditional density

$$q_{N,j}(x \mid X_j) = \frac{H_{N,j}(x;X_j)}{\mathcal{N}_{N,j}(X_j)}\exp\left\{-\frac{\beta}{N}\sum_{a=1}^j W(x_a,x)\right\}. \tag{10.129}$$

Lemma 10.12 (Uniform marked-reservoir bounds)**.** *For every compact $K \subset \mathbb{R}^{\mathsf{d}}$, there is $C = C(\beta,\mu,K) < \infty$ such that, for every $N \ge 2$, every $1 \le j < N$, and $\mu^{\otimes j}$-a.e. X_j,*

$$C^{-1} \le H_{N,j}(x;X_j) \le C \quad \text{for } \mu\text{-a.e. } x, \tag{10.130}$$

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$$C^{-1} \le \mathcal{N}_{N,j}(X_j) \le C. \tag{10.131}$$

Moreover,

$$0 \le q_{N,j}(x \mid X_j) \le C \quad \text{for } \mu\text{-a.e. } x, \tag{10.132}$$

and, for $x, x' \in K$ *with* $|x - x'| \le r$,

$$|H_{N,j}(x; X_j) - H_{N,j}(x'; X_j)| \le C\omega_K(r). \tag{10.133}$$

Proof. Suppose first that $n \ge 1$. Conditional on all reservoir coordinates except Y_ℓ, the density of Y_ℓ relative to μ is proportional to

$$\exp\left\{\beta h_\mu(y) - \frac{\beta}{N}\left(\sum_{m\neq\ell} W(y, Y_m) + \sum_{a=1}^{j} W(x_a, y)\right)\right\}. \tag{10.134}$$

The numerator in (10.134) is at most $e^{\beta H_\mu}$. By Jensen's inequality, its normalizing denominator is at least the exponential of

$$2\beta I_\mu - \frac{\beta}{N}\left(\sum_{m\neq\ell} h_\mu(Y_m) + \sum_{a=1}^{j} h_\mu(x_a)\right) \ge -\beta H_\mu, \tag{10.135}$$

because the two sums contain $n - 1 + j = N - 2$ terms. Thus, every one-coordinate conditional density, and hence every one-coordinate marginal density of $\mathsf{R}_{N,j}^{X_j}$, is bounded above by

$$e^{2\beta H_\mu}. \tag{10.136}$$

The upper bound on $H_{N,j}$ follows from $W \ge 0$ and (10.2). For the lower bound, Jensen's inequality and (10.136) give

$$\begin{aligned} H_{N,j}(x; X_j) &\ge \exp\left\{\beta h_\mu(x) - \frac{\beta n}{N}\int W(x, y)\, \mathrm{d}(\mathsf{R}_{N,j}^{X_j})^{(1)}(y)\right\} \\ &\ge \exp\left\{-\beta e^{2\beta H_\mu} H_\mu\right\}. \end{aligned} \tag{10.137}$$

The same bounds are immediate when $n = 0$. The upper bound for $\mathcal{N}_{N,j}$ follows from $H_{N,j} \le C$ and $W \ge 0$. For the lower bound, Jensen's inequality gives

$$\begin{aligned} \mathcal{N}_{N,j}(X_j) &\ge C^{-1}\int \exp\left\{-\frac{\beta}{N}\sum_{a=1}^{j} W(x_a, x)\right\}\mathrm{d}\mu(x) \\ &\ge C^{-1}\exp\left\{-\frac{\beta}{N}\sum_{a=1}^{j} h_\mu(x_a)\right\} \ge C^{-1}e^{-\beta H_\mu}. \end{aligned} \tag{10.138}$$

Equation (10.132) follows from (10.129), (10.130)–(10.131), and $W \ge 0$.

To prove (10.133), note that the marked-reservoir law is independent of the new point x, while the exponential in (10.127) belongs to $[0, 1]$. Therefore,

$$\begin{aligned} &\left|\mathbb{E}_{\mathsf{R}_{N,j}^{X_j}}\left[\exp\left\{-\frac{\beta}{N}\sum_{\ell=1}^{n} W(x, Y_\ell)\right\}\right] - \mathbb{E}_{\mathsf{R}_{N,j}^{X_j}}\left[\exp\left\{-\frac{\beta}{N}\sum_{\ell=1}^{n} W(x', Y_\ell)\right\}\right]\right| \\ &\quad\le \frac{\beta}{N}\sum_{\ell=1}^{n}\mathbb{E}_{\mathsf{R}_{N,j}^{X_j}}\left[|W(x, Y_\ell) - W(x', Y_\ell)|\right] \le C\omega_K(r). \end{aligned} \tag{10.139}$$

Also, $|h_\mu(x) - h_\mu(x')| \le \omega_K(r)$, and hence (10.133) follows from (10.127) and the boundedness of h_μ. □

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Conditional deficit from good anchors.

Choose concentric balls $B_0 \Subset B_1 \Subset B$, and fix α in the range (10.68). Set

$$\rho_N := N^{-\alpha}, \tag{10.140}$$

and fix $c_- > 0$ sufficiently small that $\beta(2\mathsf{s})^{-1}c_-^{-\mathsf{s}} > 1$. The upper inequalities in (10.68) ensure that $N^{-1/\mathsf{s}} \ll \rho_N$.

For $X_j = (x_1, \dots, x_j)$ and $1 \le a \le j$, define the truncated gradient load

$$\Lambda_{N,j,a}(X_j) := \frac{1}{N} \sum_{\substack{1 \le b \le j \\ b \ne a}} |x_b - x_a|^{-\mathsf{s}-1} \mathbf{1}_{\{|x_b - x_a| \ge 4\rho_N\}}. \tag{10.141}$$

A good anchor is an observed particle lying in the interior ball, isolated from the other observed particles at scale ρ_N, and subject to a controlled far-field gradient load. Precisely, call the index a a good anchor if

$$x_a \in B_0, \qquad \min_{b \ne a} |x_b - x_a| \ge 4\rho_N, \qquad \rho_N^{\mathsf{d}+2} \Lambda_{N,j,a}(X_j)^2 \le \eta \mathfrak{a}_N, \tag{10.142}$$

where $\eta > 0$ will be fixed sufficiently small. When $j = 1$, the other-particle sum is empty, $\Lambda_{N,1,1} = 0$, and the isolation condition is vacuous; equivalently, the minimum over the empty set is interpreted as $+\infty$. Let $G_{N,j}(X_j)$ denote the number of good anchors.

Lemma 10.13 (Good-anchor conditional deficit)**.** *After choosing $\eta > 0$ sufficiently small and then increasing N_0, for every $N \ge N_0$, every $1 \le j < N$, and $\mu^{\otimes j}$-a.e. X_j,*

$$\delta_{N,j}(X_j) \ge c\mathfrak{a}_N G_{N,j}(X_j). \tag{10.143}$$

Proof. Fix a good anchor a. For $x \in B(x_a, \rho_N)$, factor (10.129) as

$$\sqrt{q_{N,j}(x \mid X_j)} = U_{N,j,a}(x; X_j) \exp\left\{ -\frac{\beta}{2N} W(x_a, x) \right\}, \tag{10.144}$$

where

$$U_{N,j,a}(x; X_j) := \mathcal{N}_{N,j}(X_j)^{-1/2} H_{N,j}(x; X_j)^{1/2} \exp\left\{ -\frac{\beta}{2N} \sum_{b \ne a} W(x_b, x) \right\}. \tag{10.145}$$

By Lemma 10.12, $U_{N,j,a}$ is uniformly bounded. If $r = |x - x_a| \le \rho_N$, the isolation condition in (10.142) and the mean-value theorem give

$$|U_{N,j,a}(x; X_j) - U_{N,j,a}(x_a; X_j)| \le C(\omega_{B_1}(r) + r\Lambda_{N,j,a}(X_j)). \tag{10.146}$$

Indeed, the first term follows from (10.133); for the other marked particles, the segment joining x_a to x stays a fixed proportion of $|x_b - x_a|$ away from x_b, and $|\nabla W(z)| \le C|z|^{-\mathsf{s}-1}$.

Let

$$\mathcal{T}_{N,a} := \left\{ x : \ c_- N^{-1/\mathsf{s}} \le |x - x_a| \le \rho_N \right\}. \tag{10.147}$$

For large N, this annulus lies in B. Set $v_a = U_{N,j,a}(x_a; X_j)$ and $t(r) = \beta(2\mathsf{s}N)^{-1} r^{-\mathsf{s}}$. For $x \in \mathcal{T}_{N,a}$, (10.144) gives the exact decomposition

$$\sqrt{q_{N,j}(x \mid X_j)} - 1 = \left(v_a e^{-t(r)} - 1 \right) + (U_{N,j,a}(x; X_j) - v_a)\, e^{-t(r)}. \tag{10.148}$$

The local lower bound (10.103), polar coordinates around x_a, and the critical identity $\mathsf{d} = 2\mathsf{s}$ imply, with $\epsilon_N = \beta(2\mathsf{s})^{-1} N^{\alpha\mathsf{s}-1}$,

$$\begin{aligned} &\inf_{v \ge 0} \int_{\mathcal{T}_{N,a}} \left| v e^{-t(|x - x_a|)} - 1 \right|^2 \mathrm{d}\mu(x) \\ &\qquad \ge c N^{-2} \inf_{v \in \mathbb{R}} \int_{\epsilon_N}^1 \left| v e^{-t} - 1 \right|^2 t^{-3} \, \mathrm{d}t \ge c N^{-2} \log N = c\mathfrak{a}_N. \end{aligned} \tag{10.149}$$

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Here, the upper endpoint in the t-variable exceeds one by the choice of c_-, while $\epsilon_N \downarrow 0$ because $\alpha \mathsf{s} < 1$; the second inequality is Lemma 10.2 with $\lambda = 1$. On the other hand, (10.146) and $e^{-t(r)} \le 1$ give

$$\int_{\mathcal{T}_{N,a}} |U_{N,j,a}(x; X_j) - v_a|^2 \, e^{-2t(r)} \, \mathrm{d}\mu(x) \\ \le C \int_0^{\rho_N} \omega_{B_1}(r)^2 r^{\mathsf{d}-1} \, \mathrm{d}r + C\rho_N^{\mathsf{d}+2} \Lambda_{N,j,a}(X_j)^2. \tag{10.150}$$

Applying $(u+w)^2 \ge \frac{1}{2}u^2 - w^2$ to (10.148), and then using (10.149) and (10.150), gives

$$\int_{\mathcal{T}_{N,a}} (\sqrt{q_{N,j}(x \mid X_j)} - 1)^2 \, \mathrm{d}\mu(x) \ge c\mathfrak{a}_N \\ - C \int_0^{\rho_N} \omega_{B_1}(r)^2 r^{\mathsf{d}-1} \, \mathrm{d}r - C\rho_N^{\mathsf{d}+2} \Lambda_{N,j,a}(X_j)^2. \tag{10.151}$$

The first error is the translation-modulus contribution. By (10.62),

$$\int_0^{\rho_N} \omega_{B_1}(r)^2 r^{\mathsf{d}-1} \, \mathrm{d}r \le C \begin{cases} \rho_N^2, & \mathsf{d} = 1, \\ \rho_N^4 \log^2(e/\rho_N), & \mathsf{d} = 2, \\ \rho_N^{\mathsf{d}+2}, & \mathsf{d} \ge 3. \end{cases} \tag{10.152}$$

The lower inequalities in (10.68) make (10.152) equal to $o(\mathfrak{a}_N)$. The last term in (10.151) is at most $C\eta\mathfrak{a}_N$ by goodness. Taking η small proves a lower bound $c\mathfrak{a}_N$ on each good-anchor annulus.

The annuli associated with distinct good anchors are disjoint because their centers are separated by at least $4\rho_N$. Summing their contributions and using (10.106) proves (10.143). □

Product-law abundance of good anchors.

Proposition 10.14 (Uniform good-anchor count)**.** *There are $c_B > 0$, $c_0 > 0$, and N_0 such that, for every $N \ge N_0$ and every integer satisfying (10.101),*

$$\mathbb{E}_{\mu^{\otimes j}} [G_{N,j}] \ge c_B j, \qquad 1 \le j \le k. \tag{10.153}$$

Proof. For a fixed anchor a, the probability that $X_a \in B_0$ and another observed point lies within $4\rho_N$ is at most

$$Cj\rho_N^{\mathsf{d}}. \tag{10.154}$$

Conditional on $X_a = x \in B_0$, the remaining variables are independent. If $Y \sim \mu$, then boundedness of the density and the critical identity $2\mathsf{s} = \mathsf{d}$ give, uniformly in $x \in B_0$,

$$\mathbb{E}_\mu \left[|x - Y|^{-2\mathsf{s}-2} \mathbf{1}_{\{|x-Y| \ge 4\rho_N\}} \right] \le C\rho_N^{-2}, \tag{10.155}$$

$$\mathbb{E}_\mu \left[|x - Y|^{-\mathsf{s}-1} \mathbf{1}_{\{|x-Y| \ge 4\rho_N\}} \right] \le C \begin{cases} \rho_N^{-1/2}, & \mathsf{d} = 1, \\ \log(e/\rho_N), & \mathsf{d} = 2, \\ 1, & \mathsf{d} \ge 3. \end{cases} \tag{10.156}$$

The first estimate follows because the radial integrand is proportional to r^{-3}; the second follows by integrating $r^{\mathsf{d}/2-2}$. Expanding the square in (10.141) and using conditional independence, we obtain

$$\mathbb{E}_{\mu^{\otimes j}} \left[\Lambda_{N,j,a}(X_j)^2 \,\middle|\, X_a = x \right] \le \frac{C}{N^2} \left[j\rho_N^{-2} + j^2 \begin{cases} \rho_N^{-1}, & \mathsf{d} = 1, \\ \log^2(e/\rho_N), & \mathsf{d} = 2, \\ 1, & \mathsf{d} \ge 3. \end{cases} \right]. \tag{10.157}$$

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Markov's inequality and $\mathfrak{a}_N = N^{-2}\log N$ therefore show that the probability that $X_a \in B_0$ and the gradient-load condition in (10.142) fails is at most

$$\frac{C}{\log N}\left[j\rho_N^{\mathsf{d}} + j^2\rho_N^{\mathsf{d}+2}\begin{cases}\rho_N^{-1}, & \mathsf{d}=1,\\ \log^2(e/\rho_N), & \mathsf{d}=2,\\ 1, & \mathsf{d}\geq 3.\end{cases}\right]. \tag{10.158}$$

For $j \leq c_0 N/\sqrt{\log N}$, the expressions in (10.154) and (10.158) converge to zero uniformly in j. More explicitly, the potentially largest terms are bounded, up to logarithmic factors, by

$$\begin{cases} N^{1-\alpha} \quad \text{and} \quad N^{2-2\alpha}, & \mathsf{d}=1,\\ N^{1-2\alpha} \quad \text{and} \quad N^{2-4\alpha}, & \mathsf{d}=2,\\ N^{1-\alpha\mathsf{d}} \quad \text{and} \quad N^{2-\alpha(\mathsf{d}+2)}, & \mathsf{d}\geq 3.\end{cases} \tag{10.159}$$

Every exponent in (10.159) is negative by (10.68); for $\mathsf{d}\geq 3$, note that $2/(\mathsf{d}+2) > 1/\mathsf{d}$. Therefore, for all sufficiently large N, a fixed anchor is good with probability at least $\mu(B_0)/2$, uniformly in j throughout the stated window. Exchangeability yields (10.153) with $c_B = \mu(B_0)/2$. □

Proof of Proposition 10.10. By Lemma 10.13 and proposition 10.14, the functions $G_{N,j}$ satisfy (10.111)–(10.113) throughout the window (10.101), after decreasing c_0 and increasing N_0, if necessary. The accumulation conclusion (10.114) of Lemma 10.11 proves (10.102). □

Corollary 10.15 (Mesoscopic failure of all three classical local scales)**.** *Assume either that* $\mathsf{s} = \mathsf{d}/2$ *and* $\mu \geq m_B > 0$ *on some ball, or that* $\mathsf{d}/2 < \mathsf{s} < \mathsf{d}$*. For every sufficiently small fixed* $\kappa > 0$*, set*

$$k_N := \left\lfloor \kappa \mathfrak{a}_N^{-1/2} \right\rfloor. \tag{10.160}$$

Then

$$\liminf_{N\to\infty} \mathrm{TV}(\mathbb{Q}_{N,\beta}^{(k_N)}(\mu), \mu^{\otimes k_N}) > 0, \tag{10.161}$$

$$\liminf_{N\to\infty} \mathsf{H}(\mathbb{Q}_{N,\beta}^{(k_N)}(\mu) \mid \mu^{\otimes k_N}) > 0, \tag{10.162}$$

$$\liminf_{N\to\infty} \chi^2(\mathbb{Q}_{N,\beta}^{(k_N)}(\mu) \mid \mu^{\otimes k_N}) > 0. \tag{10.163}$$

Moreover,

$$\frac{k_N}{N} \asymp \begin{cases}(\log N)^{-1/2}, & \mathsf{s}=\mathsf{d}/2,\\ N^{\mathsf{d}/(2\mathsf{s})-1}, & \mathsf{d}/2<\mathsf{s}<\mathsf{d},\end{cases} \longrightarrow 0, \tag{10.164}$$

$$\frac{k_N^2}{N^2} \asymp \begin{cases}(\log N)^{-1}, & \mathsf{s}=\mathsf{d}/2,\\ N^{\mathsf{d}/\mathsf{s}-2}, & \mathsf{d}/2<\mathsf{s}<\mathsf{d},\end{cases} \longrightarrow 0. \tag{10.165}$$

Proof. The total-variation lower bound follows from Propositions 10.9 and 10.10. Pinsker's inequality and $\chi^2(\nu \mid \pi) \geq 4\,\mathrm{TV}(\nu,\pi)^2$ give the relative-entropy and chi-square conclusions. The final two limits follow from (2.38) and (10.160). □

Consequently, no estimate of the form

$$\mathrm{TV}(\mathbb{Q}_{N,\beta}^{(k)}(\mu), \mu^{\otimes k}) \leq C_{\beta,\mu}\frac{k}{N} \tag{10.166}$$

can hold uniformly in $1 \leq k \leq N$ throughout $\mathsf{d}/2 \leq \mathsf{s} < \mathsf{d}$.

Proof of Theorem 2.3. Part (i) is Corollary 10.8. Part (ii) follows from Propositions 10.9 and 10.10 and Corollary 10.15. □

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Appendix A. Weighted logarithmic cavity estimates

The statement of Lemma 7.4 is kept with the model verification in Section 7. We give its proof here so that the main one-particle argument is not interrupted by this secondary weighted extension.

Proof of Lemma 7.4. We first introduce the notation used only in this appendix. For $\theta \geq 0$, set

$$W_\theta(x) := e^{\theta L(x)} = (e + |x|)^\theta, \qquad a(x) := h_\mu(x) - 2I_\mu, \tag{A.1}$$

$$c := \mathsf{c}_{\log,\beta,\mu}, \qquad \Theta := pc, \qquad \Theta_* := p\Theta = p^2 c. \tag{A.2}$$

Define

$$G_*(x) := \left[\int_{\mathbb{R}^{\mathsf{d}}} |\widetilde{W}(x,y)|^p W_{\Theta_*}(y)\,\mathrm{d}\mu(y)\right]^{1/p}, \qquad A_*(x) := |a(x)|^2 + G_*(x)^2, \tag{A.3}$$

and

$$M_* := \int_{\mathbb{R}^{\mathsf{d}}} W_{\Theta_*}\,\mathrm{d}\mu. \tag{A.4}$$

The hypotheses (7.34) say precisely that

$$M_* < \infty, \qquad \|W_{2c}A_*\|_{L^2(\mu)} < \infty. \tag{A.5}$$

We shall prove the stronger pointwise estimate

$$\mathbb{E}_{\mathbb{Q}_{N,\beta}(\mu)}\left[e^{\lambda\Phi_N(x;\hat{\Xi}_1)} B_N(x;\hat{\Xi}_1)^2\right] \leq \frac{\mathsf{C}_{\mathrm{wt},\beta,p,\mu}}{N} W_{\lambda c}(x) A_*(x) \tag{A.6}$$

for every $N \geq 2$, every $\lambda \in \{0,1,2\}$, and μ-a.e. x, where the constant depends only on the quantities in (A.5), on p, β, and on $\mathfrak{P}_{r\beta}(\mu)$.

Step 1: Product-space reduction. Fix $N \geq 2$, $\lambda \in \{0,1,2\}$, and x. Set $s := \lambda p/2$ and $\eta_N := sc/N$. By Lemma 4.4 and Hölder's inequality with conjugate exponents $r = p/(p-2)$ and $p/2$,

$$\begin{aligned}&\mathbb{E}_{\mathbb{Q}_{N,\beta}(\mu)}\left[e^{\lambda\Phi_N(x;\hat{\Xi}_1)} B_N(x;\hat{\Xi}_1)^2\right] \\ &\quad\leq \mathfrak{P}_{r\beta}(\mu)^{1/r}\left\{\mathbb{E}_{\mu^{\otimes(N-1)}}\left[e^{s\Phi_N(x;\hat{\Xi}_1)}|B_N(x;\hat{\Xi}_1)|^p\right]\right\}^{2/p}.\end{aligned} \tag{A.7}$$

Since

$$e^{s\Phi_N(x;\hat{\Xi}_1)} = e^{sc} W_{sc}(x) \prod_{j=2}^{N} W_{\eta_N}(\xi_j), \tag{A.8}$$

let

$$Z_\eta := \int W_\eta\,\mathrm{d}\mu, \qquad \mathrm{d}\mu_\eta := Z_\eta^{-1} W_\eta\,\mathrm{d}\mu. \tag{A.9}$$

Log-convexity of $\eta \mapsto Z_\eta$, together with $Z_0 = 1$, $0 \leq \eta_N \leq \Theta/N$, and (A.5), gives

$$Z_{\eta_N}^{2(N-1)/p} \leq \max\{1, M_*\}^{2/p}. \tag{A.10}$$

Consequently, (A.7) yields

$$\begin{aligned}&\mathbb{E}_{\mathbb{Q}_{N,\beta}(\mu)}\left[e^{\lambda\Phi_N(x;\hat{\Xi}_1)} B_N(x;\hat{\Xi}_1)^2\right] \\ &\quad\leq C_{\beta,p,\mu} W_{\lambda c}(x)\left\{\mathbb{E}_{\mu_{\eta_N}^{\otimes(N-1)}}\left[|B_N(x;\hat{\Xi}_1)|^p\right]\right\}^{2/p}.\end{aligned} \tag{A.11}$$

Here and below, $C_{\beta,p,\mu}$ denotes a finite constant depending only on the quantities in (A.5), on p, β, and on $\mathfrak{P}_{r\beta}(\mu)$.

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Step 2: Moment estimate under the tilted product law. Write $k_x(y) \coloneqq \widetilde{W}(x,y)$. Since $\int k_x \,\mathrm{d}\mu = 0$, the mean under the tilted law satisfies

$$m_{\eta_N}(x) \coloneqq \int k_x(y)\,\mathrm{d}\mu_{\eta_N}(y) = Z_{\eta_N}^{-1}\int k_x(y)(W_{\eta_N}(y)-1)\,\mathrm{d}\mu(y). \tag{A.12}$$

Moreover, $Z_{\eta_N} \geq 1$ and

$$0 \leq W_{\eta_N}(y) - 1 \leq \eta_N L(y) W_{\eta_N}(y) \leq \eta_N L(y) W_\Theta(y). \tag{A.13}$$

If $c = 0$, then $\eta_N = 0$ and $m_{\eta_N} = 0$. If $c > 0$, Hölder's inequality and $\Theta_* = p\Theta$ give

$$\begin{aligned} |m_{\eta_N}(x)| &\leq \eta_N \int |k_x(y)| L(y) W_\Theta(y)\,\mathrm{d}\mu(y) \\ &\leq \eta_N G_*(x) \|L\|_{L^{p/(p-1)}(\mu)} \\ &\leq \frac{C_{\beta,p,\mu}}{N} G_*(x). \end{aligned} \tag{A.14}$$

The logarithmic moment is finite by (A.5), and the same bound is trivial when $c = 0$.

Also,

$$\left[\int |k_x(y)|^p \,\mathrm{d}\mu_{\eta_N}(y)\right]^{1/p} \leq G_*(x), \tag{A.15}$$

because $Z_{\eta_N} \geq 1$ and $W_{\eta_N} \leq W_{\Theta_*}$. Applying the scalar Rosenthal inequality under $\mu_{\eta_N}^{\otimes(N-1)}$ to the centered variables $k_x(\xi_j) - m_{\eta_N}(x)$, we have

$$\left\{\mathbb{E}_{\mu_{\eta_N}^{\otimes(N-1)}}\left[\left|\frac{1}{N}\sum_{j=2}^N (k_x(\xi_j) - m_{\eta_N}(x))\right|^p\right]\right\}^{2/p} \leq \frac{C_p}{N} G_*(x)^2. \tag{A.16}$$

Since

$$B_N(x;\hat{\Xi}_1) = \frac{\beta}{N}a(x) - \frac{\beta}{N}\sum_{j=2}^N (k_x(\xi_j) - m_{\eta_N}(x)) - \beta\frac{N-1}{N} m_{\eta_N}(x), \tag{A.17}$$

(A.14) and (A.16) give

$$\left\{\mathbb{E}_{\mu_{\eta_N}^{\otimes(N-1)}}\left[|B_N(x;\hat{\Xi}_1)|^p\right]\right\}^{2/p} \leq \frac{C_{\beta,p,\mu}}{N} A_*(x). \tag{A.18}$$

Combining (A.11) and (A.18) proves (A.6).

Step 3: Verification of the weighted cavity constants. It remains to check that (A.6) implies (3.26). Taking $\lambda = 1$ gives

$$N\|\mathbb{E}_{\mathbb{Q}_{N,\beta}(\mu)}\left[e^{\Phi_N(\cdot;\hat{\Xi}_1)} B_N(\cdot;\hat{\Xi}_1)^2\right]\|_{L^2(\mu)} \leq \mathsf{C}_{\mathrm{wt},\beta,p,\mu}\|W_c A_*\|_{L^2(\mu)} < \infty, \tag{A.19}$$

and similarly

$$N\mathbb{E}_{\mathbb{Q}_{N,\beta}(\mu)}\left[\int e^{\Phi_N(x;\hat{\Xi}_1)} B_N(x;\hat{\Xi}_1)^2\,\mathrm{d}\mu(x)\right] \leq \mathsf{C}_{\mathrm{wt},\beta,p,\mu}\int W_c A_*\,\mathrm{d}\mu < \infty. \tag{A.20}$$

Taking $\lambda = 0$ gives the unweighted B_N^2 estimate. Finally, since μ is a probability measure,

$$\left(\int e^{\Phi_N(x;\hat{\Xi}_1)}|B_N(x;\hat{\Xi}_1)|\,\mathrm{d}\mu(x)\right)^2 \leq \int e^{2\Phi_N(x;\hat{\Xi}_1)} B_N(x;\hat{\Xi}_1)^2\,\mathrm{d}\mu(x), \tag{A.21}$$

and the case $\lambda = 2$ gives the estimate defining $\mathsf{V}_{\log,\beta,\mu}$. The assumption $\|W_{2c}A_*\|_{L^2(\mu)} < \infty$ implies all the displayed integrability bounds above, and hence (3.26). □

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Appendix B. Dependence of constants

This appendix records one admissible choice of the constants in Theorem 3.4. The numerical values are not optimized; the purpose is to make the dependence on the quantitative hypotheses explicit. All essential suprema and infima are taken with respect to μ or $\mu^{\otimes 2}$, according to the number of variables.

Let R_p be an admissible constant in the scalar Rosenthal inequality, and let $\mathsf{A}_{2,p}$ be the fixed-order canonical U-statistic constant from Lemma 8.1. We also use the functions Λ_1 and Λ_2 defined in (4.52)–(4.53).

Let $\mathsf{R}_{\mathrm{cav},\beta,\mu}$ denote the cavity constant supplied by Lemma 4.8; explicit choices are given in (4.64) and (4.65). Define

$$\mathsf{U}_{\beta,\mu} \coloneqq 2\left(\beta\|h_\mu - 2I_\mu\|_{L^2(\mu)} + \mathsf{R}_{\mathrm{cav},\beta,\mu}\right), \tag{B.1}$$

$$\mathsf{C}^{\mathrm{up}}_{\chi,\beta,\mu} \coloneqq \mathsf{U}^2_{\beta,\mu} + (\mathfrak{P}_{2\beta}(\mu) - 1), \tag{B.2}$$

$$\mathsf{C}^{\mathrm{up}}_{\mathrm{TV},\beta,\mu} \coloneqq \frac{1}{2}\sqrt{\mathsf{C}^{\mathrm{up}}_{\chi,\beta,\mu}}, \qquad \mathsf{C}^{\mathrm{up}}_{\mathsf{H},\beta,\mu} \coloneqq \mathsf{C}^{\mathrm{up}}_{\chi,\beta,\mu}. \tag{B.3}$$

For the fixed-marginal branch, let $1 < \tau < 2$, $q = \tau/(\tau-1)$, and $p \geq 2q$. The constants $\mathsf{A}_{j,q}$ and B_q are those in Lemmas 8.1 and 8.2, with the admissible choice (8.4). Under the hypotheses of Theorem 3.6, use $\mathsf{M}_{\tau,\beta,\mu}$ and $\mathsf{U}_{\tau,\beta,\mu}$ from (3.49). Substitution into (8.21) gives exactly (3.50); no optimization of $\mathsf{A}_{j,q}$, B_q, or the fixed-k onset is intended.

For $\beta > 0$, recall $\mathsf{M}_{p,\mu}$ from (9.6) and $\mathsf{R}_{p,\mu}$, $\mathsf{a}_{\beta,\mu}$, and $\mathsf{J}_{\beta,\mu}$ from (9.17)–(9.19). The latter three constants are strictly positive. The lower-bound constants may be chosen as

$$\mathsf{N}_{\beta,\mu} \coloneqq \left\lceil \max\left\{2, \frac{2\beta\|h_\mu - 2I_\mu\|_{L^2(\mu)}\mathsf{U}_{\beta,\mu}}{\mathsf{J}_{\beta,\mu}}\right\}\right\rceil, \tag{B.4}$$

$$\mathsf{S}_{\beta,\mu} \coloneqq \frac{\mathsf{J}_{\beta,\mu}}{4\beta}, \tag{B.5}$$

$$\mathsf{R}_{\mathrm{TV},\beta,\mu} \coloneqq \max\left\{1, \left(\frac{2(\frac{\mathsf{A}_{2,p}}{2}\|\widetilde{W}\|_{L^p(\mu^{\otimes 2})})^{p/2}\sqrt{\mathsf{C}^{\mathrm{up}}_{\chi,\beta,\mu}}}{\mathsf{S}_{\beta,\mu}}\right)^{2/(p-2)}\right\}, \tag{B.6}$$

$$\mathsf{c}^{\mathrm{low}}_{\mathrm{TV},\beta,\mu} \coloneqq \frac{\mathsf{S}_{\beta,\mu}}{4\mathsf{R}_{\mathrm{TV},\beta,\mu}}, \tag{B.7}$$

$$\mathsf{c}^{\mathrm{low}}_{\mathsf{H},\beta,\mu} \coloneqq 2(\mathsf{c}^{\mathrm{low}}_{\mathrm{TV},\beta,\mu})^2, \tag{B.8}$$

$$\mathsf{c}^{\mathrm{low}}_{\chi,\beta,\mu} \coloneqq \frac{\mathsf{J}^2_{\beta,\mu}}{4\beta^2\|\widetilde{W}\|^2_{L^2(\mu^{\otimes 2})}}. \tag{B.9}$$

Appendix C. Probability estimates

The bounded-feature estimate controls the positive correction in the two-sided response. The martingale estimate gives the canonical-profile moments and the mixed-coordinate bounds in the smooth Gaussian comparison. That comparison is used both by the signed Laplace theorem and by the absolute-moment approximation for growing total variation.

C.1. **Positive exponentials of bounded-feature statistics.** The following estimate controls the bounded positive correction for negative response parameters in Section 5.4. It is not applied to the signed core interaction.

Lemma C.1 (Small positive exponential of a bounded-feature statistic)**.** *Let $Y_1, \dots, Y_N$ be independent copies of a centered random variable in a separable real Hilbert space, with $\|Y_i\|^2 \leq K$. If*

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$4aK < 1$ and $a \geq 0$, then

$$\mathbb{E}\left[\exp\left\{\frac{a}{2N}\sum_{i\neq j}\langle Y_i, Y_j\rangle\right\}\right] \leq (1-4aK)^{-1/2}. \tag{C.1}$$

Proof. For $u \geq 0$, introduce independent copies Y_i'. Convexity of $z \mapsto e^{u\|z\|^2/N}$ and conditional Jensen give

$$\mathbb{E}\left[e^{u\|\sum_i Y_i\|^2/N}\right] \leq \mathbb{E}\left[e^{u\|\sum_i (Y_i - Y_i')\|^2/N}\right]. \tag{C.2}$$

The independent differences are symmetric. Conditional on their values d_i, insert independent Rademacher signs ϵ_i. On the finite-dimensional span of the d_i, let G be a standard Gaussian vector. The Gaussian moment-generating identity and $\cosh z \leq e^{z^2/2}$ give

$$\begin{aligned}\mathbb{E}_\epsilon\left[e^{u\|\sum_i \epsilon_i d_i\|^2/N}\right] &= \mathbb{E}_G\left[\prod_i \cosh\left(\sqrt{2u/N}\langle G, d_i\rangle\right)\right] \\ &\leq \mathbb{E}_G\left[e^{(u/N)\sum_i \langle G, d_i\rangle^2}\right].\end{aligned} \tag{C.3}$$

Let $\lambda_j \geq 0$ be the eigenvalues of $\sum_i d_i \otimes d_i$. Their sum is at most $4NK$. If $8uK < 1$, the last expectation is

$$\prod_j (1 - 2u\lambda_j/N)^{-1/2} \leq \left(1 - \frac{2u}{N}\sum_j \lambda_j\right)^{-1/2} \leq (1-8uK)^{-1/2}. \tag{C.4}$$

The first inequality follows from $\prod_j (1-t_j) \geq 1 - \sum_j t_j$ for nonnegative t_j with sum less than one, proved by induction. All calculations take place in a space of dimension at most N, so no infinite-dimensional standard Gaussian measure is being assumed.

Finally, $\sum_{i\neq j}\langle Y_i, Y_j\rangle = \|\sum_i Y_i\|^2 - \sum_i \|Y_i\|^2 \leq \|\sum_i Y_i\|^2$. Apply (C.2)–(C.4) with $u = a/2$. □

C.2. **A martingale moment inequality.**

Lemma C.2 (A scalar martingale moment bound)**.** *Let $(D_j)_{j=1}^m$ be real martingale differences in L^p, where $2 \leq p < \infty$. Then*

$$\left\|\sum_{j=1}^m D_j\right\|_p^2 \leq (p-1)\sum_{j=1}^m \|D_j\|_p^2. \tag{C.5}$$

The initial sigma algebra may be nontrivial.

Proof. For $p = 2$, the result is orthogonality. Suppose $p > 2$. Let S be measurable with respect to a sigma algebra $\mathcal{F}$, let $D \in L^p$ satisfy $\mathbb{E}[D \mid \mathcal{F}] = 0$, and suppose $S \in L^p$. For $\epsilon > 0$, put

$$A_\epsilon(t) = \mathbb{E}[|S + tD|^p] + \epsilon, \qquad J_\epsilon(t) = A_\epsilon(t)^{2/p}, \qquad 0 \leq t \leq 1. \tag{C.6}$$

Differentiation twice is valid by Hölder and domination by powers of $|S| + |D|$. If $V = S + tD$, direct calculation gives

$$\begin{aligned}J_\epsilon''(t) &= 2(2-p)A_\epsilon(t)^{2/p-2}\left(\mathbb{E}[|V|^{p-2}VD]\right)^2 \\ &\quad + 2(p-1)A_\epsilon(t)^{2/p-1}\mathbb{E}[|V|^{p-2}D^2] \leq 2(p-1)\|D\|_p^2.\end{aligned} \tag{C.7}$$

The first term is nonpositive. By Hölder, the remaining expectation is at most $(\mathbb{E}[|V|^p])^{(p-2)/p}\|D\|_p^2$. This cancels the nonpositive power of A_ϵ. Moreover,

$$J_\epsilon'(0) = 2A_\epsilon(0)^{2/p-1}\mathbb{E}[|S|^{p-2}SD] = 0, \tag{C.8}$$

by conditional centering. Integrating (C.7) twice and sending $\epsilon \downarrow 0$ gives $\|S + D\|_p^2 \leq \|S\|_p^2 + (p-1)\|D\|_p^2$. Apply this successively with $S = \sum_{i<j} D_i$ to obtain (C.5). This is the usual L^p smoothness argument; the underlying differential inequality is also given in [Pin94, Proposition 2.1]. □

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C.3. **Gaussian quadratic forms and smooth comparison.**

C.3.1. *The centered Gaussian quadratic form.* Let (S, ρ) be a standard Borel probability space and let $c \in L^2(\rho^{\otimes 2})$ be real, symmetric, and canonical, with Hilbert–Schmidt operator C on $L^2_0(\rho)$. For a finite-rank self-adjoint operator with matrix (C_{rs}) in an orthonormal basis of a finite-dimensional containing subspace, define

$$\mathcal{Q}_C = \sum_{r,s} C_{rs}(Z_r Z_s - \delta_{rs}), \tag{C.9}$$

using the coordinates of a single isonormal Gaussian process on $L^2_0(\rho)$. The elementary fourth-moment identity for independent standard Gaussians gives

$$\mathbb{E}[(Z_r Z_s - \delta_{rs})(Z_u Z_v - \delta_{uv})] = \delta_{ru}\delta_{sv} + \delta_{rv}\delta_{su}, \qquad \mathbb{E}[\mathcal{Q}_C \mathcal{Q}_D] = 2\,\mathrm{Tr}(CD). \tag{C.10}$$

The first identity follows by independence when an index occurs only once, by $\mathbb{E}[Z^2] = 1$ when indices occur in two distinct pairs, and by $\mathbb{E}[Z^4] = 3$ when all indices agree. The second follows by expanding the finite sums and using symmetry. Orthogonal changes of basis preserve (C.9). Finite-rank approximation in Hilbert–Schmidt norm thus extends the definition to all self-adjoint Hilbert–Schmidt C, and

$$\|\mathcal{Q}_C - \mathcal{Q}_D\|_2 = \sqrt{2}\|C - D\|_{\mathrm{HS}}. \tag{C.11}$$

Diagonalizing C gives the law $\sum_j \lambda_j(C)(Z_j^2 - 1)$ in L^2. This construction also justifies (2.33).

C.3.2. *Proof of the smooth comparison.*

Proof of Lemma 5.1. Set $p = 2 + \alpha$. All kernel norms in this proof are relative to $\rho^{\otimes 2}$.

Finite partitions and mixed moments. Choose increasing finite measurable partitions whose generated sigma algebra is the Borel sigma algebra modulo ρ. Let c_m be the conditional expectation of c onto the corresponding product partition, ignoring zero-mass cells. Then c_m is symmetric and canonical, and

$$\|c_m\|_p \le \|c\|_p, \qquad \|c_m - c\|_2 \longrightarrow 0. \tag{C.12}$$

The first assertion is conditional Jensen; the second follows by density of product-partition simple functions in $L^2(\rho^{\otimes 2})$. Integrating the conditional expectation in either variable proves canonicity. An orthonormal basis $\phi_1, \dots, \phi_\ell$ of the centered functions constant on the partition cells gives

$$c_m(x, y) = \phi(x)^{\mathsf{T}} A \phi(y), \qquad \mathbb{E}[\phi(x)] = 0, \qquad \mathbb{E}[\phi(x)\phi(x)^{\mathsf{T}}] = \mathrm{Id}, \qquad \|A\|_{\mathrm{HS}} = \|c_m\|_2, \tag{C.13}$$

where A is real symmetric. There is no restriction on its sign. A zero-dimensional centered subspace gives the zero kernel.

Let $\xi = \phi(x)$, and let g be an independent standard Gaussian in $\mathbb{R}^\ell$. If V, W are independent and each has either law ξ or law g, then

$$\|V^{\mathsf{T}} A W\|_p \le C_p \|c_m\|_p. \tag{C.14}$$

For two original vectors this is the kernel norm. For one original and one Gaussian vector, conditioning on ξ gives, with Z a scalar standard Gaussian,

$$\begin{aligned} \mathbb{E}[|\xi^{\mathsf{T}} A g|^p] &= \mathbb{E}[|Z|^p] \int \left(\int |c_m(x, y)|^2 \, \mathrm{d}\rho(y) \right)^{p/2} \mathrm{d}\rho(x) \\ &\le \mathbb{E}[|Z|^p] \|c_m\|_p^p. \end{aligned} \tag{C.15}$$

For two Gaussian vectors, condition on the first and diagonalize A. Minkowski in $L^{p/2}$ bounds $\||Ag|\|_p$ by $\|Z\|_p \|A\|_{\mathrm{HS}}$. Hence,

$$\|g^{\mathsf{T}} A g'\|_p \le \|Z\|_p^2 \|A\|_{\mathrm{HS}} \le \|Z\|_p^2 \|c_m\|_p. \tag{C.16}$$

This proves (C.14) with a dimension-independent constant.

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Replacement of one coordinate at a time. For independent vectors $V_1, \dots, V_N$ with these two possible laws, set

$$U(V_1, \dots, V_N) := \frac{1}{N} \sum_{i<j} V_i^{\mathsf{T}} A V_j. \tag{C.17}$$

When replacing $V_i = \xi_i$ by g_i, condition on all the other vectors. The sum is $z + y$, where z does not involve V_i and $y = N^{-1} V_i^{\mathsf{T}} A \sum_{j \neq i} V_j$. The two candidate laws have the same conditional first and second moments:

$$\mathbb{E}[y \mid (V_j)_{j \neq i}] = 0, \qquad \mathbb{E}[y^2 \mid (V_j)_{j \neq i}] = N^{-2} \left| A \sum_{j \neq i} V_j \right|^2. \tag{C.18}$$

To estimate its p-moment, reveal V_i first and the remaining vectors afterward. The terms $V_i^{\mathsf{T}} A V_j$ are martingale differences. Lemma C.2 and (C.14) imply

$$\|y\|_p \leq C_p N^{-1/2} \|c_m\|_p. \tag{C.19}$$

The fractional Taylor remainder satisfies

$$\left| h(z+y) - h(z) - h'(z) y - \tfrac{1}{2} h''(z) y^2 \right| \leq \frac{[h'']_{C^{0,\alpha}}}{(1+\alpha)(2+\alpha)} |y|^{2+\alpha}. \tag{C.20}$$

Indeed, subtracting the quadratic term in Taylor's integral formula gives $y^2 \int_0^1 (1-r)(h''(z+ry) - h''(z))\, \mathrm{d}r$. The linear and quadratic terms cancel after conditional expectation. Summing the N replacement errors yields

$$|\mathbb{E}[h(U(\xi_1, \dots, \xi_N))] - \mathbb{E}[h(U(g_1, \dots, g_N))]| \leq C_\alpha [h'']_{C^{0,\alpha}} N^{-\alpha/2} \|c_m\|_p^p. \tag{C.21}$$

All expectations are finite: h is Lipschitz, its first two derivatives are bounded, and (C.19) supplies the Taylor-remainder integrability. This is the one-coordinate moment-matching argument of [Cha06, Section 2], with the fractional and vector-valued estimates made explicit here.

The Gaussian statistic and passage to the kernel. An orthogonal change in the coordinate index, with first vector $N^{-1/2}(1, \dots, 1)$, gives

$$U(g_1, \dots, g_N) \overset{\text{law}}{=} \frac{N-1}{2N} \mathcal{Q}^{(0)}_{C_m} - \frac{1}{2N} \sum_{r=1}^{N-1} \mathcal{Q}^{(r)}_{C_m}, \tag{C.22}$$

where C_m is the operator of c_m and the superscripts denote independent copies. Diagonalize A and use $\sum_{i<j} g_i g_j = ((\sum_i g_i)^2 - \sum_i g_i^2)/2$ in each mode. The trace constants cancel, proving the identity for either sign of A. Couple the right-hand side with $\mathcal{Q}^{(0)}_{C_m}/2$. Their difference is $-(2N)^{-1} \sum_{r=0}^{N-1} \mathcal{Q}^{(r)}_{C_m}$, whose L^2 norm is $\|c_m\|_2/\sqrt{2N}$ by (C.10). Thus,

$$|\mathbb{E}[h(U(g_1, \dots, g_N))] - \mathbb{E}[h(\mathcal{Q}_{C_m}/2)]| \leq \frac{\|h'\|_\infty \|c_m\|_2}{\sqrt{2N}}. \tag{C.23}$$

For each fixed N, canonicity and the common Gaussian construction give

$$\left\| \tfrac{1}{2} \mathsf{W}_N (c - c_m) \right\|_2 = \sqrt{\frac{N-1}{2N}} \|c - c_m\|_2, \tag{C.24}$$

$$\left\| \tfrac{1}{2} (\mathcal{Q}_C - \mathcal{Q}_{C_m}) \right\|_2 = \frac{\|c - c_m\|_2}{\sqrt{2}}. \tag{C.25}$$

These quantities tend to zero. Since h is Lipschitz, we may send $m \to \infty$ in (C.21) and (C.23), using (C.12). This proves (5.2). No quantitative rate of the partition approximation is required. □

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C.3.3. *The absolute first moment.*

Proof of Proposition 6.5. Set $p = 2 + \alpha$. The zero-kernel case is immediate; for $k = 1$, use $\mathbb{E}[|\mathcal{Q}_C|]/2 \le \|c\|_2/\sqrt{2}$. Suppose $k \ge 2$ and $\|c\|_p > 0$. For $\delta > 0$, set $h_\delta(x) = \sqrt{x^2 + \delta^2}$. Direct differentiation gives

$$0 \le h_\delta(x) - |x| \le \delta, \qquad \|h_\delta'\|_\infty \le 1, \qquad \|h_\delta''\|_\infty \le \delta^{-1}, \qquad \|h_\delta'''\|_\infty \le 3\delta^{-2}. \tag{C.26}$$

Taking the minimum of the uniform and Lipschitz bounds for the difference of h_δ'' yields $[h_\delta'']_{C^{0,\alpha}} \le C_\alpha \delta^{-1-\alpha}$. Apply Lemma 5.1 with $N = k$ and $\rho = \mu$:

$$\begin{aligned}\left| \mathbb{E}_{\mu^{\otimes k}} \left[\left| \frac{1}{k} \sum_{i<j} c(x_i, x_j) \right| \right] - \tfrac{1}{2}\mathbb{E}[|\mathcal{Q}_C|] \right| \\ \le 2\delta + C_\alpha \delta^{-1-\alpha} k^{-\alpha/2} \|c\|_p^p + \frac{\|c\|_2}{\sqrt{2k}}.\end{aligned} \tag{C.27}$$

Taking $\delta = \|c\|_p k^{-\alpha/(2(2+\alpha))}$ proves the claim, since $\|c\|_2 \le \|c\|_p$ and $\alpha/(2(2+\alpha)) \le 1/6 < 1/2$. □

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